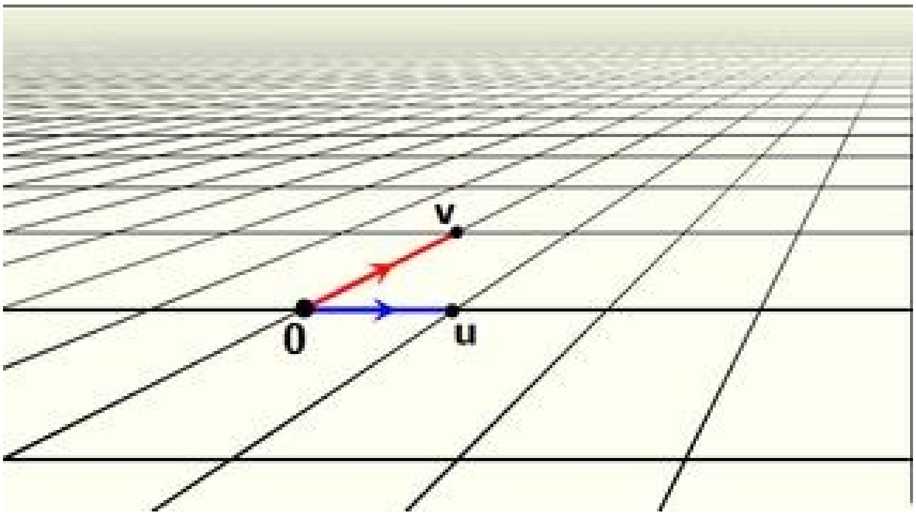
LINEAR ALGEBRA AND APPLICATIONS

Core Lecture Notes



2201NSC

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Contents

II Exercises

Core contents

0. Introduction

Welcome to Linear Algebra and Applications 2201NSC. In this part of the notes, we cover some house-keeping matters related to the running of the course, and give a brief overview to linear algebra, its importance, what we will be looking at and what we expect you to already know. In the remainder of these notes, you will find the mathematical content covered in the linear algebra part of the course, as well as the tutorial problems.

0.1 House-keeping

0.1.1 What's happening in the course?

This information can be found at Learning@Griffith (L@G, Canvas) and the Course Profile (which can also be accessed through L@G).

0.1.2 Time commitment

All 10 CP courses assume a 150-hour commitment, which translates roughly into 10 hours per week:

- 4 hours split between recorded content and face-to-face classes, to cover core material and reinforce these ideas in tutorial activities
- 4-6 hours outside of class: reviewing content (understanding theory, working through examples), doing tutorial problems

and another 30-50 hours doing assignments, revising and preparing for exams.

0.1.3 Working together

The point of this (and any) course is that you should *know more at the end than when you started*.

What you *should* do:

- help each other understand (maths, applications, context)

- share ideas for what to do

What you *should not* do:
- tell each other the answer (defeats the educative purpose)
- do the work for one another
- copy one another's work

Check the academic integrity section of the University website or ask your lecturers for clarification.

0.2 What do we need linear algebra for?

Algebra involves the use of the algebraic operations $(\times)$ and $(+)$ on mathematical objects like numbers, vectors, matrices, or functions. A *linear* operation $\mathscr{L}$ is one that obeys the relationship

$$\mathscr{L}(\alpha x+\beta y)=\alpha\mathscr{L}(x)+\beta\mathscr{L}(y)$$

which we observe in a number of common operations: scalar, dot and matrix multiplication; differentiation; and integration. *Linear Algebra begins* by considering linear problems of the type $\mathscr{L}(x)=b$, such as

$$ax=b; \qquad \mathbf{a}\cdot\mathbf{x}=b; \qquad \mathbf{Ax}=\mathbf{b}; \qquad \frac{\mathrm{d}}{\mathrm{d}t}x(t)=b(t); \qquad \int x(t)\,\mathrm{d}t=b$$

There is a wide variety of different fields where problems of this type emerge — linear algebra plays a useful and important role in helping solve them. These include:
- *Problems involving systems of linear equations*, such as interpolation of functions (e.g. cubic splines); solving for the coefficients of chemical reaction equations; solving the behaviour of electric circuits; solving economic models that balance market forces; finding locations using GPS; generating computer graphics; and medical applications such as Computer-Aided Tomography (CAT) scan.
- *Problems involving systems that change over time*, such as robotic arms and other systems of connected rigid bodies; modelling the transmission of genetically-based mutations or diseases from one generation to the next; and modelling the population levels over time of different species or subgroups of a species, based on their interactions.
- *Problems involving networks*, such as modelling (networks of) traffic flow; the robustness of an ecosystem based on the network of interactions between species; understanding the behaviour observed in social networks; and ranking pages on the internet (one of the most significant contemporary linear algebra problems).
- *Problems involving approximation*, such as finding lines of best fit; compressing images; studying fractals and unpredictability in deterministic systems; understanding and quantifying information loss in engineering (image and sound compression) and physical (entropy) applications.
- *Problems involving optimization*, such as finding the best outcome possible subject to imposed constraints; the best strategic moves in a competitive environment (game theory); the best intervention strategies (such as how best to immunize a population, or how best to harvest trees on a plantation to balance ecological and economic needs); and the best ways of assigning resources (such as choosing the best team, or matching-making on dating websites)

- *Problems involving data*. Data science uses primarily linear algebra methods in order to draw inferences from (often huge amounts of) data, and making predictions on the basis of that data. This involves identifying the factors that most influence an outcome — the same theory tells us what pieces of information we can throw away when we compress data, or the best coefficients for our lines of best fit.

Outside of these applications, there are a number of reasons from a *mathematical* perspective for studying Linear Algebra:

- Linear systems have 'nice' properties that make them *easier to solve*, which helps us solve *many real world problems*,
- Many *nonlinear systems* can be studied by building on linear theories, such as through the linearisation of non-linear ODEs, or through perturbation theory [= linear system + nonlinear perturbation].
- Many *mathematical problems* rely on linear algebra, such as solutions to DEs (regarding the *nature* of solutions), and the series representation of functions (*e.g.*, Fourier theory).

Prerequisite Knowledge It is assumed that you know or will quickly recall

- what a matrix is;
- how to add and multiply matrices;
- what a matrix transpose and symmetric matrix are;
- Gaussian elimination;
- the concepts of identity and inverse matrices;
- how to invert, calculate determinants and eigenvalues/vectors of 2×2 matrices.

0.3 Numerical calculations

Not all the calculations in the course will need to be done 'by hand'. Where appropriate, you may be able to choose which numerical software you wish to use for calculations. Choices include MATLAB/Octave, R, and Python.

0.3.1 Basic matrix entry

Let's consider how to enter the matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -3 \end{pmatrix}$$

into the three programs outlined.

In MATLAB/Octave, use square brackets $[\cdots]$ to indicate a matrix. Enter elements left to right, separating elements in the same row by spaces and separating columns with a semicolon. Thus,

```
A = [1 1 1 1 1 1; 0 0 1 1 2 0; 0 0 0 0 1 -3];
```

will create the matrix A in MATLAB's memory. MATLAB automatically determines the size of the matrix. The final semicolon, outside the square bracket, stops MATLAB from printing the result of an input in the command window.

In R, we can create the matrix A using:

```
A <- matrix(c(1,1,1,1,1,1,0,0,1,1,2,0,0,0,0,0,1,-3), 3, 6, byrow = TRUE
   )
```

Note that here we needed to stipulate the numbers of rows (3) and columns (6). The command byrow = TRUE tells R to read elements off row-by-row, and the command c tells R to combine the input data into a single object in memory. Note also that the assignment operator is <- rather than = as in MATLAB (and Python).

In Python, we need to first load the package numpy (Numerical Python), which tells Python how to handle matrices.

```
import numpy as np
A = np.array([[1,1,1,1,1,1], [0,0,1,1,2,0], [0,0,0,0,1,-3]])
```

Each row is entered in its own set of square brackets with elements separated by commas. Columns are separated by commas inside an outer pair of square brackets. The np. tells Python to use the function array defined by the numpy package.

0.3.2 Accessing elements or submatrices

Suppose we want to ask our programs to extract the $a_{25} = 2$ (row 2, column 5) element of A.

- MATLAB: A(2,5)
- R: A[2,5]
- Python: A[1,4]

Note that we stipulate row and then column in all cases. However, *Python uses the computer science convention of indexing from* 0 *instead of* 1; be very careful of this to avoid confusion!

Suppose we want to create a matrix B given by the upper left 2×3 block of A. We can do this as follows.

- MATLAB: B = A(1:2,1:3)
- R: B = A[1:2,1:3]
- Python: B = A[0:2,0:3]

For MATLAB and R we stipulate the desired row index range (separated by a colon) and then the column index range. Python also takes arguments for rows and then columns, but it asks for the beginning index and then includes all entries up to *but not including* the second index.

1. Simultaneous equations

By the end of this chapter, you should be able to

- solve a system of linear equations by hand or using software; and
- express the solution (as a unique solution or family of solutions), or identify that there is no solution.

1.1 Linear equations

1.1.1 Definition and examples of linear equations

Definition 1.1.1 — Linear Equations. A (real) linear equation in n unknowns is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where

- $a_1, a_2, \ldots, a_n$ and b are (real) coefficients; and
- $x_1, x_2, \ldots, x_n$ are (real) variables.

■ **Example 1.1 — Examples of linear questions.**

- $2x_1 + 3x_2 - x_3 - 5x_4 = 0$ is a linear equation in x_1, x_2, x_3 and x_4
- $2r + 3s = 4$ is a linear equation in r and s

■

Definition 1.1.2 — System of Linear Equations. A *system* of linear equation is a collection of m simultaneous equations in n unknowns, of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

Example 1.2 — Examples of systems of linear equations.

- Two systems with variables x and y ...

$$\begin{aligned} 2x + 3y &= 1 \\ 3x - y &= 2 \end{aligned} \qquad\qquad \begin{aligned} ax + by &= r \\ cx + dy &= s \end{aligned}$$

- ... and a system in variables x_1, x_2 and x_3

$$\begin{aligned} x_1 + x_2 + x_3 &= 3 \\ x_1 + x_2 - x_3 &= 1 \\ x_1 - x_2 + x_3 &= 1 \end{aligned}$$

■

Example 1.3 — Balancing a chemical equation. Photosynthesis converts water and carbon dioxide into oxygen and glucose:

$$x_1 CO_2 + x_2 H_2O \;\rightarrow\; x_3 O_2 + x_4 C_6H_{12}O_6$$

Since matter is neither created nor destroyed, we must have equal amounts of each element on either side, which provides us with equations for each x_i.

	C	H	O
LHS	x_1	$2x_2$	$2x_1 + x_2$
RHS	$6x_4$	$12x_4$	$2x_3 + 6x_4$

Equating left and right hand coefficients for each element gives the system of equations

$$\begin{aligned} x_1 &= 6x_4 \\ 2x_2 &= 12x_4 \\ 2x_1 + x_2 &= 2x_3 + 6x_4 \end{aligned} \qquad \text{which we re-arrange to give} \qquad \begin{aligned} x_1 - 6x_4 &= 0 \\ 2x_2 - 12x_4 &= 0 \\ 2x_1 + x_2 - 2x_3 - 6x_4 &= 0 \end{aligned}$$

■

1.1.2 Solving linear systems — the need for Gaussian elimination

We can solve the following linear system of two equations with two variables

$$2x_1 + x_2 = 3 \tag{1.1}$$
$$x_1 - x_2 = 0 \tag{1.2}$$

by re-arranging (1.2) to express x_2 in terms of x_1 ($x_2 = x_1$), substituting this result into (1.1) (to give $2x_1 + x_1 = 3 \Rightarrow x_1 = 1$) and then substituting back into (1.2) (to get $x_2 = x_1 = 1$).

This method works fine for relatively simple systems such as this one, but what about for much more complicated systems such as

$$\begin{aligned} 2x_1+x_2+x_3-4x_4+2x_5&=3\\ x_1-x_2-x_4+x_5&=0\\ x_2+2x_3-x_4+2x_5&=1\\ x_1-x_2-2x_3-3x_4-4x_5&=10\\ x_1+2x_2+3x_3-4x_4+x_5&=-2 \quad ??? \end{aligned}$$

It isn't clear that the approach we used above for the two-variable system is particularly practical (or even guaranteed to work!) for the five-variable example here. Instead, *we need a systematic approach* to answer questions such as

- does a solution always exist?
- if it exists, what is the solution?

Gaussian elimination provides such a systematic approach.

1.2 Gaussian elimination – a systematic approach

1.2.1 Outline of the Gaussian elimination method

We represent the system

$$\begin{aligned} 2x_1+x_2&=3\\ x_1-x_2&=0 \end{aligned}$$

using the *augmented matrix*

$$\left(\begin{array}{cc|c} 2 & 1 & 3\\ 1 & -1 & 0 \end{array}\right)$$

and *solve* this system using *row operations*. This approach is called *row reduction*, or *Gaussian elimination*. There are *three* allowed row operations

1. Interchanging two rows, *i.e.*,

$$\begin{array}{cc} \left(\begin{array}{ccc|c} 0 & -4 & 8 & 4\\ 1 & -1 & 1 & 0\\ 2 & 2 & 3 & 4 \end{array}\right) & \begin{array}{l} \text{R1}\\ \text{R2}\\ \text{R3} \end{array} \end{array} \longrightarrow \begin{array}{cc} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0\\ 0 & -4 & 8 & 4\\ 2 & 2 & 3 & 4 \end{array}\right) & \begin{array}{l} \text{R2}\\ \text{R1}\\ \text{R3} \end{array} \end{array}$$

2. Rescaling[1] one row, *i.e.*,

$$\begin{array}{cc} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0\\ 0 & -4 & 8 & 4\\ 2 & 2 & 3 & 4 \end{array}\right) & \begin{array}{l} \text{R1a}\\ \text{R2a}\\ \text{R3} \end{array} \end{array} \longrightarrow \begin{array}{cc} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0\\ 0 & 1 & -2 & -1\\ 2 & 2 & 3 & 4 \end{array}\right) & \begin{array}{l} \text{R1a}\\ -\text{R2b}/4\\ \text{R3} \end{array} \end{array}$$

3. Adding a multiple of one row to another row, *i.e.*,

$$\begin{array}{cc} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0\\ 0 & 1 & -2 & -1\\ 2 & 2 & 3 & 4 \end{array}\right) & \begin{array}{l} \text{R1a}\\ \text{R2b}\\ \text{R3} \end{array} \end{array} \longrightarrow \begin{array}{cc} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0\\ 0 & 1 & -2 & -1\\ 0 & 4 & 1 & 4 \end{array}\right) & \begin{array}{l} \text{R1a}\\ \text{R2b}\\ \text{R3}-2.\text{R1} \end{array} \end{array}$$

[1] So long as we do not multiply by zero.

These operations are allowed because *they produce a new set of equations with the same solution*.

Knowing the allowed steps begs the question of how we use them to get at a solution to our system. In first year, you've seen how we can approach this when the problem has a unique solution. In that case, we perform these steps until we achieve strict upper-triangular form, where all the matrix elements below the main diagonal are reduced to zeros. It is also helpful (although not essential) to reduce the diagonal elements to ones.

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & q_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & q_2 \\ \vdots & & & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & q_n \end{array}\right) \longrightarrow \left(\begin{array}{cccc|c} 1 & b_{12} & \cdots & b_{1n} & r_1 \\ 0 & 1 & \cdots & b_{2n} & r_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & r_n \end{array}\right)$$

Once we get to this point, we can calculate the solution by *back-substitution*:

- The bottom line gives
$$x_n = r_n$$
- the line above then gives
$$x_{n-1} + b_{(n-1)n} x_n = r_{n-1} \Rightarrow x_{n-1} = r_{n-1} - b_{(n-1)n} r_n$$
- and so on ...

An equivalent, alternative approach is to continue doing row-reduction until the left-side matrix is an *identity* matrix — ones down the main diagonal, zero everywhere else. At this point, the right-side column will be the *solution*. This approach effectively replaces the back-substitution with more row-reduction, to achieve the same objective.

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & q_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & q_2 \\ \vdots & & & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & q_n \end{array}\right) \longrightarrow \left(\begin{array}{cccc|c} 1 & b_{12} & \cdots & b_{1n} & r_1 \\ 0 & 1 & \cdots & b_{2n} & r_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & r_n \end{array}\right) \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & \cdots & 0 & s_1 \\ 0 & 1 & \cdots & 0 & s_2 \\ \vdots & & & & \\ 0 & 0 & \cdots & 1 & s_n \end{array}\right)$$

Because humans typically find row-reducing harder than back-substitution, we tend to stop when we reach the upper-triangular form and consider whether to keep reducing, or to back-substitute. Computers have no such problem, so persisting all the way to the identity matrix is the standard approach for programming computers to perform row reduction

■ Example 1.4 — Row-reducing two equations of two variables.

$$\begin{aligned} 2x_1 + x_2 &= 3 \\ x_1 - x_2 &= 0 \end{aligned} \quad \Rightarrow \quad \left(\begin{array}{cc|c} 2 & 1 & 3 \\ 1 & -1 & 0 \end{array}\right) \begin{array}{l} \text{R1} \\ \text{R2} \end{array}$$

$$\left(\begin{array}{cc|c} 1 & -1 & 0 \\ 2 & 1 & 3 \end{array}\right) \begin{array}{l} \text{R1a=R2} \\ \text{R2a=R1} \end{array}$$

$$\left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 3 & 3 \end{array}\right) \begin{array}{l} \text{R1a} \\ \text{R2b=R2a}-2.\text{R1a} \end{array}$$

$$\left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 1 & 1 \end{array}\right) \begin{array}{l} \text{R1a} \\ \text{R2c=R2b/3} \end{array} \quad \Rightarrow \quad \begin{aligned} x_1 - x_2 &= 0 \\ x_2 &= 1 \end{aligned}$$

$$\left[\quad \Rightarrow \quad \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right) \begin{array}{l} \text{R1b=R1a+R2c} \\ \text{R2c} \end{array} \quad \Rightarrow \quad x_1 = 1, x_2 = 1 \right]$$

■

Example 1.5 — A different reducing path still leads to the same solution (phew!!).

$$\begin{aligned} 2x_1 + x_2 &= 3 \\ x_1 - x_2 &= 0 \end{aligned} \quad \Rightarrow \quad \left(\begin{array}{cc|c} 2 & 1 & 3 \\ 1 & -1 & 0 \end{array}\right) \begin{array}{l} \text{R1} \\ \text{R2} \end{array}$$

$$\left(\begin{array}{cc|c} 1 & \frac{1}{2} & \frac{3}{2} \\ 1 & -1 & 0 \end{array}\right) \begin{array}{l} \text{R1a=R1/2} \\ \text{R2} \end{array}$$

$$\left(\begin{array}{cc|c} 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & -\frac{3}{2} & -\frac{3}{2} \end{array}\right) \begin{array}{l} \text{R1a} \\ \text{R2a=R2}-\text{R1a} \end{array}$$

$$\left(\begin{array}{cc|c} 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 1 \end{array}\right) \begin{array}{l} \text{R1a} \\ \text{R2b=}-\frac{2}{3}\text{R2a} \end{array} \quad \Rightarrow \quad \begin{aligned} x_1 + \tfrac{1}{2}x_2 &= \tfrac{3}{2} \\ x_2 &= 1 \end{aligned}$$

$$\left[\quad \Rightarrow \quad \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right) \begin{array}{l} \text{R1b=R1a-R2b/2} \\ \text{R2c} \end{array} \quad \Rightarrow \quad x_1 = 1, x_2 = 1 \right]$$

■

Example 1.6 — A larger system.

$$\begin{aligned} 3x_1 + 2x_2 - x_3 &= -2 \\ 3x_1 + x_2 - x_3 &= -5 \\ 3x_1 + 2x_2 + x_3 &= 2 \end{aligned} \quad \Rightarrow \quad \left(\begin{array}{ccc|c} 3 & 2 & -1 & -2 \\ 3 & 1 & -1 & -5 \\ 3 & 2 & 1 & 2 \end{array}\right)$$

$$\Rightarrow \quad \left(\begin{array}{ccc|c} 3 & 2 & -1 & -2 \\ 0 & -1 & 0 & -3 \\ 0 & 0 & 2 & 4 \end{array}\right) \begin{array}{l} \text{R1} \\ \text{R2-R1} \\ \text{R3-R1} \end{array}$$

Back-substitution gives

$$\begin{aligned} 2x_3 &= 4 & \Rightarrow \quad & x_3 = 2 \\ -x_2 &= -3 & \Rightarrow \quad & x_2 = 3 \\ 3x_1 + 2x_2 - x_3 = 3x_1 + 6 - 2 &= -2 & \Rightarrow \quad & x_1 = -2 \end{aligned}$$

■

Let's check that these answers satisfy our original equations:

$$\begin{aligned} 3.(-2) + 2.3 - 2 &= -6 + 6 - 2 = -2 \quad ✔ \\ 3.(-2) + 3 - 2 &= -6 + 3 - 2 = -5 \quad ✔ \\ 3.(-2) + 2.3 + 2 &= -6 + 6 + 2 = 2 \quad ✔ \end{aligned}$$

Important Tip: Always check your answer!

Once you have calculated your solution, it should satisfy the original equations. It is simple, and always a good idea, to check your answers actually do this – it is an easy way to confirm you have the right answer, or to discover that you've got something wrong. ■

1.3 Row echelon form

1.3.1 Definition of row echelon form

Gaussian elimination can also tell us when our system doesn't have a solution, or give us the full solution when it is not unique. But how do we know when to recognise those cases, and what steps we should be taking in the row reduction? In order to solve any linear system, we must first row-reduce it until we reach *row echelon form*:

Definition 1.3.1 — Row Echelon Form. A matrix is in **row echelon form** if

1. any rows of zeros are at the bottom; and
2. the first nonzero term in any row – the *leading* entry – is *to the left of any leading entries below it*.

This is the more general form of the upper-triangular matrix form described in the previous section.

■ **Example 1.7** A is not in row echelon form because row 3 has fewer leading zeros than row 2 – swapping these rows (producing matrix B) generates row-echelon form.

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -1 & 0 \\ 0 & 2 & 0 & 4 \end{pmatrix} \quad \text{R3 has } \textit{fewer} \text{ leading zeros as R2} \quad \times$$
$$B = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 2 & 0 & 4 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad \text{Swapping R2 and R3} \quad \checkmark$$

■

■ **Example 1.8** A is not in row echelon form because row 3 has as many leading zeros as row 2 – a further reduction in row 3 to eliminate this leading term (producing matrix B) generates row-echelon form.

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & -1 & 0 \\ 0 & 2 & 0 & 4 \end{pmatrix} \quad \text{R3 has } \textit{as many} \text{ leading zeros as R2} \quad \times$$
$$B = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 4 \end{pmatrix} \quad \text{R3} \to \text{R3-2R2} \quad \checkmark$$

■

■ **Example 1.9** A is in row echelon form, but B is not because row 3 has as many leading zeros as row 2 – a further reduction in row 3 to eliminate this leading term (which will recover matrix A) generates row-echelon form.

$$A = \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right) \qquad \text{all good} \qquad ✔$$

$$B = \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 & 3 & -3 \\ 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right) \qquad \text{R3 has } \textit{as many zeros} \text{ as R2} \qquad \times$$

■

1.3.2 Possible row reduction outcomes

Once your row-reduced augmented matrix is in row echelon form, you can stop row-reducing!! The solution to the reduced augmented matrix will be the same as the solution to the original – but it is possible to draw immediate conclusions about the nature of the solution from the nature of the row echelon form matrix. Let's look at some simple systems, and compare the nature of the solution to their row echelon form.

■ Example 1.10

$$\begin{array}{c} x+y=2 \\ x-y=2 \end{array} \to \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & -1 & 2 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -2 & 0 \end{array}\right)$$

- By observation, $y = 0, x = 2$ – a *unique solution*
- We have *as many* equations as variables

■

■ Example 1.11

$$\begin{array}{c} x+y=2 \\ x+y=1 \end{array} \to \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & 1 & 1 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array}\right)$$

- By observation, there is no solution – the system is *inconsistent*
- We have *as many* equations as variables

■

■ Example 1.12

$$\begin{array}{c} x+y=2 \\ -x-y=-2 \end{array} \to \left(\begin{array}{cc|c} 1 & 1 & 2 \\ -1 & -1 & -2 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right)$$

- We have *as many* equations as variables
- The same equation appears twice – we have a *family of solutions* given by that single equation

■

■ Example 1.13

$$\begin{array}{c} x+y+z=1 \\ x+y+2z=2 \end{array} \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

- We have *fewer* equations than variables – *underdetermined* system

- There is a *family of solutions* – $z = 1, y = -x$, with x a free variable (able to take on any value)

■

■ Example 1.14

$$\begin{aligned} x+y&=2 \\ x+2y&=3 \\ 2x+2y&=4 \end{aligned} \quad \to \quad \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 2 & 4 \end{array}\right) \quad \to \quad \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right)$$

- We have *more* equations than variables – *overdetermined* system
- There is *a unique solution* – $y = 1, x = 1$

■

■ Example 1.15

$$\begin{aligned} x+y&=2 \\ x+2y&=3 \\ 2x+2y&=5 \end{aligned} \quad \to \quad \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 2 & 5 \end{array}\right) \quad \to \quad \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array}\right)$$

- We have *more* equations than variables – *overdetermined* system
- There is *no solution* – there are no x, y for which $0x + 0y = 1$

■

Interpretation of row echelon forms

- Rows of the form $\left(\begin{array}{cccc|c} 0 & 0 & \cdots & 0 & 1 \end{array}\right)$
  - *Inconsistent system* – **no solution**
- Rows of the form $\left(\begin{array}{cccc|c} 0 & 0 & \cdots & 0 & 0 \end{array}\right)$
  - There is a redundant equation for each such row
- Rows of the form $\left(\begin{array}{ccccc|c} \cdots & a_1 & a_2 & \cdots & a_m & r \end{array}\right)$
  - *As many such rows* as variables ⇒ **unique solution**
  - *Fewer such rows* than variables ⇒ **family of solutions**

■

Common (but not guaranteed) outcomes

- *Overdetermined* systems usually have *no solution*
  - except when the extra equations are combinations for the first equations
- *Underdetermined* systems usually have a *family of solutions*
  - except when they are inconsistent

■

Problem 1.1 Interpret each of the following row echelon forms:

a) $\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right)$ b) $\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$

c) $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right)$ d) $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \end{array}\right)$

1.3.3 Expressing families of solutions

If the row echelon form suggests a family of solutions, you should express the solution in terms of lead and free variables

Definition 1.3.2 — Lead and free variables. If each line of a row echelon form is written in terms of the original variables, then

1. each variable appearing first in an equation is a *lead* variable; and
2. the remaining variables are *free* (or undetermined) variables.

When a system of equations has a family of solutions, we can represent this solution by expressing each lead variable as a function of the free variables, as in the examples below.

Example 1.16

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \end{array}\right) \to \begin{aligned} x_1 + x_2 + x_3 &= 1 \\ x_2 + 2x_3 &= 0 \end{aligned} \to \begin{aligned} x_1 + x_2 &= 1 - x_3 \\ x_2 &= -2x_3 \end{aligned}$$

So x_1 and x_2 are lead variables, leaving x_3 as a free variable. To complete the problem, we express each of the lead variables in terms of free variables:

$$\begin{aligned} x_2 &= -2x_3 \\ x_1 &= 1 - x_2 - x_3 = 1 - (-2x_3) - x_3 = 1 + x_3 \end{aligned}$$

Note that, once x_3 is given, x_1 and x_2 are uniquely determined. ■

Example 1.17

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \to \begin{aligned} x_1 + x_2 + x_3 + x_4 + x_5 &= 1 \\ x_3 + x_4 + 2x_5 &= 0 \\ x_5 &= -3 \end{aligned} \to \begin{aligned} x_1 + x_3 + x_5 &= 1 - x_2 - x_4 \\ x_3 + 2x_5 &= -x_4 \\ x_5 &= -3 \end{aligned}$$

So x_1, x_3 and x_5 are lead variables, leaving x_2 and x_4 as free variables. Again, we complete the problem by expressing these lead variables as functions of the free variables:

$$\begin{aligned} x_5 &= -3 \\ x_3 &= -2x_5 - x_4 = 6 - x_4 \\ x_1 &= 1 - x_2 - x_3 - x_4 - x_5 = 1 - x_2 - (6 - x_4) - x_4 - (-3) = -2 - x_2 \end{aligned}$$

Note again that, once x_2 and x_4 are given, x_1, x_3 and x_5 are uniquely determined. ■

1.3.4 Reduced row echelon form

When discussing the unique solution case, we noted that computers are programmed to reduce to the identity matrix, while people often prefer to stop sooner and back-substitute. This situation extends into the more general cases we have just been considering. We program computers to continue row reducing until we reach the extension of the identity matrix in these more general cases, *reduced row echelon form*:

Definition 1.3.3 — Reduced Row Echelon Form. A matrix is in **reduced row echelon form** if

1. it is in row echelon form;
2. all leading entries in the nonzero rows are all 1; and
3. all leading entries are the only nonzero entry in their column.

The process for continuing until we reach reduced row echelon form is called **Gauss-Jordan elimination**[2]

This form requires more row-reduction, but removes the need for back-substitution, because the lead variables only appear in one column each (because of condition 3) and they have coefficient 1 (because of condition 2). It is also preferable because the reduced row echelon form is unique, for a given system of equations.

■ **Example 1.18**

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \to \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & -2 \\ 0 & 0 & 1 & 1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \to \begin{array}{l} x_1 = -2 - x_2 \\ x_3 = 6 - x_4 \\ x_5 = -3, \end{array}$$

agreeing with Example 1.17. Here we have not had to back-substitute to get the lead variables in terms of the free variables: that automatically occurs when we express the solution from the reduced row echelon form. ■

1.3.5 Calculating reduced row echelon forms numerically

The MATLAB/Octave command rref reduces a matrix to reduced row echelon form, so can be used to solve systems of linear equations and to interpret the results if there are families of solutions, or if no solution exists. For example, to obtain the reduced row reduction above, we can use the MATLAB/Octave command (omitting the redundant lines of zeros for convenience)

```
rref([1 1 1 1 1 1;0 0 1 1 2 0;0 0 0 0 1 -3])
ans =

    1    1    0    0    0   -2
    0    0    1    1    0    6
    0    0    0    0    1   -3
```

In the statistical environment R, the pracma package provides a command rref to perform row reduction to row echelon form:

```
library(pracma)
A <- matrix(c(1,1,1,1,1,1,0,0,1,1,2,0,0,0,0,0,1,-3), 3, 6, byrow = TRUE
    )
rref(A)
     [,1] [,2] [,3] [,4] [,5] [,6]
[1,]    1    1    0    0    0   -2
[2,]    0    0    1    1    0    6
[3,]    0    0    0    0    1   -3
```

Python does not offer a simple built-in way of performing row reduction.

[2] Gaussian elimination only need go as far as row echelon form

1.4 Real-world examples of row reduction

■ **Example 1.19 — Application: Dimensional Analysis.** Dimensional analysis is a powerful technique whereby physical (and other) laws can be determined purely from the dimensions (*i.e.*, units) of the quantities being considered. Probably the most commonly used example is finding the period of a swinging pendulum.

What could the period T of a pendulum depend on? A first guess might include:

- Pendulum length l, measured in m.
- Pendulum mass m, measured in kg.
- Strength of the earth's gravitational field, measured as the acceleration due to gravity, g, in m/s^2.

The period T is a time, measured in s.

In the approach of dimensional analysis, we propose a formula of the form

$$T \propto l^a m^b g^c$$

If this formula holds, the *units* of both sides of the equation must be the same, up to some multiplicative constant. Considering the units here, we have that

$$\mathrm{s} = \mathrm{m}^a \mathrm{kg}^b (\mathrm{m}/\mathrm{s}^2)^c = \mathrm{m}^{a+c} \mathrm{kg}^b \mathrm{s}^{-2c}$$

Equating the exponents on either side gives us three equations:

$$\begin{aligned} a + c &= 0 \\ b &= 0 \\ -2c &= 1 \end{aligned}$$

Although we can tackle this by hand, let's go through the row-reduction process using the corresponding augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1/2 \end{array}\right) \begin{array}{l} \text{R1} \\ \text{R2} \\ \text{-R3/2} \end{array} \to \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1/2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1/2 \end{array}\right) \begin{array}{l} \text{R1-R3a} \\ \text{R2} \\ \text{R3a} \end{array}$$

so $a = 1/2, b = 0$ and $c = -1/2$, and the formula for the period becomes

$$T \propto \sqrt{\frac{l}{g}}$$

Note that this method is not confused by redundant inputs — including the mass (which doesn't end up mattering) as an input doesn't cause any problems, for example. The formula agrees with the standard small-angle formula for the period of a pendulum, $T = 2\pi\sqrt{\frac{l}{g}}$.

For a more sophisticated example, let's also include the height h_0 at which the pendulum is released. h_0 is clearly a length, so will also have units of metres. We propose a relation of the form

$$T \propto l^a m^b g^c h_0^d$$

which leads to the 'units' equation

$$\mathrm{s} = \mathrm{m}^a \mathrm{kg}^b (\mathrm{m/s}^2)^c \mathrm{m}^d = \mathrm{m}^{a+c+d} \mathrm{kg}^b \mathrm{s}^{-2c}$$

and subsequent system of exponent equations

$$\begin{aligned} a+c+d &= 0 \\ b &= 0 \\ -2c &= 1 \end{aligned}$$

Row reduction gives

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 1 \end{array}\right) \to \begin{array}{c} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1/2 \end{array}\right) \end{array} \begin{array}{l} \text{R1} \\ \text{R2} \\ \text{-R3/2} \end{array} \to \left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 1/2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1/2 \end{array}\right) \begin{array}{l} \text{R1-R3a} \\ \text{R2} \\ \text{R3a} \end{array}$$

which corresponds to a *family* of solutions with $c = -1/2, b = 0$ as before, but now with $a = 1/2 - d$, with free variable d. This gives us the possible formulas

$$T \propto l^{1/2-d}\, h_0^d\, g^{-1/2} = \sqrt{\frac{l}{g}} \left(\frac{h_0}{l}\right)^d$$

where d could take on any real value! We need more details about the problem to work out which value d (or, as it turns out, which combination of different values of d) gives the correct solution. ■

■ **Example 1.20 — Application: Monetisation.** As societies become more economically complicated, they typically switch from exchange of goods and services to a system of money. An important part of this monetisation process is working out a value of each commodity that is fair, based on the existing exchanges that take place. The matrix methods that we have looked at can help us solve this problem, using a Leontief input-output model (which led to the awarding of a Nobel prize in economics).

Imagine trade between groups in an unmonetised society of farmers, manufacturers and clothing producers, where each group keeps a certain fraction of its own produce, and exchanges the rest with the other groups. The table below represents such an exchange.

	Farmers	Manufacturing	Clothing
trade with F	(1/2)	1/3	1/2
trade with M	1/4	(1/3)	1/4
trade with C	1/4	1/3	(1/4)

So, for example, farmers keep 1/2 their produce, trade 1/4 on tools, 1/4 on clothes, etc. If we introduce a common currency, what would be a fair price for each commodity? ■

2. Matrices

2.1 Matrix algebra

While the ideas in the following section should be familiar from first year, here we introduce some standard notation that is used to describe general matrices.

2.1.1 Matrix notation

Definition 2.1.1 — A matrix. A *matrix* A is an $m \times n$ array of numbers:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Note that:

- the order of the subscripts is (row, column);
- We can also denote $A = (a_{ij})$;
- For real matrices (matrices with real elements), we write $A \in \mathbb{R}^{m \times n}$; and
- A matrix with $m = 1$ or $n = 1$ is usually called a *vector*. We will use lower-case bold letters to represent vectors.
  - we will denote column vectors as $\mathbf{a}, \mathbf{b}$, etc.
  - and we will denote row vectors as $\vec{\mathbf{a}}, \vec{\mathbf{b}}$, etc.

■ **Example 2.1 — Row and column vector notation.** If

$$A = \begin{pmatrix} 3 & 2 & 5 \\ -1 & 8 & 4 \end{pmatrix}$$

then we denote the *column vectors*

$$\mathbf{a}_1 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 2 \\ 8 \end{pmatrix}, \quad \mathbf{a}_3 = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

and the *row vectors*

$$\vec{\mathbf{a}}_1 = \begin{pmatrix} 3 & 2 & 5 \end{pmatrix}, \quad \vec{\mathbf{a}}_2 = \begin{pmatrix} -1 & 8 & 4 \end{pmatrix}.$$

■

2.1.2 Matrix operations

Definition 2.1.2 — Matrix equality. If $A = B$, then $a_{ij} = b_{ij}$ for each i and j

■ **Example 2.2 — Matrix equality.**

$$\text{If } B = A = \begin{pmatrix} 3 & 2 & 5 \\ -1 & 8 & 4 \end{pmatrix}, \text{ then}$$

$$b_{11} = a_{11} = 3, \quad b_{12} = a_{12} = 2, \quad \ldots, \quad b_{23} = a_{23} = 4.$$

■

Definition 2.1.3 — Matrix multiplication by a scalar. If $A \in \mathbb{R}^{m \times n}$ and $\alpha \in \mathbb{R}$ (*i.e.*, a scalar), then

$$\alpha A = \begin{pmatrix} \alpha a_{11} & \alpha a_{12} & \cdots & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{22} & \cdots & \alpha a_{2n} \\ \vdots & & & \\ \alpha a_{m1} & \alpha a_{m2} & \cdots & \alpha a_{mn} \end{pmatrix}$$

■ **Example 2.3 — Matrix multiplication by a scalar.**

$$\text{If} \quad A = \begin{pmatrix} 3 & 2 & 5 \\ -1 & 8 & 4 \end{pmatrix} \quad \text{then} \quad 2A = \begin{pmatrix} 6 & 4 & 10 \\ -2 & 16 & 8 \end{pmatrix}, \text{ and}$$

$$\text{if} \quad B = \begin{pmatrix} 3 & 3 \\ -6 & 9 \\ 0 & 3 \end{pmatrix} \quad \text{then} \quad \frac{B}{3} = \begin{pmatrix} 1 & 1 \\ -2 & 3 \\ 0 & 1 \end{pmatrix}.$$

■

Definition 2.1.4 — Matrix addition. if $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times n}$, then

$$A+B = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & & & \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix}$$
$$= \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \cdots & a_{1n}+b_{1n} \\ a_{21}+b_{21} & a_{22}+b_{22} & \cdots & a_{2n}+b_{2n} \\ \vdots & & & \\ a_{m1}+b_{m1} & a_{m2}+b_{m2} & \cdots & a_{mn}+b_{mn} \end{pmatrix}$$

■ **Example 2.4 — Matrix addition.**

$$\text{If} \quad A = \begin{pmatrix} 3 & 2 & 5 \\ -1 & 8 & 4 \end{pmatrix} \quad \text{then} \quad B = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 0 & -1 \end{pmatrix}, \text{ then}$$

$$A+B = \begin{pmatrix} 5 & 4 & 6 \\ 1 & 8 & 3 \end{pmatrix} \quad \text{and} \quad A-B = \begin{pmatrix} 1 & 0 & 4 \\ -3 & 8 & 5 \end{pmatrix}.$$

■

Definition 2.1.5 — Matrix multiplication. if $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, we define the product

$$C = AB = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & & & \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{pmatrix}$$
$$= \begin{pmatrix} \sum_i a_{1i}b_{i1} & \sum_i a_{1i}b_{i2} & \cdots & \sum_i a_{1i}b_{ip} \\ \sum_i a_{2i}b_{i1} & \boxed{\sum_i a_{2i}b_{i2}} & \cdots & \sum_i a_{mi}b_{ip} \\ \vdots & \vdots & & \vdots \\ \sum_i a_{mi}b_{i1} & \sum_i a_{mi}b_{i2} & \cdots & \sum_i a_{mi}b_{ip} \end{pmatrix}$$

- Each element of C is the sum of products between a row of A and a column of B, *e.g.*

$$c_{22} = \sum_i a_{2i}b_{i2} = a_{21}b_{12} + a_{22}b_{22} + \cdots + a_{2n}b_{n2}$$

- For this to work, we require that the number of columns of $A =$ the number of rows of B
- $C \in \mathbb{R}^{m\times p}$ — it is an $m \times p$ matrix

■ **Example 2.5 — Matrix multiplication.**

$$\text{If } A=\begin{pmatrix}3&-2\\2&4\\1&-3\end{pmatrix} \text{ and } B=\begin{pmatrix}-2&1&3\\4&1&6\end{pmatrix}, \text{ then}$$

$$AB=\begin{pmatrix}3&-2\\2&4\\1&-3\end{pmatrix}\begin{pmatrix}-2&1&3\\4&1&6\end{pmatrix}$$

$$=\begin{pmatrix}3\cdot(-2)+(-2)\cdot 4 & 3\cdot 1+(-2)\cdot 1 & 3\cdot 3+(-2)\cdot 6\\ 2\cdot(-2)+4\cdot 4 & 2\cdot 1+4\cdot 1 & 2\cdot 3+4\cdot 6\\ 1\cdot(-2)+(-3)\cdot 4 & 1\cdot 1+(-3)\cdot 1 & 1\cdot 3+(-3)\cdot 6\end{pmatrix}=\begin{pmatrix}-14&1&-3\\12&6&30\\-14&-2&-15\end{pmatrix}$$

$$\text{and}\quad BA=\begin{pmatrix}-2&1&3\\4&1&6\end{pmatrix}\begin{pmatrix}3&-2\\2&4\\1&-3\end{pmatrix}$$

$$=\begin{pmatrix}(-2)\cdot 3+1\cdot 2+3\cdot 1 & (-2)\cdot(-2)+1\cdot 4+3\cdot(-3)\\ 4\cdot 3+1\cdot 2+6\cdot 1 & 4\cdot(-2)+1\cdot 4+6\cdot(-3)\end{pmatrix}=\begin{pmatrix}-1&-1\\20&-22\end{pmatrix}$$

Note that $AB \neq BA$ in general (with isolated exceptions). ■

$$\text{If } A=\begin{pmatrix}1&2\\4&5\\3&6\end{pmatrix} \text{ and } B=\begin{pmatrix}3&4\\1&2\end{pmatrix}, \text{ then}$$

$$AB=\begin{pmatrix}1&2\\4&5\\3&6\end{pmatrix}\begin{pmatrix}3&4\\1&2\end{pmatrix}=\begin{pmatrix}1\cdot 3+2\cdot 1 & 1\cdot 4+2\cdot 2\\ 4\cdot 3+5\cdot 1 & 4\cdot 4+5\cdot 2\\ 3\cdot 3+6\cdot 1 & 3\cdot 4+6\cdot 2\end{pmatrix}=\begin{pmatrix}5&8\\17&26\\15&24\end{pmatrix}$$

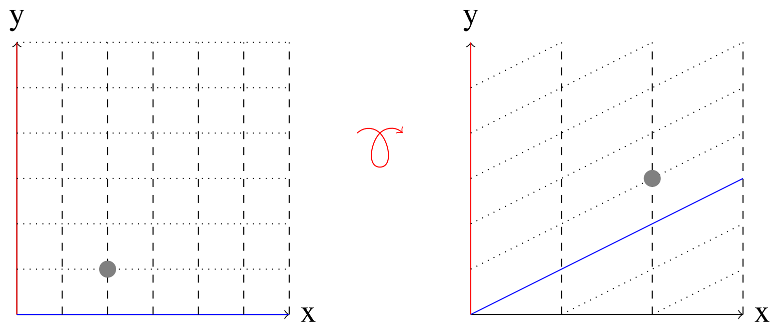
In this case, BA *does not exist!* ■

2.1.3 Motivation for matrix multiplication

The definition of matrix multiplication comes from *linear transformations*. Imagine a function

$$f:\begin{pmatrix}x\\y\end{pmatrix}\mapsto\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}2x\\x+y\end{pmatrix}$$

which transforms points and lines on the plane as shown below:



Our method for matrix multiplication comes about from devising a way that vector $\begin{pmatrix} x \\ y \end{pmatrix}$ becomes vector $\begin{pmatrix} u \\ v \end{pmatrix}$ through matrix multiplication — that is, so that we can write this linear transformation as a product $\begin{pmatrix} u \\ v \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$ for some matrix A.

Because

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2x \\ x+y \end{pmatrix} = \begin{pmatrix} 2\cdot x + 0\cdot y \\ 1\cdot x + 1\cdot y \end{pmatrix}$$

we choose to extract the coefficients of our unknowns into the matrix A (in much the same way we do when we write down the augmented matrix). So we end up with

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2x \\ x+y \end{pmatrix} = \begin{pmatrix} 2\cdot x + 0\cdot y \\ 1\cdot x + 1\cdot y \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

so we choose

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$

as the matrix representation of our transformation. We will cover more general linear transformations in chapter 3.

2.1.4 Important rules of algebra

In mathematics, whenever we have a set of 'numbers' with addition and multiplication operations between them, we investigate certain standard (algebraic) rules to see whether they hold:

Commutativity whether $a+b=b+a$
Associativity whether $(a+b)+c=a+(b+c)$ $\quad [=a+b+c]$
Distributivity whether $a(b+c)=ab+ac$
Identity whether there is a number i such that $ia=ai=a$
Inverse whether each number a has an inverse a^{-1} such that $aa^{-1}=a^{-1}a=i$

For our new numbers — matrices — all the usual rules hold except *commutativity of multiplication*

1. $A+B=B+A$
2. $AB \neq BA$ (in general)
3. $(A+B)+C=A+(B+C)$
4. $(AB)C=A(BC)$
5. $A(B+C)=AB+AC$
6. $(A+B)C=AC+BC$
7. $(\alpha\beta)A=\alpha(\beta A)$
8. $\alpha(AB)=(\alpha A)B=A(\alpha B)$
9. $(\alpha+\beta)B=\alpha A+\alpha B$
10. $\alpha(A+B)=\alpha A+\alpha B$

We call rule 1 the *law of commutativity* for addition, rule 3 the *law of associativity* for addition, rule 4 the *law of associativity* for multiplication, and rule 5 the *law of distributivity* of multiplication over addition. Note that noncommutativity of multiplication must be respected when employing distributivity (*e.g.*, rules 5 & 6) and multiplicative associativity (*e.g.*, rule 4).

2.1.5 Identities, inverses, and transposes

Definition 2.1.6 — The (multiplicative) identity. For matrix multiplication, only square matrices have identities. The $n \times n$ matrix

$$I = (\delta_{ij}) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

is the identity[a] for all square matrices in $\mathbb{R}^{n \times n}$. We often denote the $n \times n$ identity matrix by I_n — that is,

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \textit{etc.}$$

[a] The symbol δ_{ij} is called the Kronecker delta.

■ Example 2.6 — The (multiplicative) identity.

$$\begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$$

■

Definition 2.1.7 — The (multiplicative) inverse. The inverse A^{-1} of matrix A is a matrix such that:

$$A^{-1}A = AA^{-1} = I$$

Important notes regarding the inverse of a matrix:

- Only square matrices can have inverses, but *not all square matrices do*
- A square matrix is *invertible* (or *nonsingular*) if it has an inverse A^{-1}
- We will see the condition for the existence of A^{-1} in section 2.4.4.
- We can use Gaussian elimination to find A^{-1} (section 2.4)
- $(AB)^{-1} = B^{-1}A^{-1}$ (note the change of order), $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$, *etc.*
  - Proof: $(AB)B^{-1}A^{-1} = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$, *etc.*
- Matrix inverses are unique: if B and C are inverses of A, then $B = BI = B(AC) = (BA)C = IC = C$

Definition 2.1.8 — Transpose of a matrix. The transpose of A, denoted A^{T}, is the reflection of that matrix across its main diagonal:

$$A = (a_{ij}) \quad \Rightarrow \quad A^{\mathrm{T}} = (a_{ji})$$

Effectively, this interchanges the rows and columns of the matrix. Examples and properties of the transpose:

$$A = \begin{pmatrix} 3 & -2 \\ 2 & 4 \\ 1 & -3 \end{pmatrix} \quad \Rightarrow \quad A^{\mathrm{T}} = \begin{pmatrix} 3 & 2 & 1 \\ -2 & 4 & -3 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & -2 & 1 \\ 1 & 4 & 5 \\ -2 & 1 & -3 \end{pmatrix} \quad \Rightarrow \quad B^{\mathrm{T}} = \begin{pmatrix} 3 & 1 & -2 \\ -2 & 4 & 1 \\ 1 & 5 & -3 \end{pmatrix}$$

$$\left(A^{\mathrm{T}}\right)^{\mathrm{T}} = A; \quad (\alpha A)^{\mathrm{T}} = \alpha A^{\mathrm{T}};$$
$$(A+B)^{\mathrm{T}} = A^{\mathrm{T}} + B^{\mathrm{T}}; \quad (AB)^{\mathrm{T}} = B^{\mathrm{T}} A^{\mathrm{T}} (\text{note change of order}).$$

Definition 2.1.9 — Symmetric matrices. A matrix A is *symmetric* if (and only if) it is equal to its transpose, i.e. $A = A^{\mathrm{T}}$.

2.2 Applications involving matrix algebra

Example 2.7 — Calculating the Marital Population. Imagine that, in a given year, the probability a married person divorces is 30%, and the probability an unmarried person marries is 20%. If 80% of the population is married, what percentage is married after 1 year, 2 years, or 10 years[1]?

The numbers of un/married people in the next year is given by

$$\begin{aligned} \%\text{ married next year} &= [\%\text{ married}] \times [\%\text{ don't divorce}] + [\%\text{ unmarried}] \times [\%\text{ marry}] \\ \%\text{ unmarried next year} &= [\%\text{ married}] \times [\%\text{ divorce}] + [\%\text{ unmarried}] \times [\%\text{ don't marry}] \end{aligned}$$

or, given 30% divorce rate and 20% marriage rate,

$$\begin{aligned} \begin{pmatrix} \%\text{ married next yr} \\ \%\text{ unmarried next yr} \end{pmatrix} &= \begin{pmatrix} [\%\text{ married}] \times 0.7 + [\%\text{ unmarried}] \times 0.2 \\ [\%\text{ married}] \times 0.3 + [\%\text{ unmarried}] \times 0.8 \end{pmatrix} \\ &= \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} \%\text{ married} \\ \%\text{ unmarried} \end{pmatrix} \end{aligned}$$

Initially 80% of the population is married, which we represent by the initial vector

$$\mathbf{x}_0 = \begin{pmatrix} 80 \\ 20 \end{pmatrix}$$

Therefore, after 1 year, we have

$$\mathbf{x}_1 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 80 \\ 20 \end{pmatrix} = \begin{pmatrix} 56+4 \\ 24+16 \end{pmatrix} = \begin{pmatrix} 60 \\ 40 \end{pmatrix}$$

After 2 years, we have

$$\mathbf{x}_2 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^2 \begin{pmatrix} 80 \\ 20 \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 60 \\ 40 \end{pmatrix} = \begin{pmatrix} 42+8 \\ 18+32 \end{pmatrix} = \begin{pmatrix} 50 \\ 50 \end{pmatrix}$$

After 10 years, we have

$$\mathbf{x}_{10} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^{10} \begin{pmatrix} 80 \\ 20 \end{pmatrix} \approx \begin{pmatrix} 40 \\ 60 \end{pmatrix}$$

■

[1] Here for simplicity we are not taking into account deaths and births.

■ **Example 2.8 — Turtle population model.** Loggerhead Sea Turtles have the following demographic stages

Stage #	Description (age in years)	Annual survivorship	Eggs laid per year
1	Hatchlings (< 1)	0.67	0
2	Juveniles (1-21)	0.74	0
3	Novice adult (22)	0.81	127
4	Mature adult (23+)	0.81	79

From one year to the next, these populations must change through turtles dying, growing older and thus changing demographic group, or through eggs hatching to produce hatchlings. We can represent these changes through a *Leslie* matrix, which relates the population across the demographic groups to the populations at the previous stage (previous year, in this case):

$$\begin{pmatrix} \#\,\text{hatchlings}_{n+1} \\ \#\,\text{juveniles}_{n+1} \\ \#\,\text{novice adults}_{n+1} \\ \#\,\text{mature adults}_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 127 & 79 \\ 0.67 & 0.74 & 0 & 0 \\ 0 & 0.0006 & 0 & 0 \\ 0 & 0 & 0.81 & 0.81 \end{pmatrix} \begin{pmatrix} \#\,\text{hatchlings}_{n} \\ \#\,\text{juveniles}_{n} \\ \#\,\text{novice adults}_{n} \\ \#\,\text{mature adults}_{n} \end{pmatrix}$$

We thus define the *Leslie* matrix L, which allows us to calculate the populations after n years, $\mathbf{x}_n$, from the initial state $\mathbf{x}_0$:

$$L = \begin{pmatrix} 0 & 0 & 127 & 79 \\ 0.67 & 0.74 & 0 & 0 \\ 0 & 0.0006 & 0 & 0 \\ 0 & 0 & 0.81 & 0.81 \end{pmatrix} \quad \Rightarrow \quad \mathbf{x}_n = L^n \mathbf{x}_0$$

The Leslie population model makes the following predictions

Stage #	Initial population	10 years	25 years	50 years
1	200,000	114,264	74,039	35,966
2	300,000	329,212	213,669	103,795
3	500	214	139	68
4	1,500	1,061	687	334

That is, the population dwindles without intervention of some kind . . . ■

2.3 Block matrices

Recall that the matrix product of two 2×2-matrices is given by

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11}+a_{12}b_{21} & a_{11}b_{21}+a_{12}b_{22} \\ a_{21}b_{11}+a_{22}b_{21} & a_{21}b_{12}+a_{22}b_{22} \end{pmatrix}$$

The same results holds if, instead of having numbers in the places of the a_{ij} and b_{ij}, we have matrices sub-matrices A_{ij} and B_{ij} of a larger matrices A and B. That is, we can multiply matrices

$$AB = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11}+A_{12}B_{21} & A_{11}B_{12}+A_{12}B_{22} \\ A_{21}B_{11}+A_{22}B_{21} & A_{21}B_{12}+A_{22}B_{22} \end{pmatrix}$$

Here, the A_{ij} and B_{ij} must have the right shapes, so that they can be multiplied together. More precisely, since all the sub-matrix products have the form $A_{ij}B_{jk}$, the shape requirement is that the number of columns of A_{ij} has to match the number of rows of the B_{jk}, for a fixed j (and *any* i or k). If this condition is met, it follows automatically that A and B can be multiplied together.

Furthermore, there is no need to restrict this approach to $2times2$ block matrices — the rules follow the usual rules for multiplying regular matrices A and B. That is, we can calculate AB as long as the number of columns of blocks in A matches the number of rows of blocks of B.

There are some important consequences:

1. In the next section, we will use the result that

$$AB = A\begin{pmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \\ | & | & & | \end{pmatrix},$$

   which we obtain by treating the entire matrix A is a single block, and dividing B into blocks made from its columns. We can generalise this result for the case when each block of B comprises multiple columns, in which case

$$AB = A\begin{pmatrix} | & | & & | \\ B_{11} & B_{12} & \cdots & B_{1p} \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ AB_{11} & AB_{12} & \cdots & AB_{1p} \\ | & | & & | \end{pmatrix}.$$

   The block matrix B_{11} should not be confused with the upper left element of B, b_{11}.
2. If we are multiplying a matrix A by a column vector $\mathbf{b}$, then

$$A\mathbf{b} = \begin{pmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{pmatrix}\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = b_1\mathbf{a}_1 + b_2\mathbf{a}_2 + \cdots + b_n\mathbf{a}_n$$

   This second result says that we can interpret $A\mathbf{b}$ as a **linear combination** of the columns of A

   Definition 2.3.1 — Linear combination. A *linear combination* of mathematical objects (vectors, matrices, functions, etc.) $A_1, A_2, \cdots, A_n$ is any sum of the form

$$\sum_{i=1}^{n} x_i A_i = x_1 A_1 + x_2 A_2 \cdots x_n A_n$$

   for real numbers $x_1, x_2, \ldots, x_n$.

   Proposition 2.3.1 The linear system $A\mathbf{x} = \mathbf{b}$ has at least one solution iff[2] $\mathbf{b}$ is a linear combination of the columns of A.

[2] 'iff' is mathematics shorthand for 'if and only if'. A iff B means 'A if B' and 'B if A' — both statements imply the other, so both directions must be proven.

This result follows immediately from the fact that *any* product $A\mathbf{x}$ must be a linear combination of the columns of A. As we shall see later, this is a crucial idea, helping us understand which linear systems have solutions.

3. Another way of thinking about matrix multiplication AB is to consider A made up of row blocks, and B made up of column blocks. Then

$$AB = \begin{pmatrix} \text{——} & \vec{\mathbf{a}}_1 & \text{——} \\ \text{——} & \vec{\mathbf{a}}_2 & \text{——} \\ & \vdots & \\ \text{——} & \vec{\mathbf{a}}_m & \text{——} \end{pmatrix} \begin{pmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \\ | & | & & | \end{pmatrix} = \begin{pmatrix} \vec{\mathbf{a}}_1\mathbf{b}_1 & \vec{\mathbf{a}}_1\mathbf{b}_2 & \cdots & \vec{\mathbf{a}}_1\mathbf{b}_p \\ \vec{\mathbf{a}}_2\mathbf{b}_1 & \vec{\mathbf{a}}_2\mathbf{b}_2 & \cdots & \vec{\mathbf{a}}_2\mathbf{b}_p \\ \vdots & \vdots & & \vdots \\ \vec{\mathbf{a}}_m\mathbf{b}_1 & \vec{\mathbf{a}}_m\mathbf{b}_2 & \cdots & \vec{\mathbf{a}}_m\mathbf{b}_p \end{pmatrix}$$

These results are all special examples of *block matrices* and their multiplication. We can (arbitrarily) consider the matrix $A \in \mathbb{R}^{4\times4}$ as composed of blocks A_{ij}, and then perform operations on the matrices using the blocks, if they are of appropriate size. This can be quite useful if parts of the matrix are filled with zeros or simple matrix forms, and can help us interpret the action of various parts of a matrix on the result.

■ **Example 2.9 — Block multiplication.** Block multiplication can also be quite useful if parts of the matrix are filled with zeros or simple matrix forms, and can help us interpret the action of various parts of a matrix on the result. Consider the 4×4 matrix A, and the general 4×2 matrix B, defined as

$$A = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \left(\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right) \quad \text{and} \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{pmatrix} = \left(\begin{array}{c} B_{11} \\ \hline B_{21} \end{array}\right)$$

where

$$A_{11} = I_3,\ A_{12} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix},\ A_{21} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix},\ A_{22} = \begin{pmatrix} 1 \end{pmatrix},\ B_{11} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix},\ B_{21} = \begin{pmatrix} b_{41} & b_{42} \end{pmatrix}.$$

Since the blocks have appropriate sizes to be multiplied, we have that

$$AB = \left(\begin{array}{c} A_{11}B_{11} + A_{12}B_{21} \\ \hline A_{21}B_{11} + A_{22}B_{21} \end{array}\right) = \left(\begin{array}{c} IB_{11} + A_{12}B_{21} \\ \hline 0B_{11} + IB_{21} \end{array}\right) = \left(\begin{array}{c} B_{11} + A_{12}B_{21} \\ \hline B_{21} \end{array}\right).$$

If A is a fixed matrix, but we have to calculate AB for many different B, this gives us a detailed picture of how AB depends on B. It also speeds up somewhat the calculation for a specific B. If, for example

$$B = \left(\begin{array}{cc} -1 & 2 \\ -3 & 4 \\ 1 & -2 \\ \hline 3 & -4 \end{array}\right)$$

then

$$AB = \left(\begin{array}{c} B_{11} + A_{12}B_{21} \\ \hline B_{21} \end{array}\right) = \begin{pmatrix} -1+6 & 2+-8 \\ -3+0 & 4+0 \\ 1+9 & -2+-12 \\ 3 & -4 \end{pmatrix} = \begin{pmatrix} 5 & -6 \\ -3 & 4 \\ 10 & -14 \\ 3 & -4 \end{pmatrix}$$

■

2.4 Matrix inversion

2.4.1 Elementary matrices

Recall the *three* allowed row operations

1. Interchanging two rows;
2. Rescaling one row;
3. Adding a multiple of one row to another row.

We can represent each of these operations via matrix multiplication. We call these representative matrices *elementary matrices* E_i. The steps in *Gaussian elimination* can therefore be represented as

$$A \rightarrow E_1A \rightarrow E_2E_1A \rightarrow \cdots \rightarrow E_kE_{k-1}\ldots E_2E_1A$$

Examples of the elementary matrices:

1. Interchanging two rows: identity matrix with two off-diagonal 1s

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left(\begin{array}{ccc|c} 0 & -4 & 8 & 4 \\ 1 & -1 & 1 & 0 \\ 2 & 2 & 3 & 4 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & -4 & 8 & 4 \\ 2 & 2 & 3 & 4 \end{array}\right)$$

2. Rescaling one row: identity matrix with one non-unit diagonal term

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1/4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & -4 & 8 & 4 \\ 2 & 2 & 3 & 4 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -2 & -1 \\ 2 & 2 & 3 & 4 \end{array}\right)$$

3. Adding multiple of one row to another row: identity matrix with extra off-diagonal term

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -2 & -1 \\ 2 & 2 & 3 & 4 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -2 & -1 \\ 0 & 4 & 1 & 4 \end{array}\right)$$

Recall that, *for systems with unique solutions*, the row-reduction took us from our original augmented matrix $(A|\mathbf{q})$ to the upper-triangular augmented matrix $(B|\mathbf{r})$, and finally to the augmented identity matrix $(I|\mathbf{s})$ (if we completed Gauss-Jordan elimination). This process is thus equivalent to (pre-)multiplying the original augmented matrix by the appropriate sequence of elementary matrices:

$$(I|\mathbf{s}) = E_m \ldots E_1(A|\mathbf{q}) = (E_m \ldots E_1A|E_m \ldots E_1\mathbf{q}).$$

Notice that we have written the augmented matrix in two blocks, A and $\mathbf{q}$.

An immediate and perhaps unexpected consequence is that we can determine A^{-1}:

$$I = E_m \ldots E_1A = A^{-1}A \quad \Rightarrow \quad E_m \ldots E_1 = A^{-1}$$

Furthermore, since $E_m \ldots E_1(A|I) = (E_m \ldots E_1A|E_m \ldots E_1I) = (I|A^{-1})$, we can use the same Gaussian elimination method that we used to solve $A\mathbf{x} = \mathbf{q}$ in order to calculate the inverse of A!

2.4.2 Matrix inversion examples

■ Example 2.10 — Matrix inversion via Gaussian elimination.

Find the inverse of $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$

$$
\begin{aligned}
\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right) &\to \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 1 & 0 & 1 \end{array}\right) \begin{array}{l} \scriptstyle \mathrm{R1/2} \\ \scriptstyle \mathrm{R2} \end{array} \\
&\to \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 1 \end{array}\right) \begin{array}{l} \scriptstyle \mathrm{R1a} \\ \scriptstyle \mathrm{R2-R1a} \end{array} \\
&\to \left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right) \begin{array}{l} \scriptstyle \mathrm{R1a\text{-}R2a} \\ \scriptstyle \mathrm{2.R2a} \end{array} \\
&\Rightarrow A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}
\end{aligned}
$$

$$
\text{Check: } \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2-1 & -2+2 \\ 1-1 & -1+2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad ✔
$$

■

■ Example 2.11 — Matrix inversion via Gaussian elimination.

Show that the inverse of $A = \begin{pmatrix} 3 & 2 & -1 \\ 3 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix}$ is $A^{-1} = \begin{pmatrix} -\frac{1}{2} & \frac{2}{3} & \frac{1}{6} \\ 1 & -1 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$

$$
\begin{aligned}
\left(\begin{array}{ccc|ccc} 3 & 2 & -1 & 1 & 0 & 0 \\ 3 & 1 & -1 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 & 0 & 1 \end{array}\right) &\to \left(\begin{array}{ccc|ccc} 3 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & -1 & 0 & 1 \end{array}\right) \begin{array}{l} \mathrm{R1} \\ \mathrm{R2\text{-}R1} \\ \mathrm{R3\text{-}R1} \end{array} \\
&\to \left(\begin{array}{ccc|ccc} 3 & 0 & -1 & -1 & 2 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array}\right) \begin{array}{l} \mathrm{R1 + 2R2a} \\ -\mathrm{R2a} \\ \mathrm{R3a/2} \end{array} \\
&\to \left(\begin{array}{ccc|ccc} 3 & 0 & 0 & -\frac{3}{2} & 2 & \frac{1}{2} \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array}\right) \begin{array}{l} \mathrm{R1a{+}R3b} \\ \mathrm{R2b} \\ \mathrm{R3b} \end{array} \\
&\to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & \frac{2}{3} & \frac{1}{6} \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array}\right) \begin{array}{l} \mathrm{R1b/3} \\ \mathrm{R2b} \\ \mathrm{R3b} \end{array} \quad \text{as required}
\end{aligned}
$$

■

2.4.3 Calculating inverses numerically

The MATLAB/Octave, R, and Python commands to invert a matrix are all `inv`. To find the inverses in the previous two examples; in MATLAB/Octave, we obtain

```
inv([2 1;1 1])
ans =

    1.0000   -1.0000
   -1.0000    2.0000
```

while in R we obtain (first loading pracma or matlib)

```
A <- matrix(c(2, 1, 1, 1), 2, 2, byrow = TRUE)
inv(A)
[,1] [,2]
[1,]    1   -1
[2,]   -1    2
```

In Python using numpy, you must also instruct Python to use the linear algebra toolbox, *viz.*

```
A = np.array([[2,1],[1,1]])
np.linalg.inv(A)
array([[ 1., -1.],
       [-1.,  2.]])
```

2.4.4 When is a matrix invertible?

We use the term **non-singular** as a synonym for *invertible* – we shall see where this name comes from later in the course.

Theorem 2.4.1 — Equivalent conditions for invertibility (nonsingularity). The following statements about $A \in \mathbb{R}^{n\times n}$ are equivalent:

A. A is invertible (nonsingular)
B. The only solution to $A\mathbf{x} = 0$ is $\mathbf{x} = 0$
C. A is row-equivalent to I (*i.e.*, there is a sequence of elementary matrices such that $E_m \dots E_1 A = I$).

Proof. To prove this theorem, we will show that A $\Rightarrow$ B $\Rightarrow$ C $\Rightarrow$ A

A $\Rightarrow$ B Assume A is invertible, so A^{-1} exists. If $A\mathbf{x} = 0$, then $\mathbf{x} = (A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}0 = 0$, so $\mathbf{x} = 0$ is the only possible solution.

B $\Rightarrow$ C Assume $A\mathbf{x} = 0$ only has one solution, $\mathbf{x} = 0$. We know *unique solutions* correspond to row echelon forms $(A|\mathbf{q}) \to (I|\mathbf{s})$. For the case $\mathbf{x} = 0$, $\mathbf{s} = 0$. Therefore there must be elementary matrices transforming $A \to I$

C $\Rightarrow$ A There exist $E_m \dots E_1$ such that $E_m \dots E_1 A = I$. We note that the E_i must be invertible[3]. Therefore we can write $A = E_1^{-1} \dots E_m^{-1} I = E_1^{-1} \dots E_m^{-1}$.

■

Corollary 2.4.2 — Uniqueness of solutions for linear systems. The linear system $A\mathbf{x} = \mathbf{b}$ comprising n equations in n unknowns has a unique solution iff A is nonsingular.

Proof. We prove the equivalent two-part corollary

1. A is nonsingular $\Rightarrow A\mathbf{x} = \mathbf{b}$ has a unique solution; and
2. A is singular[4] $\Rightarrow A\mathbf{x} = \mathbf{b}$ has no solution or many solutions

First, note that if A is nonsingular, then A^{-1} exists, so the only possible solution to $A\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = (A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{b}$. Second, if A is singular, then $A\mathbf{x} = 0$ has a non-zero solution $\hat{\mathbf{x}}$. Therefore $A\mathbf{x} = \mathbf{b} \Rightarrow A(\mathbf{x} + \hat{\mathbf{x}}) = A\mathbf{x} + A\hat{\mathbf{x}} = \mathbf{b} + 0 = \mathbf{b}$. Thus if A is singular, $A\mathbf{x} = \mathbf{b}$ has either no solution, or (at least) two solutions. ■

[3] This can be proved by considering each possible elementary matrix in turn, or by considering the fact that each elementary process is reversible.

[4] Opposite of nonsingular; a singular matrix has no inverse matrix.

The title of this section is 'When is a matrix invertible?' We haven't really given a satisfactory answer to that question – so far, all we have proven is that being invertible is *equivalent* to some other mathematical statements. What we really want is some way of *calculating* whether a matrix is invertible or not, directly from its elements. This is what we will achieve in the next section.

2.5 Determinants

2.5.1 Motivation

Our theorem and corollary from the previous section shows that A is nonsingular if and only if $A\mathbf{x} = \mathbf{b}$ has a unique solution. We would like a *measurable property of the matrix itself* that distinguishes singular from nonsingular matrices. To achieve this, we introduce a quantity known as the *determinant*:

- We denote the determinant of a (square) matrix A by $\det A$ or by $|A|$
- $\det A$ is calculated from the matrix elements a_{ij}
- $\det A \neq 0$ iff A is nonsingular (invertible).

How do we calculate $\det A$?

2.5.2 Derivation for 1x1-matrices

The problem $ax = b$ for scalars a, x, and b will have a unique solution iff $a \neq 0$. We can think of this as a 1x1 matrix problem $Ax = b$, where $x = A^{-1}b = b/a_{11}$. By defining $\det A = a_{11}$ for a 1x1 matrix, we have a quantity matching our requirements – if $\det A = 0$ the matrix is not invertible, but it is invertible in any other case. We therefore define

$$\boxed{\det A = a_{11}}$$

for 1x1 matrices.

2.5.3 Derivation for 2x2-matrices

The matrix $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ is nonsingular if it can be row reduced to I. If we perform the row reductions

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \to \begin{array}{l} \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{11}a_{22} - a_{12}a_{21} \end{pmatrix} \end{array} \begin{array}{l} \mathrm{R1} \\ a_{11}\mathrm{R2} - a_{21}\mathrm{R1} \end{array}$$

we will be able to row reduce A to I as long as $a_{11}a_{22} - a_{12}a_{21} \neq 0$. This is a good starting candidate for the determinant. But in this argument, we have implicitly assumed that $a_{11} \neq 0$. What if $a_{11} = 0$? As long as $a_{21} \neq 0$, then

$$\begin{pmatrix} 0 & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \to \begin{pmatrix} a_{21} & a_{22} \\ 0 & a_{12} \end{pmatrix}$$

and we will be able to row reduce A to I as long as $a_{21} \neq 0$ and $a_{12} \neq 0$. But if either of these values is zero, then our $a_{11}a_{22} - a_{12}a_{21} = 0$ (since in this case we assumed $a_{11} = 0$). This is exactly the same condition as before! We therefore define

$$\boxed{\det A = a_{11}a_{22} - a_{12}a_{21}}$$

for 2×2 matrices. You can remember this as 'product of diagonals minus product of antidiagonals'.

2.5.4 Derivation for 3x3 matrices

The matrix $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ is nonsingular if it can be row reduced to I. If we perform the row reductions

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \to \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{11}a_{22} - a_{21}a_{12} & a_{11}a_{23} - a_{21}a_{13} \\ 0 & a_{11}a_{32} - a_{31}a_{12} & a_{11}a_{33} - a_{31}a_{13} \end{pmatrix}$$

we will be able to row reduce A to I as long as

$$\begin{aligned}
&\det \begin{pmatrix} a_{11}a_{22} - a_{21}a_{12} & a_{11}a_{23} - a_{21}a_{13} \\ a_{11}a_{32} - a_{31}a_{12} & a_{11}a_{33} - a_{31}a_{13} \end{pmatrix} \\
&\equiv \begin{vmatrix} a_{11}a_{22} - a_{21}a_{12} & a_{11}a_{23} - a_{21}a_{13} \\ a_{11}a_{32} - a_{31}a_{12} & a_{11}a_{33} - a_{31}a_{13} \end{vmatrix} \\
&= [a_{11}a_{22} - a_{21}a_{12}][a_{11}a_{33} - a_{31}a_{13}] - [a_{11}a_{32} - a_{31}a_{12}][a_{11}a_{23} - a_{21}a_{13}] \\
&= \cdots \\
&= a_{11}[a_{11}a_{22}a_{33} - a_{22}a_{31}a_{13} - a_{21}a_{12}a_{33} - a_{11}a_{32}a_{23} + a_{32}a_{21}a_{13} + a_{31}a_{12}a_{23}] \neq 0
\end{aligned}$$

In analogy with the 2x2 case, for a 3x3 matrix we could propose

$$\boxed{\det A = a_{11}a_{22}a_{33} - a_{22}a_{31}a_{13} - a_{21}a_{12}a_{33} - a_{11}a_{32}a_{23} + a_{32}a_{21}a_{13} + a_{31}a_{12}a_{23}}$$

But what about the cases where $a_{11} = 0$, etc. ?

1. If $a_{11} = 0, a_{21} \neq 0, a_{31} \neq 0$, we swap R1 and R2 and perform the row reduction

$$\begin{pmatrix} a_{21} & a_{22} & a_{23} \\ 0 & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \to \begin{pmatrix} a_{21} & a_{22} & a_{23} \\ 0 & a_{12} & a_{13} \\ 0 & a_{21}a_{32} - a_{31}a_{22} & a_{21}a_{33} - a_{31}a_{23} \end{pmatrix}$$

   We will be able to row reduce A to I as long as

$$\begin{aligned}
&\begin{vmatrix} a_{12} & a_{13} \\ a_{21}a_{32} - a_{31}a_{22} & a_{21}a_{33} - a_{31}a_{23} \end{vmatrix} \neq 0 \\
&\Rightarrow \quad a_{21}a_{12}a_{33} - a_{31}a_{12}a_{23} - a_{32}a_{21}a_{13} + a_{22}a_{31}a_{13} \neq 0
\end{aligned}$$

   which occurs precisely when our proposed definition for $\det A \neq 0$ (with $a_{11} = 0$).

2. If $a_{11} = a_{21} = 0, a_{31} \neq 0$, then we swap R1 and R3

$$\begin{pmatrix} a_{31} & a_{32} & a_{33} \\ 0 & a_{22} & a_{23} \\ 0 & a_{12} & a_{13} \end{pmatrix}$$

   and note that will be able to row reduce A to I as long as

$$\begin{vmatrix} a_{22} & a_{23} \\ a_{12} & a_{13} \end{vmatrix} \neq 0 \quad \Rightarrow \quad a_{22}a_{13} - a_{12}a_{23} \neq 0$$

   which occurs precisely when our proposed definition for $\det A \neq 0$ (with with $a_{11} = a_{21} = 0$).

3. Finally, if $a_{11} = a_{21} = 0 = a_{31} = 0$, then A is not row-reducible to I; but our proposed definition for $\det A = 0$ in this case!

So in all cases, our proposed definition of $\det A$ for 3x3 matrices indicates that the matrix is invertible iff $\det A \neq 0$. We therefore define

$$\boxed{\det A = a_{11}a_{22}a_{33} - a_{22}a_{31}a_{13} - a_{21}a_{12}a_{33} - a_{11}a_{32}a_{23} + a_{32}a_{21}a_{13} + a_{31}a_{12}a_{23}}$$

for 3×3 matrices.

2.5.5 General definition

Already for the 3×3 matrix, the definition of the matrix determinant is getting complicated. Remembering a formula like this is difficult, and won't get any easier as our matrices get bigger. Is there easier way to go about calculating the determinant?

Observe that the determinant expression can be calculated using the following approach: choose the top row of the matrix; multiply each element by the determinant of what is left of the matrix when the row and column containing the element is crossed out; alternately add or subtract these contributions along the row. See the examples in the boxes below.

■ Example 2.12 — Calculating the 2x2 determinant.

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \qquad \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$\det A = a_{11}a_{22} - a_{12}a_{21}$$

■

■ Example 2.13 — Calculating the 3x3 determinant.

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \qquad \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \qquad \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{aligned} \det A &= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{32}a_{21} - a_{22}a_{31}) \\ &= a_{11}a_{22}a_{33} - a_{22}a_{31}a_{13} - a_{21}a_{12}a_{33} - a_{11}a_{32}a_{23} + a_{32}a_{21}a_{13} + a_{31}a_{12}a_{23} \end{aligned}$$

■

To make this approach to defining the determinant more precise, for any matrix A we define the *minor* M_{ij} to be the matrix comprising the elements remaining once the row and column containing a_{ij} are removed:

$$\begin{pmatrix} 11 & 12 & 13 & 14 & 15 \\ 21 & 22 & 23 & 24 & 25 \\ 31 & 32 & 33 & 34 & 35 \\ 41 & 42 & 43 & 44 & 45 \\ 51 & 52 & 53 & 54 & 55 \end{pmatrix} \Rightarrow \quad M_{23} = \begin{pmatrix} 11 & 12 & 14 & 15 \\ 31 & 32 & 34 & 35 \\ 41 & 42 & 44 & 45 \\ 51 & 52 & 54 & 55 \end{pmatrix}$$

and we define the *cofactor* C_{ij} as

$$C_{ij} = (-1)^{i+j} \det M_{ij}$$

Definition 2.5.1 — Determinant of a matrix. The determinant of a square matrix $A \in \mathbb{R}^{n\times n}$ is given by

$$\det A = \begin{cases} a_{11} & \text{if } n = 1 \\ \sum_{i=1}^{n} a_{1i} C_{1i} & \text{if } n > 1 \end{cases}$$

where

$$C_{ij} = (-1)^{i+j} \det M_{ij}$$

is the cofactor of a_{1i}, and M_{ij} is the minor matrix obtained by removing row i and column j.
Non-square matrices do not have a determinant.

Confirmation for the 2x2 matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$\det A = a_{11}C_{11} + a_{12}C_{12} = a_{11}a_{22} - a_{12}a_{21} \quad ✔$$

Confirmation for the 3x3 matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{aligned} \det A &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}a_{22}a_{33} - a_{11}a_{32}a_{23} - a_{12}a_{21}a_{33} + a_{12}a_{31}a_{23} + a_{13}a_{21}a_{32} - a_{13}a_{31}a_{22} \quad ✔ \end{aligned}$$

Note: we are not forced to sum the $a_{ij}C_{ij}$ along the top row!

$$\det A = \sum_{i=1}^{n} a_{ki}C_{ki} \quad \text{along any row } k$$
$$\det A = \sum_{i=1}^{n} a_{ik}C_{ik} \quad \text{down any column } k$$

This is useful, because sometimes it is faster to choose a particular row or column – for example, if there are many zeros in it. So to calculate

$$\begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}$$

we would expand down the first column:

$$= 0 - 0 + 0 - 2\begin{vmatrix} 2 & 3 & 0 \\ 4 & 5 & 0 \\ 1 & 0 & 3 \end{vmatrix}$$

and then sum down the last column:

$$= -2\left(0 - 0 + 3\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix}\right) = -2 \cdot 3 \cdot (10 - 12) = 12$$

2.5.6 Calculating determinants numerically

The MATLAB/Octave, R, and Python commands to invert a matrix are all det. To find the determinant of the matrix $A = \begin{pmatrix} 3 & 2 & -1 \\ 3 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix}$, in MATLAB/Octave we obtain

```
det([3 2 -1; 3 1 -1; 3 2 1])
ans = -6
```

while in R we obtain

```
A <- matrix(c(3, 2, -1, 3, 1, -1, 3, 2, 1), 3, 3, byrow = TRUE)
det(A)
[1] -6
```

and in Python

```
A = np.array([[3,2,-1],[3,1,-1],[3,2,1]])
np.lingalg.det(A)
-6.0
```

2.5.7 Useful properties of determinants

Some useful properties of determinants to remember:

1. A is singular iff $\det A = 0$

2. $\det(A) = \det(A^{\mathrm{T}})$
3. If A is an upper or lower triangular matrix[5], $\det A$ is the product of the diagonal elements
4. $\det A = 0$ if:
   - A has one row [column] consisting entirely of zeros; or
   - A has two identical rows [columns]; or
   - one row [column] of A is a linear combination of the others.
5. $\det(AB) = (\det A)(\det B)$
6. Elementary matrices:
   - for row operations I and III, $\det E_j = 1$:
   - for row operation II, $\det E_k =$ scaling factor

You might also be wondering if the determinant has any other uses – is a matrix with determinant 1 any different from one with determinant -1, for example? Yes, and we will see how in chapter 3.

2.5.8 Cross product

Another convenient use of the determinant is in evaluating the cross product of two 3D vectors:

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

and therefore that

$$\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

This scalar triple product represents the volume contained within a parallelepiped with sides $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$[6]. Consequently, if any of the sides is a linear combination of the others, then this volume will be zero because all three vectors are coplanar. This matches our earlier observation that the determinant of a matrix will be zero if one column or row is a linear combination of the others.

We won't be discussing the cross product in depth during the course because, somewhat surprisingly, there is no sensible version of a cross product in $\mathbb{R}^n$ for general n; only $n = 3$ and $n = 7$ work! For example, Cross Products of Vectors in Higher Dimensional Euclidean Spaces by W. S. Massey.

2.6 Cramer's rule

Cramer's rule gives us the unique solution, if it exists, to the system $A\mathbf{x} = \mathbf{b}$:

$$x_i = \frac{\det A_i}{\det A}$$

[5] An important special case is diagonal matrices, which show up in the spectral theorem: 5.6.2.

[6] This is because $\mathbf{b} \times \mathbf{c}$ represents the area A of the parallelogram whose adjacent sides are the vectors $\mathbf{b}$ and $\mathbf{c}$, and the volume of the parallelepiped is this base area A times by the height $h = |\mathbf{a}| \cos\theta$, if θ is the angle between $\mathbf{a}$ and the base parallelogram formed by $\mathbf{b}$ and $\mathbf{c}$.

where A_i is the matrix A with the i-th column replaced by the elements of $\mathbf{b}$. For example, using Cramer's rule to solve

$$\begin{aligned} x+2y+z&=5 \\ 2x+2y+z&=6 \\ x+2y+3z&=9 \end{aligned}$$

gives:

$$x=\frac{\begin{vmatrix} 5 & 2 & 1 \\ 6 & 2 & 1 \\ 9 & 2 & 3 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 2 & 3 \end{vmatrix}}=\frac{-4}{-4}=1 \qquad y=\frac{\begin{vmatrix} 1 & 5 & 1 \\ 2 & 6 & 1 \\ 1 & 9 & 3 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 2 & 3 \end{vmatrix}}=\frac{-4}{-4}=1$$

$$z=\frac{\begin{vmatrix} 1 & 2 & 5 \\ 2 & 2 & 6 \\ 1 & 2 & 9 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 2 & 3 \end{vmatrix}}=\frac{-8}{-4}=2$$

where the substitution of the elements of $\mathbf{b}$ into the matrix A has been highlighted. In general, when $M\mathbf{x}=\mathbf{y}$, Cramer's rule involves replacing columns of the representative matrix by the solution, and calculating determinants, as follows:

$$x_i=\frac{\begin{vmatrix} M_{11} & M_{12} & \dots & M_{1(i-1)} & y_1 & M_{1(i+1)} & \cdots & M_{1n} \\ M_{21} & M_{22} & \dots & M_{2(i-1)} & y_2 & M_{2(i+1)} & \cdots & M_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ M_{n1} & M_{n2} & \dots & M_{n(i-1)} & y_n & M_{n(i+1)} & \cdots & M_{nn} \end{vmatrix}}{|M|}$$

Why does Cramer's rule work? It certainly seems a rather mysterious results. To derive Cramer's rule, first we note that

$$a_{i1}C_{j1}+a_{i2}C_{j2}+\cdots+a_{in}C_{jn}=\begin{cases} \det A & \text{if } i=j \\ 0 & \text{if } i\neq j \end{cases}$$

where the C_{ji} are the co-factors defined in §2.5.5. This result holds because, when $i=j$, the resulting expression is just the definition of $\det A$, but when $i\neq j$, the resulting expression is the determinant of a matrix with a repeated row, which we know to be 0. Next, we define the *adjoint* matrix

$$\operatorname{adj} A=\begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & & & \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{pmatrix}$$

Note that this is the transpose of what we might have expected. It follows for nonsingular matrices that

$$A\,(\text{adj}\,A) = (\det A)\,I \quad \Rightarrow \quad A^{-1} = \frac{1}{\det A}\,\text{adj}\,A$$

Indeed, this result gives us an alternative way of finding the inverse of a matrix, by calculating this matrix of co-factors (rather than by using Gaussian elimination). Note that the cofactors require calculation of determinants of $(n-1)\text{x}(n-1)$ matrices, so it is often faster numerically to invert a matrix using Gaussian elimination.

But to establish Cramer's rule, we note that if $\mathbf{x}$ is a solution to $A\mathbf{x} = \mathbf{b}$, then

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det A}\,\text{adj}\,A\;\mathbf{b}$$

and therefore

$$x_i = \frac{b_1C_{1i} + b_2C_{2i} + \cdots + b_nC_{ni}}{\det A} = \frac{\det A_i}{\det A}$$

because $(b_1C_{1i} + b_2C_{2i} + \cdots + b_nC_{ni})$ is the determinant of a matrix A_i whose cofactors down the i-th column are the cofactors of A, but whose elements down the i-th column are the elements of $\mathbf{b}$. To obtain A_i, we set $A_i = A$, but replace the i-th column by $\mathbf{b}$

$$\begin{pmatrix} | & | & & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{b} & \cdots & \mathbf{a}_n \\ | & | & & | & & | \end{pmatrix}$$

This shows the origin of Cramer's rule.

2.7 Numerical solution of $A\mathbf{x} = \mathbf{b}$

It is worth noting that there are a number of ways we can now consider solving the equation $A\mathbf{x} = \mathbf{b}$:

- If we know that A is invertible, we can calculate $\mathbf{x}$ by calculating A^{-1}, and then finding $A^{-1}\mathbf{b}$
- If we do not know whether A is invertible, we can perform the row reduction on the augmented matrix $(A|\mathbf{b})$

If A is invertible, MATLAB and Octave both allow the shorthand command `A \ b` equivalent to $A^{-1}b$. For example,

```
A = [2 -1; 1 1];
b = [3; 0];
inv(A)*b
ans =

    1
   -1
A\b
ans =

    1
   -1
```

Note that ** performs matrix multiplication by default in MATLAB/Octave* – these programs try to treat everything as matrices! The advantage of this command is that it is more computationally efficient than calculating A^{-1} and then multiplying as two separate steps[7].

If we want to instead perform the row reduction directly, we can issue the MATLAB/Octave command rref [A b]. This puts the augmented matrix into reduced row echelon form; note that concatenating matrices in MATLAB is as simple as including them as blocks within a matched set of square brackets (so long as the sizes are compatible).

```
rref([A b])
ans =

    1    0    1
    0    1   -1
```

To solve in R, we must load the matlib library, then use

```
A <- matrix(c(2,-1,1,1), 2, 2, byrow = TRUE)
b <- matrix(c(3,0), 2, 1, byrow = TRUE)
Solve(A, b, fractions = TRUE)
x1     =   1
   x2  =  -1
```

To verify, we can perform the calculation using the inverse explicitly. Note that *matrix multiplication in R is indicated by combining %*% together.*

```
inv(A)%*%b
       [,1]
[1,]      1
[2,]     -1
```

Python can solve linear systems using the solve command in numpy. Note that *matrix multiplication in Python is indicated by the @ symbol*.

```
A = np.array([[2,  -1],[1,1]])
b = np.array([[3],[0]])
np.linalg.solve(A,b)
Out: array([[ 1.],
            [-1.]])
np.linalg.inv(A)@b
Out: array([[ 1.],
            [-1.]])
```

[7]This can be a significant consideration when dealing with very large matrices, as might be encountered in many applications involving 'big data'.

3. Linear transformations

Problems of the form $A\mathbf{x} = \mathbf{b}$ arise across a vast range of mathematical applications. Furthermore, many problems that don't look at all like matrix problems, at first glance, can be solved by converting them into matrix problems. This includes many important problems relating to differentiation, integration, and more general function transformations.

The reason so many problems can be related to matrix problems is because they share some key properties with matrix multiplication – they are all **linear transformations**. In this chapter, we will define this very important property, and connect it with transformations defined by matrix multiplication.

3.1 What is a linear transformation?

Definition 3.1.1 — Linear Transformation. A mapping T is said to be a **linear transformation** if for any $\mathbf{u}, \mathbf{v}$ and any $\alpha \in \mathbb{R}$,

$$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \;\; \textbf{and} \;\; T(\alpha\mathbf{u}) = \alpha T(\mathbf{u})$$

Equivalently, the transformation T is linear if for any $\mathbf{u}, \mathbf{v}$ and any $\alpha, \beta \in \mathbb{R}$,

$$T(\alpha\mathbf{u}+\beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v})$$

Theorem 3.1.1 For any linear transformation, $T(\mathbf{0}) = \mathbf{0}$.

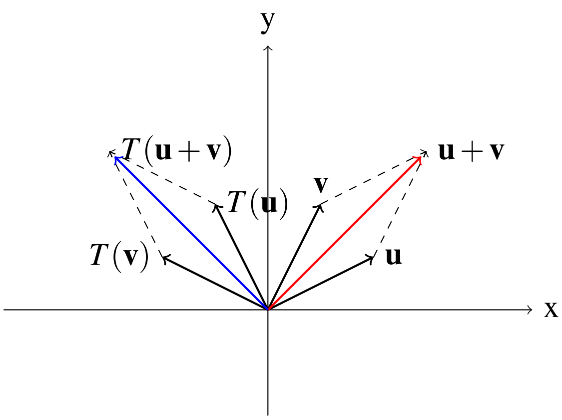
Proof. From the definition of a linear transformation,

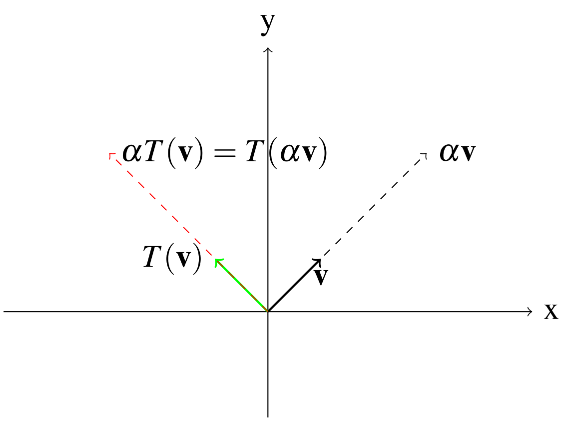
$$\alpha T(\mathbf{0}) = T(\alpha\mathbf{0}) = T(\mathbf{0}),$$

for any α. This is only possible for any α if $T(\mathbf{0}) = \mathbf{0}$. ■

Important Note: For this definition to make any sense, the operation of adding $\mathbf{u}$ and $\mathbf{v}$, and the operation of multiplying $\mathbf{u}$ by scalar α, must be well-defined. Furthermore, the resulting quantities $\mathbf{u}+\mathbf{v}$ and $\alpha\mathbf{u}$ must always exist, for any choice of $\mathbf{u}$, $\mathbf{v}$ and α. We know this is the case for vectors, but this definition applies to *any set of mathematical objects* for which these operations can be defined, including matrices and functions. Such a set is called a **vector space**: we will look at vector spaces later in the course. ■

■ **Example 3.1** Suppose T is an anti–clockwise rotation by $90°$ in the $x-y$ plane. Using the parallelogram construction we can see that $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u})+T(\mathbf{v})$ and similarly for $T(\alpha\mathbf{v}) = \alpha T(\mathbf{v})$.





■

■ **Example 3.2** $T:\mathbb{R}^2 \to \mathbb{R}^2$ given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+1 \\ y \end{pmatrix}$ is **not** a linear transformation.

Say $\mathbf{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Then $\mathbf{u}+\mathbf{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \Rightarrow T(\mathbf{u}+\mathbf{v}) = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$. But

$$T(\mathbf{u}) = \begin{pmatrix} 3 \\ 1 \end{pmatrix},\ T(\mathbf{v}) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \Rightarrow T(\mathbf{u})+T(\mathbf{v}) = \begin{pmatrix} 5 \\ 2 \end{pmatrix} \neq T(\mathbf{u}+\mathbf{v})$$

■

Note: You can prove T is *not* a linear transformation using a single counterexample, but to prove it *is* a linear transformation you must prove it in general. ■

■ **Example 3.3** Consider another transformation on $\mathbb{R}^2$ given by $T(\mathbf{v}) = 3\mathbf{v}$. We check using the above properties to confirm that T is linear.
(a) Check whether $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.
Let

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad \text{and} \quad \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \therefore \mathbf{u}+\mathbf{v} = \begin{pmatrix} u_1+v_1 \\ u_2+v_2 \end{pmatrix}$$

$$\therefore T(\mathbf{u}+\mathbf{v}) = T\left(\begin{pmatrix} u_1+v_1 \\ u_2+v_2 \end{pmatrix}\right) = 3\begin{pmatrix} u_1+v_1 \\ u_2+v_2 \end{pmatrix} = \begin{pmatrix} 3u_1+3v_1 \\ 3u_2+3v_2 \end{pmatrix} = 3\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + 3\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = T(\mathbf{u}) + T(\mathbf{v})$$

(b) Check whether $T(\alpha\mathbf{u}) = \alpha T(\mathbf{u})$.

$$T(\alpha\mathbf{u}) = T\left(\begin{pmatrix} \alpha u_1 \\ \alpha u_2 \end{pmatrix}\right) = 3\begin{pmatrix} \alpha u_1 \\ \alpha u_2 \end{pmatrix} = \alpha 3\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \alpha T(\mathbf{u})$$

■

■ **Example 3.4** Consider the mapping from $\mathbb{R}^2$ to $\mathbb{R}^3$ given by

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = (x_2, x_1, x_1+x_2)^T = \begin{pmatrix} x_2 \\ x_1 \\ x_1+x_2 \end{pmatrix}.$$

We will show that this is a linear transformation using the equivalent definition given above, $T(\alpha\mathbf{u}+\beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v})$. Let

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad \text{and} \quad \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \therefore \alpha\mathbf{u} = \begin{pmatrix} \alpha u_1 \\ \alpha u_2 \end{pmatrix} \quad \text{and} \quad \beta\mathbf{v} = \begin{pmatrix} \beta v_1 \\ \beta v_2 \end{pmatrix} \text{and} \therefore \alpha\mathbf{u}+\beta\mathbf{v} = \begin{pmatrix} \alpha u_1+\beta v_1 \\ \alpha u_2+\beta v_2 \end{pmatrix}$$

$$\begin{aligned} T(\alpha\mathbf{u}+\beta\mathbf{v}) &= \begin{pmatrix} \alpha u_2+\beta v_2 \\ \alpha u_1+\beta v_1 \\ \alpha u_1+\beta v_1+\alpha u_2+\beta v_2 \end{pmatrix} = \begin{pmatrix} \alpha u_2 \\ \alpha u_1 \\ \alpha\alpha u_1+\alpha\alpha u_2 \end{pmatrix} + \begin{pmatrix} \beta\beta v_2 \\ \beta v_1 \\ \beta v_1+\beta v_2 \end{pmatrix} \\ &= \alpha\begin{pmatrix} u_2 \\ u_1 \\ u_1+u_2 \end{pmatrix} + \beta\begin{pmatrix} v_2 \\ v_1 \\ v_1+v_2 \end{pmatrix} = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}). \end{aligned}$$

■

The examples above relate to transformations of vectors defined through matrix multiplication. However, we can define linear transformations in much broader contexts, as the following two examples show:

■ **Example 3.5** The definite integral operator $I : C[a,b] \to \mathbb{R}$ defined by $I(f) = \int_a^b f(t)\,\mathrm{d}t$ is a linear transformation ($C[a,b]$ is the set of functions that are continuous on $[a,b]$). To show this, we prove that $I(rf+sg) = rI(f)+sI(g)$, using the rules of integration:

$$I(rf+sg) = \int_a^b [rf(t)+sg(t)]\,\mathrm{d}t = r\int_a^b f(t)\,\mathrm{d}t + s\int_a^b g(t)\,\mathrm{d}t = rI(f)+sI(g)$$

■

■ **Example 3.6** The derivative operator $\frac{\mathrm{d}}{\mathrm{d}x} : P^2 \mapsto P^1$, where P^n is the space of polynomials of degree n, is a linear transformation. Here, the operator $\frac{\mathrm{d}}{\mathrm{d}x}$ simply takes the derivative of the quadratic polynomial

$$\frac{\mathrm{d}}{\mathrm{d}x}(c_1 + c_2x + c_3x^2) = c_2 + 2c_3x$$

How can you show that this is a linear transformation? You need to consider two polynomials in P^2, say $p_1 = a_1 + a_2x + a_3x^2$ and $p_2 = c_1 + c_2x + c_3x^2$, and show that $\frac{\mathrm{d}}{\mathrm{d}x}$ satisfies the definition of a linear transformation. ■

3.2 Matrix representation of linear transformations

Now consider Example 3.4 again, where

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = (x_2, x_1, x_1 + x_2)^T = \begin{pmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{pmatrix},$$

and let A be the matrix where $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}$. Then we can write

$$T(\mathbf{x}) = \begin{pmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A\mathbf{x}$$

In general, if A is any $m \times n$ matrix, we can define a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ given by $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$ (can you check if this is linear?).

■ **Example 3.7** Suppose $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. How would you describe the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by $T(\mathbf{v}) = A\mathbf{v}$ geometrically? (Hint: Take $\mathbf{v} = (v_1, v_2)^T$. What effect does T have on $\mathbf{v}$?)

$$T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 \\ -v_2 \end{pmatrix}$$

So this transformation reflects points in the x–axis. ■

The standard basis in $\mathbb{R}^n$ is the set of vectors

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ .. \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ .. \\ 0 \\ 0 \end{pmatrix}, \quad \ldots, \quad \mathbf{e}_{n-1} = \begin{pmatrix} 0 \\ 0 \\ .. \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ .. \\ 0 \\ 1 \end{pmatrix}$$

■

Theorem 3.2.1 If T is a linear transformation mapping from $\mathbb{R}^n$ into $\mathbb{R}^m$, there is an $m \times n$ matrix such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. In fact, the jth column vector of A (relative to the standard basis $\mathbf{e}_j$) is given by $\mathbf{a}_j = T(\mathbf{e}_j), \quad j = 1, 2, \ldots, n$.

Proof. Define $\mathbf{a}_j = T(\mathbf{e}_j), \quad j = 1, 2, \ldots, n$ and let $A = (a_{ij}) = (\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$ Then $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \ldots x_n\mathbf{e}_n$ for $\mathbf{x} \in \mathbb{R}^n$, and therefore

$$\begin{aligned}
T(\mathbf{x}) &= T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \ldots x_n\mathbf{e}_n) \\
&= x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \ldots x_nT(\mathbf{e}_n) \quad \text{since } T \text{ is a linear transformation} \\
&= x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n \\
&= x_1\begin{pmatrix} a_{11} \\ a_{21} \\ .. \\ .. \\ a_{m1} \end{pmatrix} + x_2\begin{pmatrix} a_{12} \\ a_{22} \\ .. \\ .. \\ a_{m2} \end{pmatrix} + \ldots + x_n\begin{pmatrix} a_{1n} \\ a_{2n} \\ .. \\ .. \\ a_{mm} \end{pmatrix} \\
&= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \\
&= A\mathbf{x}.
\end{aligned}$$

■

■ **Example 3.8** Consider the linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$ given by

$$T(\mathbf{x}) = (x_1 + x_2, x_2 + x_3)^T \quad \text{where} \quad \mathbf{x} = (x_1, x_2, x_3)^T \in \mathbb{R}^3$$

Find a matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^3$, relative to the standard basis.

$$T(\mathbf{e}_1) = T\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1+0 \\ 0+0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$T(\mathbf{e}_2) = T\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0+1 \\ 1+0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$T(\mathbf{e}_3) = T\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0+0 \\ 0+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

So take these vectors as the columns of the matrix A. (check that $T(\mathbf{x}) = A\mathbf{x}$).

$$\therefore A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

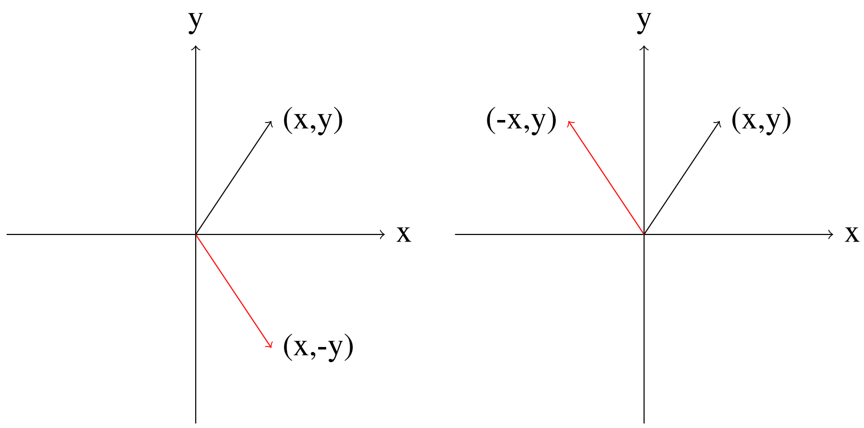
■

Theorem 3.2.1 tells us that to construct the matrix A corresponding to a particular linear transformation, relative to the standard basis of $\mathbb{R}^n$, we just see what T does to each of the basis elements of $\mathbb{R}^n$, $\mathbf{e}_1$, $\mathbf{e}_2$, ..., $\mathbf{e}_n$ in turn and these will provide the columns of A (ie. $\mathbf{a}_j = T(\mathbf{e}_j), \quad j = 1, 2, \ldots, n$). ■

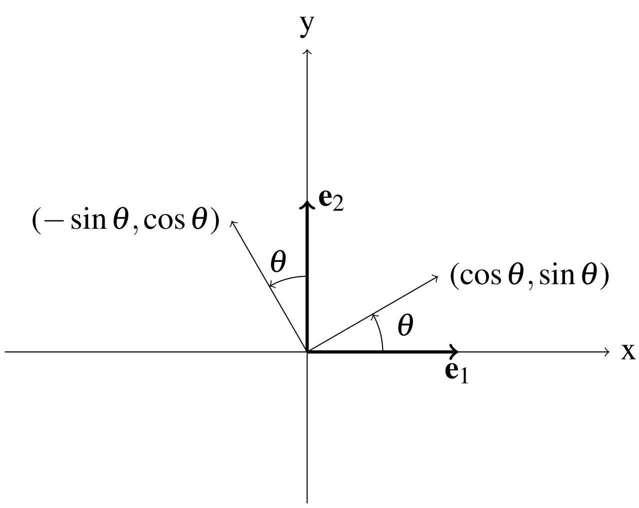
3.3 Rotation and reflection matrices in 2D

We can use Theorem 3.2.1 to find the common rotation and reflection matrices in $\mathbb{R}^2$.

3.3.1 Reflection matrices



(a) Reflection in x-axis (b) Reflection in y-axis



(c) Rotation through an angle θ

(a) Reflection in the x–axis
When points are reflected in the x–axis the mapping is $(x,y)^T \mapsto (x,-y)^T$ and so $(1,0)^T \mapsto (1,0)^T$ and $(0,1)^T \mapsto (0,-1)^T$. This gives $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ from Theorem 3.2.1.

(b) Reflection in the y–axis
When points are reflected in the y–axis the mapping is $(x,y)^T \mapsto (-x,y)^T$ and so $(1,0)^T \mapsto (-1,0)^T$ and $(0,1)^T \mapsto (0,1)^T$. This gives $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ from Theorem 3.2.1.

3.3.2 Rotation matrices

In this case we can see from the diagram that $(1,0)^T \mapsto (\cos\theta, \sin\theta)^T$ and $(0,1)^T \mapsto (-\sin\theta, \cos\theta)^T$, which gives $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.
For example, for a 90° rotation,

$$A = \begin{pmatrix} \cos\frac{\pi}{2} & -\sin\frac{\pi}{2} \\ \sin\frac{\pi}{2} & \cos\frac{\pi}{2} \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}.$$

Note the connection between rotations and the properties of the unit circle, which is commonly parametrised as $(x,y) = (\cos\theta, \sin\theta)$ with θ being measured anticlockwise from the positive x direction. This gives the first column of the rotation matrix. If we instead parametrise using the angle measured from the positive y direction we obtain the second column of the rotation matrix.

3.3.3 Application example: computer graphics and animation

At its simplest level computer graphics manipulates pictures stored as a set of vertices (say in a vector $\mathbf{x}$) by moving the vertices and then redrawing the picture. If the transformation is linear then this can be seen as a series of matrix multiplications.
The four main transformations used are:

1. Dilations ($c > 1$) and contractions ($0 < c < 1$) where $T(\mathbf{x}) = c\mathbf{x}$.
2. Reflections about an axis (eg. $T(\mathbf{x}) = M\mathbf{x}$ where $M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ or $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$).
3. Rotations ($T(\mathbf{x}) = M\mathbf{x}$ where $M = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$).
4. Translations ($T(\mathbf{x}) = \mathbf{x} + a$, where a is a constant.)
   Note: Here T is not linear and so cannot be expressed using a matrix M, but this can be overcome by introducing a new system of coordinates called homogeneous coordinates.

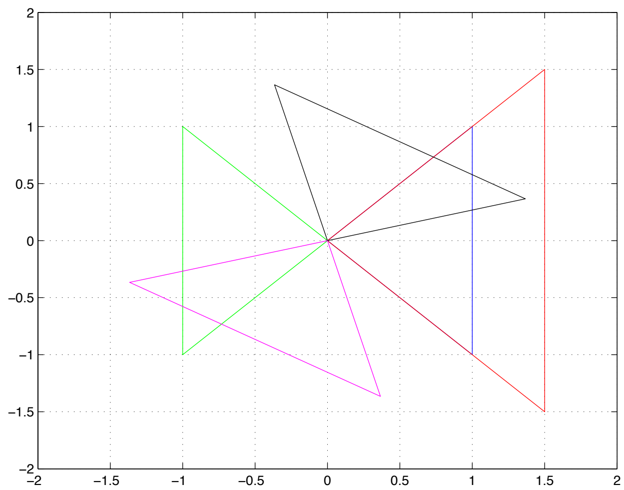
Let's look at a very simple example using MATLAB. Here we start with a triangle with vertices (0,0), (1,1) and (1,-1) represented by the matrix $A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}$. First we draw the triangle in blue (*i.e.*, plot the first row of A against the second row: you might like to revisit the syntax given in section 0.3.2).

```
>> A=[0 1 1 0; 0 1 -1 0]

A =

     0     1     1     0
     0     1    -1     0

>> plot(A(1,:),A(2,:),'b');
```

```
>> grid;
>> axis([-2 2 -2 2]);
>> axis equal
>> hold on
```

The optional command 'b' tells MATLAB to plot the result in blue, grid produces gridlines on the figure, axis() sets the range and domain of the plot, axis equal ensures that the axes are drawn with 1:1 scaling, and hold on tells MATLAB to display any future plot outcomes on the same axes (rather than the default behaviour, which is to produce a new figure for each plot command). Note that the rules of matrix multiplication mean there's no need to explicitly loop over each column of A, which is nice! Next, we dilate A to make B (red) and plot it.

```
>> B=1.5*A;
>> plot(B(1,:),B(2,:),'r');
```



Then we reflect A in the y–axis (green).

```
>> RF=[-1 0; 0 1]

RF =

    -1     0
     0     1

>> C=RF*A;
>> plot(C(1,:),C(2,:),'g');
```

Then we rotate A by 60° (black).

```
>> R60=[cos(pi/3) -sin(pi/3); sin(pi/3) cos(pi/3)]

R60 =

    0.5000   -0.8660
    0.8660    0.5000

>> D=R60*A;
>> plot(D(1,:),D(2,:),'k');
```

Finally, we reflect A in the y-axis and then rotate it by 60° (magenta). Notice the order of the matrices – the reflection matrix (RF) is on the right of the rotation matrix (R60).

```
>> E=R60*RF*A;
>> plot(E(1,:),E(2,:),'m');
```

The same can be achieved in Python using the commands:

```
import numpy as np
import matplotlib.pyplot as plt
A = np.array([[0,1,1,0],[0,1,-1,0]])
B = 1.5*A
RF = np.array([[-1,0],[0,1]])
C = RF@A
R60 = np.array([[np.cos(np.pi/3),-np.sin(np.pi/3)],\
                [np.sin(np.pi/3),np.cos(np.pi/3)]])
D = R60@RF@A
plt.plot(A[0,:],A[1,:],'k-')
plt.plot(B[0,:],B[1,:],'r-')
plt.plot(C[0,:],C[1,:],'g-')
plt.plot(D[0,:],D[1,:],'m-')
plt.grid(1)
plt.axis([-2,2,-2,2])
plt.axis('equal')
```

We've introduced a new package, matplotlib.pyplot, which provides Python with a set of plotting commands that are modelled after MATLAB's. Note that a single backslash can be used to break a line of code in Python (as in the definition of R60). The MATLAB equivalent is ellipses (...).

3.4 Results relating to linear transformations

3.4.1 Inverse linear transformations

Theorem 3.4.1 Let $T:\mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation with matrix A. Then the inverse transformation T^{-1} exists if and only if $\det(A) \neq 0$. Furthermore, $T^{-1}:\mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation with matrix A^{-1}.

■ **Example 3.9**

If $T:\mathbb{R}^3 \to \mathbb{R}^3$ has matrix $A = \begin{pmatrix} 2 & 3 & 1 \\ -1 & 2 & 3 \\ 4 & 1 & 6 \end{pmatrix}$ show that T^{-1} exists and find it.

Now det(A)=63 ($\neq 0$), so A is invertible. Hence, A^{-1} exists and T^{-1} is given by $T^{-1}(\mathbf{x}) = A^{-1}\mathbf{x}$. Also,

$$A^{-1} = \begin{pmatrix} \frac{1}{7} & -\frac{17}{63} & \frac{1}{9} \\ \frac{2}{7} & \frac{8}{63} & -\frac{1}{9} \\ -\frac{1}{7} & \frac{10}{63} & \frac{1}{9} \end{pmatrix}.$$

So $T^{-1}(x_1,x_2,x_3) = (\frac{1}{7}x_1 - \frac{17}{63}x_2 + \frac{1}{9}x_3, \frac{2}{7}x_1 + \frac{8}{63}x_2 - \frac{1}{9}x_3, -\frac{1}{7}x_1 + \frac{10}{63}x_2 + \frac{1}{9}x_3)$. ■

3.4.2 Composition (product) of linear transformations

Theorem 3.4.2 Let $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^p$ be linear transformations with matrices A and B, respectively. Then the product transformation $T_2T_1 : \mathbb{R}^n \to \mathbb{R}^p$ is a linear transformation with matrix BA.

Proof.

$$(T_2T_1)(\mathbf{x}) = T_2(T_1(\mathbf{x})) = T_2(A\mathbf{x}) = B(A\mathbf{x}) = (BA)\mathbf{x}.$$

■

■ **Example 3.10** Let $T_1 : M_n(\mathbb{R}) \to M_n(\mathbb{R})$ and $T_2 : M_n(\mathbb{R}) \to \mathbb{R}$ be linear transformations, where $M_n(\mathbb{R})$ is the set of real $n \times n$ matrices and tr(A)=trace of A = sum of elements on the diagonal.

$$T_1(A) = A + A^T, \;\; T_2(A) = \mathrm{tr}(A).$$

Then $T_2T_1 : M_n(\mathbb{R}) \to \mathbb{R}$ is defined by

$$(T_2T_1)(A) = T_2(T_1(A)) = T_2(A + A^T) = \mathrm{tr}(A + A^T)$$

For example, if $A = \begin{pmatrix} 2 & -1 \\ -3 & 6 \end{pmatrix}$, then

$$T_1(A) = A + A^T = \begin{pmatrix} 4 & -4 \\ -4 & 12 \end{pmatrix}, \text{ so that } T_2(T_1(A)) = \mathrm{tr}\left(\begin{pmatrix} 4 & -4 \\ -4 & 12 \end{pmatrix}\right) = 16.$$

■

3.4.3 Summary of properties of linear transformations

1. For any linear transformation we can find a matrix so that $T(\mathbf{v}) = A\mathbf{v}$.
2. If the transformation is invertible, the inverse transformation has the matrix A^{-1}.
3. The product of two transformations $T_1(\mathbf{v}) = A_1\mathbf{v}$ and $T_2(\mathbf{v}) = A_2\mathbf{v}$ corresponds to the product A_2A_1 of their matrices.

3.5 Application: Linear Recurrence Relations

A useful example of linear transformations is the linear recurrence relation. Recurrence relations are used to define a sequence of values, where each value is defined by some relationship involving previous terms in the sequence.

A simple example is the recurrence relation $x_n = 2x_{n-1}$, defining a sequence of values x_n whose values double from one to the next. For example, if

$$x_0 = 3 \implies x_1 = 6, x_2 = 12, x_3 = 24, x_4 = 48, \ldots$$

While the terms x_n grow exponentially, the relation $x_n = 2x_{n-1}$ is a *linear* relationship between subsequent members of the sequence.

A more complicated example is the Fibonacci sequence[1]. The Fibonacci sequence begins with $x_0 = 0$ and $x_1 = 1$; subsequent members are defined by summing the two

[1] Named for Fibonacci, who popularised the sequence in Europe, it was actually described centuries earlier by Acharya Pingala in India. Pingala also developed Pascal's triangle!

previous ones, *i.e.*, $x_n = x_{n-1} + x_{n-2}$. The Fibonacci relation is therefore also a linear one, even though the terms grow in an exponential fashion.

A convenient alternative way of expressing this linear relationship is *via* a matrix representation, transforming the pair of elements $(x_{n-1}, x_{n-2})^T$ into $(x_n, x_{n-1})^T$:

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ x_{n-2} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{n-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Applying the matrix multiplication many times allows us to calculate values of the Fibonacci sequence for large n. We will see in the next chapter how we can quickly estimate x_n for large n, without needing to perform all these matrix multiplications.

4. Eigenvalues and eigenvectors

Eigenvalues and **eigenvectors** are fundamental properties of square matrices that occur in many mathematical applications (*e.g.*, models of epidemics, the behaviour of economic variables, the buckling of structural columns, quantum physics, face recognition, Google's PageRank algorithm). In this section we will motivate the value of eigenvalues and eigenvectors in tackling various linear algebra problems, and consider useful results in some special cases.

> Although only square matrices have eigenvalues and eigenvectors, it is possible to extend the theory to rectangular matrices A ($m \times n, m \neq n$) by considering $A^{\mathrm{T}}A$, which is then square (and symmetric). More on this in later chapters. ■

4.1 Motivation

4.1.1 Returning to some earlier problems

Recall the marital population problem we encountered earlier in the notes: if 80% of the population is married, and in any year:

- the probability a married person divorces is 30%; and
- the probability an unmarried person marries is 20%;

what percentage is married in future years?

To solve, we set up the matrix problem

$$\begin{aligned}\begin{pmatrix}\text{\% married next yr}\\ \text{\% unmarried next yr}\end{pmatrix} &= \begin{pmatrix}[\text{\% married}]\times 0.7+[\text{\% unmarried}]\times 0.2\\ [\text{\% married}]\times 0.3+[\text{\% unmarried}]\times 0.8\end{pmatrix}\\ &= \begin{pmatrix}0.7 & 0.2\\ 0.3 & 0.8\end{pmatrix}\begin{pmatrix}\text{\% married}\\ \text{\% unmarried}\end{pmatrix}\end{aligned}$$

Initially 80% of the population is married, so we define the initial state

$$\mathbf{x}_0 = \begin{pmatrix} 80 \\ 20 \end{pmatrix}$$

Therefore, after 1 year, we have

$$\mathbf{x}_1 = A\,\mathbf{x}_0 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 80 \\ 20 \end{pmatrix} = \begin{pmatrix} 60 \\ 40 \end{pmatrix}$$

After 2 years, we have

$$\mathbf{x}_2 = A^2\,\mathbf{x}_0 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^2 \begin{pmatrix} 80 \\ 20 \end{pmatrix} = \begin{pmatrix} 50 \\ 50 \end{pmatrix}$$

After 10 years, we have

$$\mathbf{x}_{10} = A^{10}\,\mathbf{x}_0 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^{10} \begin{pmatrix} 80 \\ 20 \end{pmatrix} \approx \begin{pmatrix} 40 \\ 60 \end{pmatrix}$$

After 20, 30 years, we still have

$$\mathbf{x}_{20} = A^{20}\,\mathbf{x}_0 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^{20} \begin{pmatrix} 80 \\ 20 \end{pmatrix} \approx \begin{pmatrix} 40 \\ 60 \end{pmatrix}$$

$$\mathbf{x}_{30} = A^{30}\,\mathbf{x}_0 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^{30} \begin{pmatrix} 80 \\ 20 \end{pmatrix} \approx \begin{pmatrix} 40 \\ 60 \end{pmatrix}$$

If we start with a *different initial population*, we also head toward the same endpoint:

$$A^{20} \begin{pmatrix} 20 \\ 80 \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^{20} \begin{pmatrix} 20 \\ 80 \end{pmatrix} \approx \begin{pmatrix} 40 \\ 60 \end{pmatrix}$$

$$A^{20} \begin{pmatrix} 99 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^{20} \begin{pmatrix} 99 \\ 1 \end{pmatrix} \approx \begin{pmatrix} 40 \\ 60 \end{pmatrix}$$

$\begin{pmatrix} 40 \\ 60 \end{pmatrix}$ is a **steady-state vector** for this process – no matter where we start, we always seem to end up with the same fraction of married and unmarried people. Why?

Is it the same for the Loggerhead Sea Turtle model? Recall the *Leslie* matrix that we set up to describe the population evolution from one year to the next, so that the population after n years is given by $\mathbf{x}_n = L^n\,\mathbf{x}_0$. Let's consider the populations after 25 and 26 years:

$$\mathbf{x}_{25} = L^{25}\,\mathbf{x}_0 = \begin{pmatrix} 0 & 0 & 127 & 79 \\ 0.67 & 0.74 & 0 & 0 \\ 0 & 0.0006 & 0 & 0 \\ 0 & 0 & 0.81 & 0.81 \end{pmatrix}^{25} \begin{pmatrix} 200000 \\ 300000 \\ 500 \\ 1500 \end{pmatrix} \approx \begin{pmatrix} 67574 \\ 198492 \\ 123 \\ 630 \end{pmatrix}$$

$$\mathbf{x}_{26} = L^{26}\,\mathbf{x}_0 = \begin{pmatrix} 0 & 0 & 127 & 79 \\ 0.67 & 0.74 & 0 & 0 \\ 0 & 0.0006 & 0 & 0 \\ 0 & 0 & 0.81 & 0.81 \end{pmatrix}^{26} \begin{pmatrix} 200000 \\ 300000 \\ 500 \\ 1500 \end{pmatrix} \approx \begin{pmatrix} 65418 \\ 192159 \\ 119 \\ 610 \end{pmatrix}$$

Now there isn't a fixed endpoint: instead, the state is *shrinking at a constant rate*:

$$\begin{pmatrix} 65418 \\ 192159 \\ 119 \\ 610 \end{pmatrix} \approx 0.96809 \begin{pmatrix} 67574 \\ 198492 \\ 123 \\ 630 \end{pmatrix}$$

Starting with a *different initial population* leads to the same ultimately rate of decrease:

$$\begin{pmatrix} 0 & 0 & 127 & 79 \\ 0.67 & 0.74 & 0 & 0 \\ 0 & 0.0006 & 0 & 0 \\ 0 & 0 & 0.81 & 0.81 \end{pmatrix}^{25} \begin{pmatrix} 100000 \\ 100000 \\ 1000 \\ 1000 \end{pmatrix} \approx \begin{pmatrix} 52728 \\ 154883 \\ 96 \\ 492 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 127 & 79 \\ 0.67 & 0.74 & 0 & 0 \\ 0 & 0.0006 & 0 & 0 \\ 0 & 0 & 0.81 & 0.81 \end{pmatrix}^{26} \begin{pmatrix} 200000 \\ 300000 \\ 500 \\ 1500 \end{pmatrix} \approx \begin{pmatrix} 51046 \\ 149941 \\ 93 \\ 476 \end{pmatrix}$$

$$\begin{pmatrix} 51046 \\ 149941 \\ 93 \\ 476 \end{pmatrix} \approx 0.96809 \begin{pmatrix} 52728 \\ 154883 \\ 96 \\ 492 \end{pmatrix}$$

Furthermore, the vectors that describes these declining states are *proportional to one another as well*, even through their initial states were quite different!

$$\begin{pmatrix} 51046 \\ 149941 \\ 93 \\ 476 \end{pmatrix} \left[\approx 0.7554 \begin{pmatrix} 67574 \\ 198492 \\ 123 \\ 630 \end{pmatrix} \right]$$

How can we explain all this?

4.1.2 Some special vectors

The vectors $\begin{pmatrix} 40 \\ 60 \end{pmatrix}$ and $\begin{pmatrix} 65418 \\ 192159 \\ 119 \\ 610 \end{pmatrix}$ are special to each problem:

$$A \begin{pmatrix} 40 \\ 60 \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 40 \\ 60 \end{pmatrix} = \begin{pmatrix} 40 \\ 60 \end{pmatrix}$$

$$L \begin{pmatrix} 67574 \\ 198492 \\ 123 \\ 630 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 127 & 79 \\ 0.67 & 0.74 & 0 & 0 \\ 0 & 0.0006 & 0 & 0 \\ 0 & 0 & 0.81 & 0.81 \end{pmatrix} \begin{pmatrix} 67574 \\ 198492 \\ 123 \\ 630 \end{pmatrix} = \begin{pmatrix} 65418 \\ 192159 \\ 119 \\ 610 \end{pmatrix}$$

$$\approx 0.96809 \begin{pmatrix} 67574 \\ 198492 \\ 123 \\ 630 \end{pmatrix}$$

These vectors satisfy a special relationship $A\mathbf{v} = \lambda\mathbf{v}$ or $L\mathbf{v} = \lambda\mathbf{v}$:
- The vector $\mathbf{v}$ is called an **eigenvector**
- The scalar λ is the corresponding **eigenvalue**

4.2 Calculating eigenvalues and eigenvectors

Example 4.1 Consider the transformation $T(\mathbf{u}) = A\mathbf{u}$ for $A = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$. Calculate the effect of T on the vectors: (a) $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, (b) $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, (c) $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$, (d) $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, and (e) $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

(a) $T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

(b) $T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$

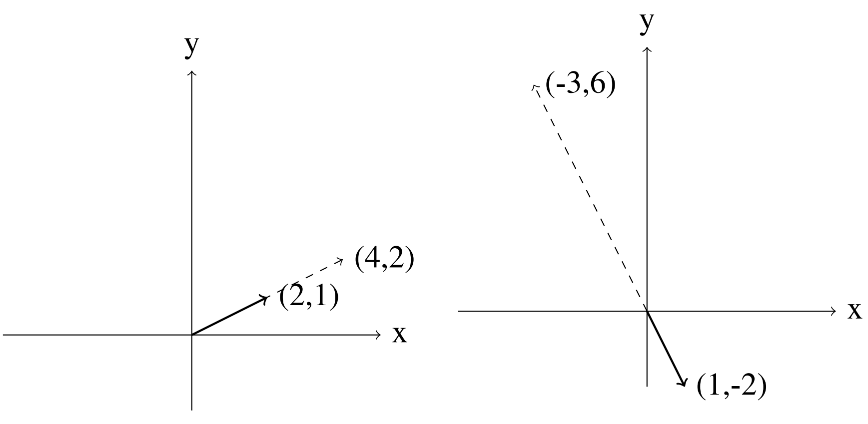
(c) $T\begin{pmatrix} -1 \\ -1 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}$

(d) $T\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

(e) $T\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$

■

For (d) and (e) we plot the original and the transformed vectors.



(d) (e)

Notice that the vector in (d) is just rescaled by 2,

$$T\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} = 2\begin{pmatrix} 2 \\ 1 \end{pmatrix},$$

and in (e) the vector is rescaled by 3 and its direction is reversed

$$T\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \end{pmatrix} = -3\begin{pmatrix} 1 \\ -2 \end{pmatrix},$$

so in each of (d) and (e) this means that

$$A\mathbf{v} = \lambda \mathbf{v}$$

where λ is a scalar (number) called the eigenvalue and $\mathbf{v}$ is the eigenvector.

In first year, and earlier in these notes, we noted that we can consider the effect of (pre-)multiplying vector $\mathbf{v}$ by matrix A as a **transformation** from the vector $\mathbf{v}$ into the vector $A\mathbf{v}$. The vector might be rotated, reflected, shrunk, reversed etc. Eigenvectors are the vectors that are **not** rotated but simply shrunk, stretched and/or reversed – they are stable directions connected to the process of multiplying by matrix A.

Definition 4.2.1 — Eigenvalues and eigenvectors. Let A be an $n \times n$ matrix. A scalar λ is said to be an **eigenvalue** of A if there exists a nonzero vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$. The vector $\mathbf{v}$ is said to be an **eigenvector** belonging to the **eigenvalue** λ.

We need to solve the system $A\mathbf{v} = \lambda\mathbf{v}$ or $(A - \lambda I)\mathbf{v} = \mathbf{0}$ and so, based on our earlier work on systems of linear equations, we can say that λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$.

Definition 4.2.2 — Characteristic polynomial. The polynomial $\det(A - \lambda I) = 0$ is known as the **characteristic polynomial** of A.

4.2.1 Method for finding eigenvalues and eigenvectors

1. Write the characteristic equation $\det(A - \lambda I) = 0$.
2. Solve the characteristic equation for the eigenvalues λ_i.
3. For each eigenvalue λ_i find the eigenvector(s) by solving $(A - \lambda_i I)\mathbf{v}_i = \mathbf{0}$.

Since $\det(A - \lambda I) = 0$, the solution for $\mathbf{v}_i$ is *not* unique: we must use Gaussian elimination to determine the family[a] of possible eigenvectors $\mathbf{v}_i$. ■

[a] Some fields will demand that the eigenvectors have a particular normalisation, *e.g.*, in quantum physics we always work with unit eigenvectors. This reduces the family to a unique solution.

■ **Example 4.2** Find the eigenvalues and eigenvectors for $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.

1. $\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4)$
2. Setting $\det(A - \lambda I) = 0$, we find that the eigenvalues of A are $\lambda_1 = 2$ and $\lambda_2 = 4$.
3. $(A - \lambda I)\mathbf{v} = \begin{pmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ for $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$
   (a) For $\lambda_1 = 2$,

$$(A - 2I)\mathbf{v}_1 = \begin{pmatrix} 3-2 & 1 \\ 1 & 3-2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From this $v_1 + v_2 = 0 \;\Rightarrow v_2 = -v_1$, and letting $v_1 = r$, we have

$$\mathbf{v}_1 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} r \\ -r \end{pmatrix} = r\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

So any non–zero multiple of $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector of A associated with $\lambda_1 = 2$. If we normalise,

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

(b) For $\lambda_2 = 4$,

$$(A-4I)\mathbf{v}_2 = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From this $-v_1 + v_2 = 0 \;\Rightarrow v_1 = v_2$, and letting $v_2 = s$, we have

$$\mathbf{v}_2 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} s \\ s \end{pmatrix} = s \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

So any non–zero multiple of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector of A associated with $\lambda_2 = 4$. Normalising,

$$\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

■

1. When calculating the eigenvectors, you could reduce the matrices to row echelon form, but it is not really necessary in the 2×2 case (the second row will be a multiple of the first, so becomes a zero row under Gaussian elimination).
2. Eigenvectors are **not unique**. For example, in (a) above, any vector along the line $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector.
3. There are some interesting results that relate the properties of the matrix (determinant, trace) to the eigenvalues. If A is an $n \times n$ matrix with n eigenvalues:
   (a) The sum of the eigenvalues is equal to the **trace** (sum of diagonal entries) of the matrix. In the above example, sum of eigenvalues = $2+4=6$ and the sum of the diagonal entries = $3+3=6$ (this makes a good check for your calculations!).
   (b) The product of the eigenvalues = determinant of A. In the above example, $\det(A) = 3^2 - 1^2 = 8$ and product of the eigenvalues = $2\times 4 = 8$.

Theorem 4.2.1 The eigenvalues of a 2×2 matrix are the solutions to

$$\lambda^2 - \operatorname{trace}(A)\lambda + \det(A) = 0$$

Proof. Consider $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$.

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} a_{11}-\lambda & a_{12} \\ a_{21} & a_{22}-\lambda \end{pmatrix} \\ \therefore \det(A-\lambda I) &= (a_{11}-\lambda)(a_{22}-\lambda) - a_{12}a_{21} \\ &= \lambda^2 - (a_{11}+a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21} \\ &= \lambda^2 - \text{trace}(A)\lambda + \det(A) \end{aligned}$$

■

Now, if A has eigenvalues λ_1 and λ_2, then

$$(\lambda-\lambda_1)(\lambda-\lambda_2) = 0 \Rightarrow \lambda^2 - (\lambda_1+\lambda_2)\lambda + \lambda_1\lambda_2 = 0$$

and so from Theorem 4.2.1 we see that

$$\lambda_1 + \lambda_2 = \text{trace}(A) \ \text{ and } \ \lambda_1\lambda_2 = \det(A).$$

The examples we have just discussed are ones where the matrix has two distinct real eigenvectors. This is not always the case, as we will see below.

■ **Example 4.3** Find the eigenvalues and eigenvectors for $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

(Note: from the above result we know that $\lambda_1 + \lambda_2 = 0$ and $\lambda_1\lambda_2 = \det(A) = 1$).

$$\det(A-\lambda I) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 = 0 \Rightarrow \lambda = \pm i$$

So the real matrix A has complex eigenvalues!

$$(A-\lambda I)\mathbf{v} = \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

For $\lambda_1 = i$, $\begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and letting $v_1 = r$,

$$\therefore -iv_1 - v_2 = 0 \Rightarrow v_2 = -iv_1 \Rightarrow \mathbf{v_1} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} r \\ -ri \end{pmatrix} = r\begin{pmatrix} 1 \\ -i \end{pmatrix}$$

So the eigenvectors associated with $\lambda = i$ are $\mathbf{v}_1 = r\begin{pmatrix} 1 \\ -i \end{pmatrix}$.

For $\lambda_2 = -i$, the eigenvectors are $\mathbf{v}_2 = r\begin{pmatrix} 1 \\ i \end{pmatrix}$ or if we let $v_2 = s$ we could write $\mathbf{v}_2 = s\begin{pmatrix} -i \\ 1 \end{pmatrix}$ (check these results). ■

Why does this real 2×2 matrix have two complex eigenvalues? The reason is that this matrix is a rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ where $\theta = 90^\circ$ and so you cannot ever have any real eigenvalues, since the transformed vector $A\mathbf{v}$ always points in a different direction from the original vector $\mathbf{v}$.

Complex eigenvalues of real matrices: if a real matrix has a complex eigenvalue $\lambda = a + bi$ ($a,\ b \in \mathbb{R}$) then the complex conjugate $\lambda^* = a - bi$ is also an eigenvalue of that matrix.

Example 4.4 Find the eigenvalues and eigenvectors for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

(Note: from the above result we know that $\lambda_1 + \lambda_2 = 2$ and $\lambda_1 \lambda_2 = \det(A) = 1$).

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 = 0 \Rightarrow \lambda = 1$$

$$(A - \lambda I)\mathbf{v} = \begin{pmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

For $\lambda = 1$, $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\therefore v_2 = 0 \Rightarrow \mathbf{v} = \begin{pmatrix} r \\ 0 \end{pmatrix} = r \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ when we let $v_1 = r$.

In this case there is only one linearly independent eigenvector. Geometrically, A is a **shear** transformation, which maps the unit square to a parallelogram. The only direction left unchanged is along the x-axis, so all eigenvectors are nonzero vectors parallel to the x-axis.

Calculating eigenvalues and eigenvectors by hand becomes difficult in most cases for matrices larger than 2×2. The following 3×3 example works because the cubic factorises, but that is not normally the case (in fact, there is no general algebraic solution for 5×5 or larger matrices). The usual practice would be to use MATLAB commands, as explained below. This example is included to show you the connection with 3×3 determinants.

Example 4.5 Find the eigenvalues and eigenvectors for $A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 3 & 0 & 2 \end{pmatrix}$.

$$\begin{aligned}
\det(A - \lambda I) &= \begin{pmatrix} 1-\lambda & 0 & 2 \\ 0 & 2-\lambda & 1 \\ 3 & 0 & 2-\lambda \end{pmatrix} \\
&= (1-\lambda) \begin{vmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} - 0 \begin{vmatrix} 0 & 1 \\ 3 & 2-\lambda \end{vmatrix} + 2 \begin{vmatrix} 0 & 2-\lambda \\ 3 & 0 \end{vmatrix} \\
&= (1-\lambda)(2-\lambda)^2 - 6(2-\lambda) \\
&= (2-\lambda)[(1-\lambda)(2-\lambda) - 6] \\
&= (2-\lambda)[2 - 3\lambda + \lambda^2 - 6] \\
&= (2-\lambda)(\lambda^2 - 3\lambda - 4) \\
&= (2-\lambda)(\lambda+1)(\lambda-4) = 0 \therefore \lambda = 2, -1, 4.
\end{aligned}$$

$$(A - \lambda I)\mathbf{v} = \mathbf{0} \Rightarrow \begin{pmatrix} 1-\lambda & 0 & 2 \\ 0 & 2-\lambda & 1 \\ 3 & 0 & 2-\lambda \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

For $\lambda_1 = 2$,

$$\begin{pmatrix} -1 & 0 & 2 \\ 0 & 0 & 1 \\ 3 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow v_3 = 0, v_1 = 0, \ \ \mathbf{v}_1 = s \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \ \text{ taking } v_2 = s.$$

For $\lambda_2 = 4$,

$$\begin{pmatrix} -3 & 0 & 2 \\ 0 & -2 & 1 \\ 3 & 0 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow v_1 = \frac{2}{3}v_3, v_2 = \frac{1}{2}v_3, \ \ \mathbf{v}_2 = t \begin{pmatrix} \frac{2}{3} \\ \frac{1}{2} \\ 1 \end{pmatrix} = \frac{t}{6} \begin{pmatrix} 4 \\ 3 \\ 6 \end{pmatrix} \ \text{ taking } v_3 = t.$$

For $\lambda_3 = -1$,

$$\begin{pmatrix} 2 & 0 & 2 \\ 0 & 3 & 1 \\ 3 & 0 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow v_1 = -v_3, v_2 = -\frac{1}{3}v_3, \ \ \mathbf{v}_3 = u \begin{pmatrix} -1 \\ -\frac{1}{3} \\ 1 \end{pmatrix} = \frac{u}{3} \begin{pmatrix} -3 \\ -1 \\ 3 \end{pmatrix} \ \text{ taking } v_3 = u.$$

■

4.2.2 Eigenvalues of triangular matrices

The matrix in Example 4.4 is an upper triangular matrix (ie. all the non–zero elements are on or above the diagonal) and the eigenvalues are the elements on the diagonal. So in this case the elements on the diagonal are 1 (and 1) and the eigenvalue is 1. This is true for all matrices that are either upper or lower triangular.

Let's look at a less trivial example and consider $A = \begin{pmatrix} 4 & 4 & 6 \\ 0 & 2 & 7 \\ 0 & 0 & 1 \end{pmatrix}$.

In this case the eigenvalues are easy to find since it is upper triangular – they are just 4, 2 and 1 (the numbers on the diagonal). It is possible to prove this in general, but it is easy to see why it is the case, since to find the eigenvalues we must find the determinant $\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 4 & 6 \\ 0 & 2-\lambda & 7 \\ 0 & 0 & 1-\lambda \end{vmatrix}$ and set it to 0 and if you expand along the first column you can see that, because of the zeros, the determinant = $(4-\lambda)(2-\lambda)(1-\lambda)$ giving the elements on the diagonal as the eigenvalues.

4.2.3 Finding eigenvalues and eigenvectors using MATLAB

MATLAB has efficient algorithms for finding eigenvalues and eigenvectors that do not use determinants (*e.g.*, the Jacobi method).

> **Note** MATLAB returns normalised eigenvectors (eigenvectors with length 1) which at first glance may not appear to agree with the eigenvectors you have calculated by hand. Remember that eigenvectors are not unique. Any nonzero scalar multiple of an eigenvector is also an eigenvector. ■

The MATLAB method for calculating determinants has two steps;

1. construct the matrix A; and
2. use the MATLAB command `eig`, which has two outputs `[V,D] = eig(A)`.

The two matrix outputs are:
`D` = diagonal matrix containing the eigenvalues of A; and
`V` = matrix whose columns are the eigenvectors of A,
where each column has been normalised, and the order of the eigenvectors matches the order of the eigenvalues.

Example 4.6 — Example 4.2 revisited.
In MATLAB, to calculate eigenvalues/vectors of $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, we type

```
>> A=[3 1; 1 3]
>> [V,D]=eig(A)
```

and the MATLAB output is

```
V =
    -0.7071      0.7071
     0.7071      0.7071
D =
       2       0
       0       4
```

This tells us that the eigenvector corresponding to the eigenvalue 2 is $\begin{pmatrix} -0.7071 \\ 0.7071 \end{pmatrix}$ and the eigenvector corresponding to the eigenvalue 4 is $\begin{pmatrix} 0.7071 \\ 0.7071 \end{pmatrix}$. This compares with our previous work that said that the eigenvector corresponding to the eigenvalue 2 is $r\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and the eigenvector corresponding to the eigenvalue 4 is $s\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Taking the case where $\lambda = 4$ first you can see that $\begin{pmatrix} 0.7071 \\ 0.7071 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (to 4 decimal places) (ie. it is a normalised version of our answer $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$).

The same applies to the eigenvector for $\lambda = 2$ except that the sign is reversed and this is because previously we chose to write $v_2 = -v_1$ giving us an eigenvector of the form $r\begin{pmatrix} 1 \\ -1 \end{pmatrix}$. However, we could equally well have written $v_1 = -v_2$ giving us an eigenvector of the form $s\begin{pmatrix} -1 \\ 1 \end{pmatrix}$, which then matches the MATLAB output once normalising is taken into account. ■

Example 4.7 — Example 4.5 revisited. In MATLAB, to calculate eigenvalues/vectors of $A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 3 & 0 & 2 \end{pmatrix}$, we type

```
>> A=[1 0 2; 0 2 1; 3 0 2]
>> [V,D]=eig(A)
```

and the MATLAB output is

```
V =
```

```
         0   -0.6882   -0.5121
    1.0000   -0.2294   -0.3841
         0    0.6882   -0.7682

D =
     2     0     0
     0    -1     0
     0     0     4
```

So eigenvalue 2 corresponds to eigenvector $\begin{pmatrix}0\\1\\0\end{pmatrix}$, which agrees with Example 5. Eigenvalue -1 corresponds to $\begin{pmatrix}-0.6882\\-0.2294\\0.6882\end{pmatrix} = 0.6882\begin{pmatrix}-1\\-\frac{1}{3}\\1\end{pmatrix}$, the normalised form of $\begin{pmatrix}-1\\-\frac{1}{3}\\1\end{pmatrix}$.
Similarly, eigenvalue 4 corresponds to $\begin{pmatrix}-0.5121\\-0.3841\\0.7682\end{pmatrix} = -0.7682\begin{pmatrix}\frac{2}{3}\\\frac{1}{2}\\1\end{pmatrix}$, the normalised form of $\begin{pmatrix}\frac{2}{3}\\\frac{1}{2}\\1\end{pmatrix}$ that we found previously. ■

■ **Example 4.8 — Repeated eigenvalues.**
Find the eigenvalues and eigenvectors for $A = \begin{pmatrix}1&0&0\\0&2&1\\0&1&2\end{pmatrix}$.

$$A-\lambda I = \begin{pmatrix}1-\lambda&0&0\\0&2-\lambda&1\\0&1&2-\lambda\end{pmatrix} \therefore \det(A-\lambda I) = (1-\lambda)[(2-\lambda)^2-1] = (1-\lambda)(\lambda-3)(\lambda-1)$$

So eigenvalues are $\lambda = 3$ and $\lambda = 1$ (repeated).
For $\lambda_1 = 3$,

$$\begin{pmatrix}-2&0&0\\0&-1&1\\0&1&-1\end{pmatrix}\begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow v_1 = 0,\ v_2 - v_3 = 0 \Rightarrow v_2 = v_3 \therefore V_1 = s\begin{pmatrix}0\\1\\1\end{pmatrix} \text{ writing } v_2 = s$$

For $\lambda_2 = 1$,

$$\begin{pmatrix}0&0&0\\0&1&1\\0&1&1\end{pmatrix}\begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow v_2+v_3 = 0 \Rightarrow v_2 = -v_3 \therefore V_2 = r\begin{pmatrix}1\\0\\0\end{pmatrix} + t\begin{pmatrix}0\\-1\\1\end{pmatrix} \text{ writing } v_1 = r, v_2 = t$$

This tells us that any vector in this plane will be an eigenvector associated with $\lambda = 1$. Notice that in this case we still have three linearly independent eigenvectors for our 3×3 matrix despite the repeated eigenvalue. So what does MATLAB give us in this case?

```
V =
    1.0000         0         0
         0   -0.7071    0.7071
         0    0.7071    0.7071
```

```
D =
        1        0        0
        0        1        0
        0        0        3
```

The eigenvector for $\lambda = 3$ is $\begin{pmatrix} 0 \\ 0.7071 \\ 0.7071 \end{pmatrix}$ which is the normalised form of $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ as expected and the eigenvectors for $\lambda = 1$ are $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ -0.7071 \\ 0.7071 \end{pmatrix}$ which again match the values above. ■

Repeated eigenvalues do occur in practice, particularly when the matrix arises from a problem involving a lot of symmetry. ■

■ **Example 4.9 — Sea turtle example.** For the matrix L in the Sea Turtle example, the eigenvalues and eigenvectors don't have a simple analytic form. MATLAB gives us

```
V =
   0.74679 + 0.00000i    0.74679 - 0.00000i    0.24806 + 0.00000i    0.32227 + 0.00000i
  -0.64818 - 0.14885i   -0.64818 + 0.14885i    0.96874 + 0.00000i    0.94664 + 0.00000i
  -0.00062 + 0.00228i   -0.00062 - 0.00228i    0.00102 + 0.00000i    0.00059 + 0.00000i
   0.00106 - 0.00208i    0.00106 + 0.00208i    0.00343 + 0.00000i    0.00301 + 0.00000i
D =
   0.00674 + 0.16840i                   -0                    -0                    -0
                    0    0.00674 - 0.16840i                    0                     0
                    0                    0    0.56844 + 0.00000i                     0
                    0                    0                     0    0.96809 + 0.00000i
```

The eigenvector for $\lambda_4 = 0.96809$ is $\mathbf{v}_4 = \begin{pmatrix} 0.32227 & 0.94664 & 0.00059 & 0.00301 \end{pmatrix}^T$, which was the vector we identified in the long-time limit behaviour of the turtle population:

$$\begin{pmatrix} 65418 \\ 192159 \\ 119 \\ 610 \end{pmatrix} \approx 202991\mathbf{v}_4, \quad \begin{pmatrix} 51046 \\ 149941 \\ 93 \\ 476 \end{pmatrix} \approx 158394\mathbf{v}_4$$

■

4.2.4 Finding eigenvalues and eigenvectors using R & Python

To calculate eigenvalues and eigenvectors in R, use the command `eigen()`. This produces a list containing `values` and `vectors`, which can be extracted using the `$` operator.

```
> A <- matrix(c(13, -4, 2, -4, 11, -2, 2, -2, 8), 3, 3, byrow=TRUE)
> ev <- eigen(A);
> ev$values
[1] 17  8  7
> ev$vectors
           [,1]       [,2]      [,3]
[1,]  0.7453560  0.6666667 0.0000000
[2,] -0.5962848  0.6666667 0.4472136
[3,]  0.2981424 -0.3333333 0.8944272
```

The eigenvalues are listed in descending order and the eigenvectors are given as columns of ev$vectors in corresponding order.

In Python, we can solve the same eigensystem using

```
> A = np.array([[13,-4,2],[-4,11,-2],[2,-2,8]])
> eigenvalues, eigenvectors = np.linalg.eig(A)
> eigenvalues
array([17.,  8.,  7.])
> eigenvectors
array([[ 0.74535599, -0.66666667,  0.        ],
       [-0.59628479, -0.66666667,  0.4472136 ],
       [ 0.2981424 ,  0.33333333,  0.89442719]])
```

Note that R and Python produce different vectors (± 1) for the second eigenvector – this is because they have chosen different normalisation coefficients.

4.2.5 Summary

If A is an $n \times n$ matrix, we solve the characteristic equation $\det(A - \lambda I) = 0$ to find the eigenvalues. When this is expanded an n–th degree polynomial is obtained and this has n roots (some of which may be repeated) and some of which may be complex conjugate pairs ($a + bi$ and $a - bi$). This follows from the Fundamental Theorem of Algebra.

In some cases where eigenvalues are repeated there will be fewer than n linearly independent eigenvectors; however, this is not always the case. This is important because we need n linearly independent eigenvectors to be able to diagonalise a matrix, as we will see below.

4.3 Bases and linear independence

4.3.1 Motivation

For the matrix at the heart of our Marital Population problem,

$$\begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \text{ and}$$

$$\begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

So we have eigenvalues $\lambda_1 = 1, \lambda_2 = 1/2$ with their corresponding eigenvalues $\mathbf{v}_1, \mathbf{v}_2$.

If we can write some vector $\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$, what is $A\mathbf{x}$?

$$\begin{aligned} A\mathbf{x} &= A(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) \\ &= c_1(A\mathbf{v}_1) + c_2(A\mathbf{v}_2) \\ &= c_1\lambda_1\mathbf{v}_1 + c_2\lambda_2\mathbf{v}_2 \\ &= c_1\mathbf{v}_1 + \frac{c_2}{2}\mathbf{v}_2 \end{aligned}$$

This looks a little like the Matrix Representation theorem:

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n),$$

except that we're using the eigenvectors $\mathbf{v}_i$ of A instead of the standard basis $\mathbf{e}_i$. Furthermore, the output is *also* related to $\lambda_i \mathbf{v}_i$ (rather than $T(\mathbf{e}_i)$).

This eigenvector representation is a real advantage when multiplying many times because, for example,

$$A^2\mathbf{v}_i = A(A\mathbf{v}_i) = A(\lambda_i \mathbf{v}_i) = \lambda_i(A\mathbf{v}_i) = \lambda_i^2 \mathbf{v}_i$$

and $A^n\mathbf{v}_i = \lambda_i^n \mathbf{v}_i$ more generally. In the case of the Marital Population problem,

$$\begin{aligned} A^n\mathbf{x} &= A^n(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) \\ &= c_1(A^n\mathbf{v}_1) + c_2(A^n\mathbf{v}_2) \\ &= c_1\lambda_1^n\mathbf{v}_1 + c_2\lambda_2^n\mathbf{v}_2 \\ &= c_1\mathbf{v}_1 + \frac{c_2}{2^n}\mathbf{v}_2 \end{aligned}$$

By contrast, the Matrix Representation theorem is of little help for this calculation, because in general, $T(\mathbf{e}_i)$ has no special relationship with $\mathbf{e}_i$.

Over many years, from initial condition $\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$, we see that

$$\mathbf{x}_n = A^n\mathbf{x}_0 = c_1\mathbf{v}_1 + \frac{c_2}{2^n}\mathbf{v}_2 \ \to \ c_1\mathbf{v}_1$$

as $n \to \infty$, because the coefficient $\frac{c_2}{2^n} \to 0$.

This still leaves us with a few questions:

- Can we always write any $\mathbf{x}$ as a linear combination of eigenvectors?
- Why was c_1 always equal to 1 for the marital population problem?

4.3.2 Bases

Can we always write $\mathbf{x}$ as a linear combination of the eigenvectors of a matrix? To answer this question, we need to introduce the idea of a **basis**.[1]

Definition 4.3.1 — Basis. The vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ in $\mathbb{R}^m$ form a **basis** for $\mathbb{R}^m$ iff every vector $\mathbf{x} \in \mathbb{R}^m$ is a unique linear combination of the $\mathbf{v}_i$.

That is, for any $\mathbf{x} \in \mathbb{R}^m$, there must be a unique $(c_1, c_2, \ldots, c_n)^T$ such that

$$\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n = \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = V\mathbf{c}$$

The key reasons for this definition are to ensure that

- *all* $\mathbf{x} \in \mathbb{R}^m$ must have a representation $(c_1, c_2, \ldots, c_n)^T$ in the basis
- *no* $\mathbf{x} \in \mathbb{R}^m$ can have two (or more) such representations

How many vectors do we need to form a basis in $\mathbb{R}^m$?

[1] plural of 'basis' is 'bases', often pronounced 'base-ees'.

The vectors $\begin{pmatrix}1\\2\\4\end{pmatrix}, \begin{pmatrix}2\\1\\3\end{pmatrix}, \begin{pmatrix}3\\1\\2\end{pmatrix}, \begin{pmatrix}4\\0\\1\end{pmatrix}$ cannot form a basis for $\mathbb{R}^3$. To see this, we row-reduce the problem $V\mathbf{c} = \mathbf{x}$, to find the coefficients of the $\mathbf{v}_i$:

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & x \\ 2 & 1 & 1 & 0 & y \\ 4 & 3 & 2 & 1 & z \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & ? \\ 0 & 1 & 0 & 1 & ? \\ 0 & 0 & 1 & 1 & ? \end{array}\right]$$

Whenever there are fewer rows than columns, as is the case here, then we get either no solution (if there is a zero row with a non-zero RHS); or a family of solutions. *We can never get a unique solution.* Our conclusion is that we can't have more vectors than the dimension of the space.

The vectors $\begin{pmatrix}1\\2\\4\end{pmatrix}, \begin{pmatrix}2\\1\\3\end{pmatrix}$ also cannot form a basis for $\mathbb{R}^3$. Again, we row-reduce the problem $V\mathbf{c} = \mathbf{x}$:

$$\left[\begin{array}{cc|c} 1 & 2 & x \\ 2 & 1 & y \\ 4 & 3 & z \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & ? \\ 0 & 1 & ? \\ 0 & 0 & ? \end{array}\right]$$

Whenever there are fewer rows than columns, as is the case here, we must get some rows with zeros on the left-hand side. We may get no solution, or a family of solutions, depending on the value of the '?' in the bottom row. But, once again *we can never get a unique solution*. Our conclusion, like before, is that we can't have fewer vectors than the dimension of the space.

It follows that **a basis for $\mathbb{R}^n$ requires n vectors**. Too few vectors, and there are guaranteed to be vectors in $\mathbb{R}^n$ that have no representation. Too many vectors, and *none* of the representations for vectors in $\mathbb{R}^n$ will be unique (and some vectors may still have *no* representation at all).

4.3.3 Linear independence

But are we guaranteed to span $\mathbb{R}^n$ if we have n different vectors? Do the vectors $\mathbf{v}_1 = \begin{pmatrix}1\\2\\4\end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix}2\\1\\3\end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix}1\\-1\\-1\end{pmatrix}$ form a basis for $\mathbb{R}^3$? No, they can't. The problem here is that $\mathbf{v}_3$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$: $\mathbf{v}_3 = \mathbf{v}_2 - \mathbf{v}_1$. This means that

$$\begin{vmatrix} | & | & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ | & | & | \end{vmatrix} = 0 \quad \Longrightarrow \quad V\mathbf{c} = \mathbf{x} \text{ has either no solution or a family of solutions}$$

We need to avoid this situation, ensuring that *none* of the $\mathbf{v}_i$ can be expressed as a linear combination of the others. This leads us to the concept of **linear independence**.

Definition 4.3.2 The vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are *linearly independent* if and only if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0} \;\Rightarrow\; c_1 = c_2 = \cdots = c_n = 0.$$

That is, no $\mathbf{v}_k$ is a linear combination of the others. It is also useful to define the opposing notion, **linear dependence**:

Definition 4.3.3 If there *are* non-zero $c_1, c_2, \cdots, c_n$ for which $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$, then the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are *linearly dependent*.

To establish linear independence, we need to show that $V\mathbf{c} = \mathbf{0}$ has only the trivial solution, $\mathbf{c} = \mathbf{0}$. This can be checked using row reduction. Form the augmented matrix

$$\left(\begin{array}{ccc|c} | & & | & 0 \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n & 0 \\ | & & | & 0 \end{array}\right)$$

and obtain reduced row echelon form. If there are fewer than n non-zero rows, then the system has a free variable and the vectors are linearly dependent. Otherwise, the vectors are linearly independent.

For the special case of n vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ in $\mathbb{R}^n$, linear independence can be determined directly from the determinant of the matrix whose columns are the vectors:

$$\begin{vmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{vmatrix} \neq 0 \quad \Longleftrightarrow \quad \mathbf{v}_1, \ldots, \mathbf{v}_n \text{ are linearly independent}$$

Combining these results gives us a simple test to see whether a set of vectors forms a basis in $\mathbb{R}^n$.

Theorem 4.3.1 A set of vectors is a basis for $\mathbb{R}^n$ iff it is made up of n linearly independent vectors.

Proof. We've seen that any basis for $\mathbb{R}^n$ must contain precisely n vectors: let's call them $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, and define $V = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{pmatrix}$.

If the $\mathbf{v}_i$ form a basis, then each vector has a unique representation, including the zero vector $\mathbf{0}$. Thus $V\mathbf{c} = \mathbf{0}$ has a unique solution, $\mathbf{c} = 0$, which means that the $\mathbf{v}_i$ are linearly independent.

If the $\mathbf{v}_i$ are linearly independent, then $\det(V) \neq 0$, so V^{-1} exists. Thus, each $\mathbf{x} \in \mathbb{R}^n$ has a unique representation in the basis, given by coefficients $\mathbf{c} = V^{-1}\mathbf{x}$, which means that the $\mathbf{v}_i$ form a basis. ■

4.3.4 Eigenvectors as a basis

We are now in a position to determine whether the eigenvectors of a matrix can be used to represent every vector in $\mathbb{R}^n$.

Theorem 4.3.2 If the eigenvectors of an $n \times n$ matrix are linearly independent, they form a basis for $\mathbb{R}^n$.

This follows immediately from the previous result. Moreover, in some cases we can predict if the eigenvectors will be linearly independent. The following is one of the most common and important:

Theorem 4.3.3 If an $n \times n$ matrix has n distinct real eigenvalues, then the corresponding eigenvectors form a basis for $\mathbb{R}^n$.

Proof. We will prove by contradiction that if the first $r(< n)$ eigenvectors are linearly independent, then the first $r+1$ eigenvectors must also be linearly independent. Since this is true for $r = 1$, it implies that it must be true for all r, meaning that *all* the eigenvectors must be linearly independent.

Assume the first r eigenvectors are linear independent, but that the first $r+1$ eigenvectors are *not* linearly independent, i.e.

$$\mathbf{v}_{r+1} = c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r$$

Therefore

$$\begin{aligned} A\mathbf{v}_{r+1} &= A(c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r) \\ \lambda_{r+1}\mathbf{v}_{r+1} &= c_1\lambda_1\mathbf{v}_1 + \cdots \lambda_r c_r\mathbf{v}_r \\ \mathbf{0} &= (\lambda_{r+1} - \lambda_1)c_1\mathbf{v}_1 + \cdots + (\lambda_{r+1} - \lambda_r)c_r\mathbf{v}_r \end{aligned}$$

Since $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are linearly independent, each coefficient in the final equation must be zero. Furthermore, the eigenvalues are distinct, so $\lambda_{r+1} - \lambda_i \neq 0$ for each i. Hence, $c_1 = \cdots = c_r = 0$. But this would imply $\mathbf{v}_{r+1} = \mathbf{0}$, which is impossible since an eigenvector must be non-zero. Therefore, $\mathbf{v}_{r+1}$ cannot be written as a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_r$, and hence $\mathbf{v}_1, \ldots, \mathbf{v}_{r+1}$ are linearly independent. ■

4.3.5 Changing bases

Once we establish a new basis for the vectors in a problem, it's important to be able to represent a vector in the new basis.

■ **Example 4.10** Write the vector $\mathbf{x} = (5,5)^\mathrm{T}$ in terms of $\mathbf{v}_1 = (2,3)^\mathrm{T}$ and $\mathbf{v}_2 = (1,4)^\mathrm{T}$.

We solve by inverting $V\mathbf{c} = \mathbf{x} \implies \mathbf{c} = V^{-1}\mathbf{x}$:

$$\begin{aligned} c_1\mathbf{v}_1 + c_2\mathbf{v}_2 &= \mathbf{x} \\ \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} &= \begin{pmatrix} 5 \\ 5 \end{pmatrix} \\ \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} &= \frac{1}{5}\begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}\begin{pmatrix} 5 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \end{aligned}$$

We say that $\mathbf{x}$ has coordinates $(3,-1)^\mathrm{T}$ with respect to the basis $\{\mathbf{v}_1, \mathbf{v}_2\}$

As a quick check, we see that $3 \cdot (2,3)^\mathrm{T} - (1,4)^\mathrm{T} = (6,9)^\mathrm{T} - (1,4)^\mathrm{T} = (5,5)^\mathrm{T}$ ✔ ■

■ **Example 4.11** Write $(1,0,0)^\mathrm{T}$ in terms of the basis $\mathbf{v}_1 = (1,-1,0)^\mathrm{T}, \mathbf{v}_2 = (-2,1,1)^\mathrm{T}, \mathbf{v}_3 = (1,1,1)^\mathrm{T}$.

We solve by inverting $V\mathbf{c} = \mathbf{x} \implies \mathbf{c} = V^{-1}\mathbf{x}$:

$$
\begin{aligned}
c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 &= \mathbf{x} \\
\begin{pmatrix} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} &= \begin{pmatrix} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}^{-1}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & -1/2 & 1/2 \\ -1/3 & -1/6 & 1/2 \\ 1/3 & 1/6 & 1/2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 \\ -1/3 \\ 1/3 \end{pmatrix}
\end{aligned}
$$

We say that $\mathbf{x}$ has coordinates $(0, -1/3, 1/3)^T$ with respect to the basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

As a quick check, $-1/3 \cdot (-2,1,1)^T + 1/3 \cdot (1,1,1)^T = (2/3, -1/3, -1/3)^T + (1/3, 1/3, 1/3) = (1,0,0)^T$ ✔ ■

4.3.6 Similar matrices

Apart from re-writing vectors in terms of a new basis, we might also want to re-write *transformations* in terms of the new basis. In the Matrix Representation Theorem the basis used was the standard basis. If another basis had been chosen, then a different matrix would have been constructed to represent the transformation. How can we take an existing transformation (from the standard basis), and re-write it in terms of a new basis?

Consider the transformation $T : \mathbb{R}^2 \mapsto \mathbb{R}^2$ represented by $A = \begin{pmatrix} 7 & 2 \\ 3 & 8 \end{pmatrix}$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. We want to find the matrix representation with respect to the basis $\{\mathbf{v}_1 = (2,3)^T, \mathbf{v}_2 = (1,4)^T\}$ that we considered in Example 4.10.

When we apply the transformation T to the basis vectors, we obtain

$$
\begin{aligned}
T(\mathbf{v}_1) = A\mathbf{v}_1 &= \begin{pmatrix} 7 & 2 \\ 3 & 8 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 20 \\ 30 \end{pmatrix} = 10\mathbf{v}_1 \\
T(\mathbf{v}_2) = A\mathbf{v}_2 &= \begin{pmatrix} 7 & 2 \\ 3 & 8 \end{pmatrix}\begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 15 \\ 35 \end{pmatrix} = 5\mathbf{v}_1 + 5\mathbf{v}_2
\end{aligned}
$$

So the matrix representing T with respect to basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ is

$$B = \begin{pmatrix} 10 & 5 \\ 0 & 5 \end{pmatrix}$$

But how does B represent this transformation with respect to the new basis? If we define $\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$, and $T(\mathbf{x}) = d_1\mathbf{v}_1 + d_2\mathbf{v}_2$, then these coefficients (c_1, c_2) and (d_1, d_2) are related via the matrix B as $\mathbf{d} = B\mathbf{c}$, i.e.

$$\begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \begin{pmatrix} 10 & 5 \\ 0 & 5 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \tag{4.1}$$

Can we work out more directly how to calculate B from A? We know that in the standard basis

$$\begin{aligned} \mathbf{x} &= c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = V\mathbf{c} \\ T(\mathbf{x}) = A\mathbf{x} &= d_1\mathbf{v}_1 + d_2\mathbf{v}_2 = V\mathbf{d} \end{aligned}$$

Combining these results gives us

$$V\mathbf{d} = A\mathbf{x} = AV\mathbf{c} \implies \mathbf{d} = V^{-1}AV\mathbf{c}$$

where we know that V^{-1} must exist, because the $\mathbf{v}_i$ vectors form a basis. Comparing this to (4.1), we see the relation between A and B:

$$B = V^{-1}AV$$

We call any matrices A and B **similar** if they are related in this fashion. If A and B are similar, they represent the same transformation, expressed using different bases, and can be calculated using the basis vectors in the form of the matrix V and its inverse.

Definition 4.3.4 Let A and B be $n \times n$ matrices. B is said to be **similar** to A if there exists an invertible matrix S such that $B = S^{-1}AS$.

Similarity is an **equivalence** relation: if B is similar to A, then A is similar to B: $SBS^{-1} = S(S^{-1}AS)S^{-1} = (SS^{-1})A(SS^{-1}) = A$, so $A = (S^{-1})^{-1}BS^{-1}$. The matrices have to be square for the relationship to be possible. A further property of similar matrices is that they must have the same eigenvalues.

Theorem 4.3.4 If A and B are similar, the two matrices have the same characteristic polynomials and, hence, the same eigenvalues.

Proof. Let $p_A(\lambda)$ and $p_B(\lambda)$ denote the characteristic polynomials of A and B, respectively. Since A and B are similar, a non-singular matrix S exists such that $B = S^{-1}AS$. Thus,

$$\begin{aligned} p_B(\lambda) &= \det(B - \lambda I) = \det(S^{-1}AS - \lambda I) \\ &= \det(S^{-1}AS - \lambda S^{-1}IS) = \det(S^{-1}(A - \lambda I)S) \\ &= \det(S^{-1})\det(A - \lambda I)\det(S) = \det(S^{-1}S)\det(A - \lambda I) = p_A(\lambda). \end{aligned}$$

■

4.4 Diagonalising matrices

Let us consider an example of rewriting a transformation using its own *eigenvectors* as a basis. Consider the transformation $T : \mathbb{R}^2 \mapsto \mathbb{R}^2$ represented, with respect to the standard basis $\mathbf{e}_1, \mathbf{e}_2$, by

$$A = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}.$$

We want to find the matrix representation of T with respect to the basis formed by the eigenvectors of A,

$$\mathbf{v}_1 = (2,3)^{\mathrm{T}}, \mathbf{v}_2 = (1,-1)^{\mathrm{T}}.$$

When we apply the transformation T to the basis vectors, we obtain

$$T(\mathbf{v}_1) = A\mathbf{v}_1 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \mathbf{v}_1$$
$$T(\mathbf{v}_2) = A\mathbf{v}_2 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix} = \frac{1}{2}\mathbf{v}_2$$

So the matrix representing T with respect to basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ is

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}$$

When the new basis is the eigenvectors of A, the new representation D is a diagonal matrix! We call this process **diagonalising** the matrix A. Diagonal matrices are *much* easier to work with, so diagonalisation is a standard approach to a number of important problems, as we will soon see.

Theorem 4.4.1 An $n \times n$ matrix A with n linearly independent eigenvectors can be rewritten as $A = VDV^{-1}$, where D is the diagonal matrix of eigenvalues and V is the matrix with eigenvectors corresponding to these eigenvalues as columns. In this case we say that V **diagonalises** A.

Example 6 (revisited)

From MATLAB,

```
>> A=[1 0 0; 0 2 1; 0  1 2]

A =
      1      0      0
      0      2      1
      0      1      2

>> [V,D]=eig(A)

V =
    1.0000         0         0
         0   -0.7071    0.7071
         0    0.7071    0.7071


D =
      1      0      0
      0      1      0
      0      0      3

>> V*D*inv(V)

ans =
      1      0      0
      0      2      1
      0      1      2
```

and we can see that the final matrix is A as expected.

4.4.1 Properties of diagonal matrices

We will write a diagonal matrix D as $\mathrm{diag}(a_1,a_2,\ldots,a_n)$, where

$$\mathrm{diag}(a_1,a_2,\ldots,a_n) = \begin{pmatrix} a_1 & 0 & 0 & \ldots & 0 \\ 0 & a_2 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & a_n \end{pmatrix}$$

Then the following statements are true for diagonal matrices.

1. $\mathrm{diag}(a_1,a_2,\ldots,a_n)+\mathrm{diag}(b_1,b_2,\ldots,b_n) = \mathrm{diag}(a_1+b_1,a_2+b_2,\ldots,a_n+b_n)$
2. $\mathrm{diag}(a_1,a_2,\ldots,a_n)\mathrm{diag}(b_1,b_2,\ldots,b_n) = \mathrm{diag}(a_1b_1,a_2b_2,\ldots,a_nb_n)$
3. $\det(\mathrm{diag}(a_1,a_2,\ldots,a_n)) = a_1a_2\ldots a_n \Rightarrow D$ is invertible iff $a_i \neq 0,\ i=1,2,\ldots,n$.
4. If D is invertible, $\mathrm{diag}(a_1,a_2,\ldots,a_n)^{-1} = \mathrm{diag}(a_1^{-1},a_2^{-1},\ldots,a_n^{-1})$.
5. The eigenvalues of D are $a_1,a_2,\ldots,a_n$.

The proofs of 1 and 2 follow from the definition and we have seen results 3 and 5 before. To prove 4, observe that

$$\begin{pmatrix} a_1 & 0 & 0 & \ldots & 0 \\ 0 & a_2 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & a_n \end{pmatrix} \begin{pmatrix} a_1^{-1} & 0 & 0 & \ldots & 0 \\ 0 & a_2^{-1} & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & a_n^{-1} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & 1 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{pmatrix} = I$$

4.4.2 Powers of a matrix

If we can write $A = VDV^{-1}$ then this allows us to calculate powers of A in a really easy way, which is useful[2].

$$A^2 = AA = (VDV^{-1})(VDV^{-1}) = VDDV^{-1} = VD^2V^{-1}$$

and then

$$A^3 = A^2A = (VD^2V^{-1})(VDV^{-1}) = VD^2DV^{-1} = VD^3V^{-1}$$

and we could prove (by induction), that for any positive integer power n, $A^n = VD^nV^{-1}$. In fact, this is true for any integer. (For $n=0$ it is clearly true.)

We can also find A^{-1}:

$$A^{-1}A = I$$

$$A^{-1}VDV^{-1} = I$$
$$A^{-1}VD = V$$
$$A^{-1}V = VD^{-1}$$
$$A^{-1} = VD^{-1}V^{-1}.$$

[2]In addition to some examples we've already seen, it allows us to efficiently calculate the results of feeding square matrix arguments into several functions defined using power series representations, *e.g.*, exponentiation with a matrix exponent.

Then

$$A^{-2} = (A^{-1})^2$$
$$= (VD^{-1}V^{-1})(VD^{-1}V^{-1})$$
$$= V(D^{-1})^2V^{-1}.$$

Continuing in this way,

$$A^{-k} = V(D^{-1})^kV^{-1} = VD^{-k}V^{-1}.$$

Theorem 4.4.2 Suppose A is invertible and diagonalizable, with $A = VDV^{-1}$. Then, for any $n \in \mathbb{Z}$, $A^n = VD^nV^{-1}$.

Example

Consider $A = \begin{pmatrix} -1 & -2 \\ -2 & 2 \end{pmatrix}$.

Now $\lambda_1 + \lambda_2 = \text{trace}(A) = 1$ and $\lambda_1\lambda_2 = -6 \Rightarrow \lambda_1 = -2, \lambda_2 = 3$. The eigenvectors are $\mathbf{v}_1 = s\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ so take $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ here and $\mathbf{v}_2 = r\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ so take $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

$$\therefore D = \begin{pmatrix} -2 & 0 \\ 0 & 3 \end{pmatrix} \text{ and } V = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \Rightarrow V^{-1} = \frac{1}{-5}\begin{pmatrix} -2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}.$$

Recall that if $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then $A^{-1} = \frac{1}{\det(A)}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

This gives

$$A = VDV^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} -2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} -4 & -2 \\ 3 & -6 \end{pmatrix} = \begin{pmatrix} -1 & -2 \\ -2 & 2 \end{pmatrix}.$$

So now we can easily calculate A^4.

$$A^4 = VD^4V^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} (-2)^4 & 0 \\ 0 & 3^4 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 145 & -130 \\ -130 & 340 \end{pmatrix} = \begin{pmatrix} 29 & -26 \\ -26 & 68 \end{pmatrix}.$$

We can also find A^{-1} (you can check this using the usual method).

$$A^{-1} = VD^{-1}V^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} -1/2 & 0 \\ 0 & 1/3 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} -1/3 & -1/3 \\ -1/3 & 1/6 \end{pmatrix}.$$

4.5 Applications

Let's look at some of the uses of the ideas we have developed in this chapter.

4.5.1 Marital population

We are now in a position to complete our understanding of the Marital Population problem, considering the long-time limit behaviour of $\mathbf{x}_n = A^n \mathbf{x}_0$. Because $A = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}$ has two distinct eigenvalues, its eigenvectors form a basis of $\mathbb{R}^n$, so we can write

$$A = VDV^{-1} \implies A^n = VD^nV^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2^{-n} \end{pmatrix} \begin{pmatrix} 0.2 & 0.2 \\ 0.6 & -0.4 \end{pmatrix}$$

As $n \to \infty$,

$$A^n \to \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0.2 & 0.2 \\ 0.6 & -0.4 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 0.2 & 0.2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0.4 & 0.4 \\ 0.6 & 0.6 \end{pmatrix}$$

For an initial condition representing percentages of the population, the long-time limit is always

$$A^n \begin{pmatrix} p \\ 100 - p \end{pmatrix} = \begin{pmatrix} 0.4 & 0.4 \\ 0.6 & 0.6 \end{pmatrix} \begin{pmatrix} p \\ 100 - p \end{pmatrix} = \begin{pmatrix} 40 \\ 60 \end{pmatrix}$$

This explains why the system always ends up at the same final state, *irrespective of the initial condition*. We have now proved our observation, which is the primary aim of mathematical modelling.

4.5.2 Fibonacci

In the previous chapter, we considered the recurrence relation for the Fibonacci sequence

$$0, 1, 1, 2, 3, 5, 8, \ldots,$$

given by

$$x_n = x_{n-1} + x_{n-2}.$$

This recurrence can be written in matrix form as

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ x_{n-2} \end{pmatrix}.$$

Repeated application gives

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{n-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

We can therefore study the Fibonacci sequence by examining the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

The eigenvalues are found from the characteristic equation

$$\begin{vmatrix} 1 - \lambda & 1 \\ 1 & -\lambda \end{vmatrix} = -\lambda(1 - \lambda) - 1 = \lambda^2 - \lambda - 1 = 0.$$

Hence,

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \qquad \lambda_2 = \frac{1-\sqrt{5}}{2}.$$

The first eigenvalue,

$$\lambda_1 = \frac{1+\sqrt{5}}{2},$$

is the golden ratio. The second satisfies

$$\lambda_2 = -\frac{1}{\lambda_1},$$

which is consistent with

$$\lambda_1 \lambda_2 = \det(A) = -1.$$

Corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ \sqrt{5} - 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1-\sqrt{5} \\ 2 \end{pmatrix}.$$

These satisfy

$$\lambda_1 \mathbf{v}_1 - \mathbf{v}_2 = \begin{pmatrix} 1+\sqrt{5} \\ 2 \end{pmatrix} - \begin{pmatrix} 1-\sqrt{5} \\ 2 \end{pmatrix} = \begin{pmatrix} 2\sqrt{5} \\ 0 \end{pmatrix}.$$

Therefore,

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\lambda_1}{2\sqrt{5}} \mathbf{v}_1 - \frac{1}{2\sqrt{5}} \mathbf{v}_2.$$

Applying A^{n-1} to both sides gives

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = A^{n-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Since

$$A^{n-1} \mathbf{v}_1 = \lambda_1^{n-1} \mathbf{v}_1$$

and

$$A^{n-1} \mathbf{v}_2 = \lambda_2^{n-1} \mathbf{v}_2,$$

we obtain

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = \frac{\lambda_1^n}{2\sqrt{5}} \mathbf{v}_1 - \frac{\lambda_2^{n-1}}{2\sqrt{5}} \mathbf{v}_2.$$

Now,

$$\lambda_2 = \frac{1-\sqrt{5}}{2} \approx -0.618,$$

so

$$|\lambda_2| < 1.$$

Therefore,

$$\lambda_2^n \to 0 \qquad \text{as} \qquad n \to \infty.$$

For large n, the contribution from the second eigenvalue becomes very small, giving

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} \approx \frac{\lambda_1^n}{2\sqrt{5}} \begin{pmatrix} 2 \\ \sqrt{5}\; -1 \end{pmatrix}.$$

Since

$$\sqrt{5}\; - 1 = \frac{2}{\lambda_1},$$

this can be written as

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} \approx \frac{1}{\sqrt{5}} \begin{pmatrix} \lambda_1^n \\ \lambda_1^{n-1} \end{pmatrix}.$$

Hence, the nth Fibonacci number is approximately

$$x_n \approx \frac{\lambda_1^n}{\sqrt{5}} = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n.$$

Although x_n is always an integer, this approximation is generally not an integer because it neglects the small contribution associated with λ_2. Since $|\lambda_2| < 1$, however, this neglected term rapidly approaches zero as n increases. Consequently, the approximation becomes extremely close to an integer for large n.

A further consequence is that the ratio of successive Fibonacci numbers approaches the golden ratio:

$$\frac{x_n}{x_{n-1}} \to \lambda_1 = \frac{1+\sqrt{5}}{2} \qquad \text{as} \qquad n \to \infty.$$

4.5.3 Car rental

A car rental company has three locations in Brisbane from which cars can be rented: the domestic airport, the international airport, and the CBD. The following table shows the probability that a client drops a car off at a particular location, given the location where the car was collected. What is the expected distribution of available cars on any given day, across the three locations?

Pick-up location	Drop-off location Domestic	International	CBD
Domestic	0.8	0.1	0.1
International	0.2	0.6	0.2
CBD	0.3	0.2	0.5

First, consider a day on which x_1 cars are picked up from the domestic airport, x_2 from the international airport, and x_3 from the CBD. Where would we expect them to ultimately end up?

Let's consider each drop-off point in turn. The expected number of cars returned to the domestic airport, y_1, would be 0.8 of those picked up from there, as well as 0.2 of those picked up from the international airport, and 0.3 of those picked up from the CBD — that is,

$$y_1 = 0.8x_1 + 0.2x_2 + 0.3x_3.$$

In a similar way, we find that the expected number of cars returned to the international airport and the CBD, denoted y_2 and y_3 respectively, would be

$$y_2 = 0.1x_1 + 0.6x_2 + 0.2x_3$$
$$y_3 = 0.1x_1 + 0.2x_2 + 0.5x_3.$$

We can express this succinctly in a matrix, using the notation $\mathbf{x}_0$ to represent the vector containing the number of picked up cars, and $\mathbf{x}_1$ to represent the vector containing the number of cars dropped off, at the three locations:

$$\mathbf{x}_1 = \begin{pmatrix} 0.8 & 0.2 & 0.3 \\ 0.1 & 0.6 & 0.2 \\ 0.1 & 0.2 & 0.5 \end{pmatrix} \mathbf{x}_0 = A\mathbf{x}_0$$

After n iterations of this process, the expected distribution of these cars at the three locations will be given by $\mathbf{x}_n = A^n\mathbf{x}_0$. Let's estimate this value from the eigenvalues of A. Using MATLAB, we find the eigenvalues are

```
> [V,D]=eigs([0.8 0.2 0.3;0.1 0.6 0.2;0.1 0.2 0.5])
V =

    0.86645    0.80902   -0.30902
    0.37907   -0.50000   -0.50000
    0.32492   -0.30902    0.80902

D =

Diagonal Matrix

    1.00000          0          0
          0    0.56180          0
          0          0    0.33820
```

The eigenvalues are therefore

$$\lambda_1 = 1, \mathbf{v}_1 \approx \begin{pmatrix} 0.86645 \\ 0.37907 \\ 0.32492 \end{pmatrix}; \lambda_2 \approx 0.5618, \mathbf{v}_2 \approx \begin{pmatrix} 0.80902 \\ -0.50000 \\ -0.30902 \end{pmatrix}; \lambda_3 \approx 0.3382, \mathbf{v}_3 \approx \begin{pmatrix} -0.30902 \\ -0.50000 \\ 0.80902 \end{pmatrix}$$

Since the three eigenvalues are different, these vectors must form a basis in $\mathbb{R}^3$, so coefficients can be found to represent *any* initial distribution of cars, i.e.

$$\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3.$$

Furthermore, after many iterations, the expected distribution becomes

$$\begin{aligned} \mathbf{x}_n &= A^n\mathbf{x}_0 = c_1\lambda_1^n\mathbf{v}_1 + c_2\lambda_2^n\mathbf{v}_2 + c_3\lambda_3^n\mathbf{v}_3 \\ \therefore \lim_{n\to\infty}\mathbf{x}_n &= c_1\left[\lim_{n\to\infty}\lambda_1^n\right]\mathbf{v}_1 + c_2\left[\lim_{n\to\infty}\lambda_2^n\right]\mathbf{v}_2 + c_3\left[\lim_{n\to\infty}\lambda_3^n\right]\mathbf{v}_3 \\ &= c_1\mathbf{v}_1 \end{aligned}$$

because λ_2^n and λ_3^n both $\to 0$ as n increases. The fractional expected distribution at each location is therefore given by

$$\frac{c_1\mathbf{v}_1}{c_1 0.86645 + c_1 0.37907 + c_1 0.32492} = \frac{\mathbf{v}_1}{0.86645+0.37907+0.32492} \approx \begin{pmatrix} 0.55174 \\ 0.24139 \\ 0.20690 \end{pmatrix},$$

so approximately 55% of the cars end up at the domestic airport, 24% at the international airport, and 21% in the CBD.

Why did we normalise by the sum of elements of $c_1\mathbf{v}_1$, rather than by the length of $c_1\mathbf{v}_1$? It is because we interpret the elements of the $\mathbf{x}$ vector as the probabilities of finding a car returned to a particular location. The elements themselves need to sum to one (rather than the elements squared summing to one, which is what we'd be doing if we normalised by dividing by the length of $\mathbf{x}$) for us to interpret them as probabilities. ■

4.5.4 Ranking competitors

Consider a tournament in which five players compete against one another. Each player plays every other player once, with the results summarised as follows:

Player	beat Player			
	P2	P3	P4	P5
P1	Y	N	Y	Y
P2		Y	Y	Y
P3			Y	N
P4				Y

Ranking these players presents some problems. For example, P1 beat P2, and P2 beat P3, but P3 beat P1. So who should come out on top?

To solve this, let's introduce a ranking score r_i for each player, which we will use as a rank for each player. To determine each player's rank, we define it in terms of the ranks of the players they have beaten. More specifically, it seems reasonable that a player's rank should be related to the sum of the ranks of the players they have beaten. Taking the average rank of players they have beaten exaggerates the importance of victories against top players over (possibly many) losses to other players. But making the rank equal to the sum leads to a potentially unsolvable problem, as we will soon see. To avoid this problem, we amend the rule slightly: the player's rank should be *proportional* to the sum of the ranks of the players they have beaten.

For the case above, we would define the ranks for the five players through the equations

$$\begin{aligned}
r_1 &= \alpha(r_2 + r_4 + r_5) \\
r_2 &= \alpha(r_3 + r_4 + r_5) \\
r_3 &= \alpha(r_1 + r_4) \\
r_4 &= \alpha r_5 \\
r_4 &= \alpha r_3
\end{aligned}$$

which we can rewrite as the matrix equation

$$\mathbf{r} = \alpha \begin{pmatrix} 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \mathbf{r} = \alpha A \mathbf{r}$$

Now perhaps we can see the advantage to this approach. Insisting that the ranking be given by the sum (i.e. setting $\alpha = 1$) means we must solve the problem $\mathbf{r} = A\mathbf{r}$, which may only have solution $\mathbf{r} = \mathbf{0}$. However, by looking for a solution to the problem $\mathbf{r} = \alpha A \mathbf{r}$, we turn this into a problem involving the eigenvectors of A.

Matrices constructed in this fashion have the property that they have a real largest eigenvalue whose eigenvector contains only positive values[3] — this is the vector we can use to represent the rank. In this case, a call to MATLAB reveals the largest eigenvalue is $\lambda_1 \approx 1.6593$, with corresponding eigenvector

$$\mathbf{v}_1 = \begin{pmatrix} 0.60572 \\ 0.55368 \\ 0.46734 \\ 0.16974 \\ 0.28165 \end{pmatrix}$$

This provides the relative ranking values for the players, telling us that their order of ranking is (from top to bottom) P1, P2, P3, P5 then P4.

A variant of this approach is the basis for Google's method for ranking web pages. The matrix A is built by considering which web pages link to which other pages: the rank of a page is proportional to the sum of the ranks of the pages that link to it. Google constructs the $N \times N$ matrix A (where N is the number of existing webpages that both link to others and are linked to themselves), and then solves this single, vast, linear algebra problem in order to determine page rankings.

Note that in this section we are not referring the the matrix property known as rank, which we will discuss in chapter 6.

[3]This follows from an important result called the Perron–Frobenius theorem, which is (only just) beyond the scope of the course. It plays an important role in characterising the behaviour of Markov chains such as the car rental problem in §4.5.3.

5. Inner product spaces

We now look at the concept of an inner product, which is a generalisation of the concept of the dot product that acts on $\mathbb{R}^2$ or $\mathbb{R}^3$ that you have seen previously. The dot product gives a measure of how 'close' two vectors are to one another, in the sense of pointing in a similar direction. In $\mathbb{R}^2$ or $\mathbb{R}^3$, we associate the size of the dot product with the angle between the vectors. The inner product allows us to achieve a similar aim, comparing vectors and having a sense of how 'close' they are.

5.1 The dot product

Recall the definition of the dot product from first year:

Definition 5.1.1 The **dot product** of two vectors $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \in \mathbb{R}^3$ is a scalar given by

$$\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^{\mathrm{T}} \mathbf{y} = (x_1 \ x_2 \ x_3) \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = x_1 y_1 + x_2 y_2 + x_3 y_3.$$

An alternative definition is $\mathbf{x} \cdot \mathbf{y} = ||\mathbf{x}|| \, ||\mathbf{y}|| \cos \theta$, where θ is the smallest angle between the two vectors $\mathbf{x}$ and $\mathbf{y}$ and $||\mathbf{v}||$ is the length of $\mathbf{v}$, given by $||\mathbf{v}|| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + v_3^2}$. This alternative definition allows us to calculate the angle between two vectors, and gauge the extent to which they point in similar directions

■ **Example 5.1** If $\mathbf{x} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$ and $\mathbf{y} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$,

$$\mathbf{x}\cdot\mathbf{y} = \mathbf{x}^{\mathrm{T}}\mathbf{y} = (3\ -2\ 1)\begin{pmatrix}4\\3\\2\end{pmatrix} = 3(4)+(-2)(3)+1(2) = 8$$

Now

$$\|\mathbf{x}\| = \sqrt{3^2+(-2)^2+1^2} = \sqrt{14} \quad \text{and} \quad \|\mathbf{y}\| = \sqrt{4^2+3^2+2^2} = \sqrt{29}$$

$$\therefore \cos\theta = \frac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|} = \frac{8}{\sqrt{14}\sqrt{29}} \Rightarrow \theta \approx 66.6^\circ$$

■

From the definition above, the angle between any two non–zero vectors **x** and **y** is given by

$$\cos\theta = \frac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}$$

which implies that non–zero vectors **x** and **y** are perpendicular (**orthogonal**) if and only if $\mathbf{x}\cdot\mathbf{y} = 0$. ■

5.1.1 Numerically calculating the dot product

For MATLAB/Octave,

Dot product of vectors $\mathbf{u},\mathbf{v}$	`dot(u,v)`
Length of a vector $\mathbf{v}$	`norm(v)`
To normalise a vector $\mathbf{v}$	`w=v/norm(v)`
To find the angle between $\mathbf{u}$ and $\mathbf{v}$	`acos(dot(u,v)/(norm(u)*norm(v)))`

The MATLAB command `length(v)` finds the number of entries in a 1D vector **v** and is quite different from the command `norm(v)`. ■

For R,

Dot product of vectors $\mathbf{u},\mathbf{v}$	`dot(u,v)`
Length of a vector $\mathbf{v}$	`norm(v,type="2")`
To normalise a vector $\mathbf{v}$	`w=v/norm(v,type="2")`
To find the angle between $\mathbf{u}$ and $\mathbf{v}$	`acos(dot(u,v)/` `(norm(u,type="2")*norm(v,type="2")))`

Note that there's no need to tell R that you're breaking an instruction over multiple lines – the unmatched bracket is enough to alert it of the linebreak.

For Python,

Dot product of vectors $\mathbf{u}, \mathbf{v}$	`np.dot(u,v)`
Length of a vector $\mathbf{v}$	`np.linalg.norm(v)`
To normalise a vector $\mathbf{v}$	`w=v/np.linalg.norm(v)`
To find the angle between $\mathbf{u}$ and $\mathbf{v}$	`np.arccos(np.dot(u,v)/ \` `(np.linalg.norm(u)*np.linalg.norm(v)))`

Why not just calculate these directly using the definitions? In general, the built-in functions[a] are going to be more efficient or more reliable than code you write yourself (in the sense that the authors have thought carefully about dealing with numerical problems such as over/under-flow, data types, *etc.*). ■

[a]All three languages we are discussing in this course are built atop the same underlying library of functions (`LAPACK`, `LINPACK`), so they're all the same thing at the most basic level.

5.2 Inner products

We wish to generalise these definitions, particularly for situations where we might not have a geometrical picture to guide us (more on that later). From now on, we will replace the terms dot product by *inner product* and length by *norm*.

Definition 5.2.1 An **inner product** on a vector space[a] V is an operation on V that assigns to each pair of vectors $\mathbf{x}, \mathbf{y} \in V$ a real number satisfying:

1. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ with equality if and only if $\mathbf{x} = \mathbf{0}$
2. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$ for all $\mathbf{x}, \mathbf{y} \in V$
3. $\langle \alpha\mathbf{x} + \beta\mathbf{y}, \mathbf{z} \rangle = \alpha \langle \mathbf{x}, \mathbf{z} \rangle + \beta \langle \mathbf{y}, \mathbf{z} \rangle$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $\alpha, \beta \in \mathbb{R}$.

[a]We will discuss the precise definition of a vector space in chapter 6. For now, $\mathbb{R}^m$ is an example of a vector space.

Definition 5.2.2 The **norm** of $\mathbf{x}$ is defined in terms of an inner product by $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$.

We can also define the idea of a unit vector, using the norm as a measure of a vector's length:

Definition 5.2.3 A **normalised** or **unit** vector $\mathbf{v}$ is a vector whose norm $\|\mathbf{v}\| = 1$. To *normalise* a vector, we divide it by its norm, i.e.

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \quad \|\hat{\mathbf{v}}\| = 1.$$

Finally, we also use these definitions to extend our concept of *orthogonality*:

Definition 5.2.4 Two vectors[a] $\mathbf{x}$ and $\mathbf{y}$ are said to be **orthogonal** iff $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. The relationship is also denoted $\mathbf{x} \perp \mathbf{y}$.

[a]Note that, for orthogonality defined via an inner product, we allow the case that one or other vector is $\mathbf{0}$.

Why do we have these requirement for an inner product? There are specific reasons for each condition:

1. We want the inner product to be positive, or zero only when $\mathbf{x} = \mathbf{0}$, so that we can take the square root to get a measure of the size of the vector (*i.e.*, the norm). We also want only the zero vector to have size zero: that way, showing that $\|\mathbf{x}\| = 0$ is a way of showing that $\mathbf{x} = \mathbf{0}$.
2. We want the definition to be symmetric, so we can consider it as a function of a given *pair* of vectors. Neither vector is more significant than the other, and there is no effect of ordering. This corresponds with our idea of the angle between the vectors: each vector is thought of as being θ from the other and the magnitude $(\pm\theta)$ is irrelevant.
3. *Linearity* is an important property of the dot product, used in various proofs, so we wish to retain it for the more general inner product.

5.2.1 The dot product as an inner product

The dot product is arguably the most important example of an inner product in $\mathbb{R}^n$, and is identified as the **standard inner product**.

Definition 5.2.5 For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the **standard inner product** of x and y, denoted by is $\langle \mathbf{x}, \mathbf{y} \rangle$, is

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^{\mathrm{T}}\mathbf{y} = (x_1\ x_2 \ldots x_n) \begin{pmatrix} y_1 \\ y_2 \\ .. \\ .. \\ y_n \end{pmatrix} = x_1y_1 + x_2y_2 + \ldots + x_ny_n.$$

Definition 5.2.6 The **Euclidean norm** of $\mathbf{x}$ is the norm defined via the standard inner product:

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{x_1^2 + x_2^2 + \ldots x_n^2}$$

As a few simple examples:

1. The vector $\mathbf{0}$ is orthogonal to every vector in $\mathbb{R}^n$.
2. The vectors $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 6 \end{pmatrix}$ are orthogonal in $\mathbb{R}^2$, since $(3\ 2)\begin{pmatrix} -4 \\ 6 \end{pmatrix}$ = 3(-4) + 2(6) = 0.
3. The vectors $\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ are orthogonal in $\mathbb{R}^3$ (check this).
4. If $\mathbf{x} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, then $\|\mathbf{x}\| = \sqrt{3^2 + 4^2} = 5$, and the normalised vector parallel to $\mathbf{x}$ is $\begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}$.

Proof that the dot product is an inner product. We need to check each of the properties given in Definition 5.2.5. Given $\mathbf{x} = (x_1, x_2, \ldots, x_n)^{\mathrm{T}}$, $\mathbf{y} = (y_1, y_2, \ldots, y_n)^{\mathrm{T}}$, and

$\mathbf{z} = (z_1, z_2, \ldots, z_n)^{\mathrm{T}} \in \mathbb{R}^n$, we see that:

1. $\langle \mathbf{x}, \mathbf{x} \rangle = x_1^2 + x_2^2 + \ldots + x_n^2 \geq 0$.
   If $\mathbf{x} = \mathbf{0}$ then $x_i = 0 \; \forall i = 1, 2, \ldots, n \Rightarrow x_1^2 + x_2^2 + \ldots + x_n^2 = 0 \Rightarrow \langle \mathbf{x}, \mathbf{x} \rangle = 0$
   If $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ then $x_1^2 + x_2^2 + \ldots + x_n^2 = 0 \Rightarrow x_i = 0 \; \forall i = 1, 2, \ldots, n \Rightarrow \mathbf{x} = \mathbf{0}$.

2. $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 \ldots + x_n y_n = y_1 x_1 + y_2 x_2 \ldots + y_n x_n = \langle \mathbf{y}, \mathbf{x} \rangle$

3. $\langle \alpha \mathbf{x} + \beta \mathbf{y}, \mathbf{z} \rangle = (\alpha x_1 + \beta y_1) z_1 + (\alpha x_2 + \beta y_2) z_2 + \ldots + (\alpha x_n + \beta y_n) z_n = \alpha (x_1 z_1 + x_2 z_2 \ldots + x_n z_n) + \beta (y_1 z_1 + y_2 z_2 \ldots + y_n z_n) = \alpha \langle \mathbf{x}, \mathbf{z} \rangle + \beta \langle \mathbf{y}, \mathbf{z} \rangle$

■

5.2.2 Other examples of inner products

A simple alternative definition for an inner product can be defined on $\mathbb{R}^n$ by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} w_i x_i y_i$$

where $w_i \in \mathbb{R}^+$ (positive real numbers) are referred to as weights.

As an example, consider the inner product in $\mathbb{R}^2$ defined by $\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1 y_1 + x_2 y_2$. This inner product gives more weight – more importance – to the contributions from the first coordinates. Under this inner product, we have

1. The vectors $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 6 \end{pmatrix}$ are *not* orthogonal, since $2 \cdot 3 \cdot (-4) + 2 \cdot 6 = -24 + 12 \neq 0$.
2. If $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, then $\|\mathbf{x}\| = \sqrt{2 \cdot 2^2 + 1^2} = 3$, and the normalised vector parallel to $\mathbf{x}$ is $\begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}$.
3. Notice that if $\mathbf{y} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, then $\|\mathbf{y}\| = \sqrt{2 \cdot 1^2 + 2^2} = \sqrt{6} \neq \|\mathbf{x}\|$. Although $\mathbf{x}$ and $\mathbf{y}$ contain the same two components, their order is different. As the inner product weights the first component differently from the second, this results in different norms for $\mathbf{x}$ and $\mathbf{y}$.

The proof that $\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1 y_1 + x_2 y_2$ is an inner product follows the same steps as the proof for the standard inner product.

Weighted inner products of this form arise in the study of curved surfaces, for example. Such surfaces, with some generalisation, can be used to model everything from navigation on Earth through to the deformation of space-time by mass and energy (general relativity).

Moving away from point spaces, we can also define inner products over spaces of other mathematical objects. Consider the set of functions that are continuous on the interval $[a, b]$, denoted $C[a, b]$. In $C[a, b]$,

$$\langle f, g \rangle = \int_a^b f(x) g(x) \, \mathrm{d}x$$

defines an inner product, calculated for any $f, g \in C[a, b]$.

■ **Example 5.2** On $C[0,1]$ with $\langle f,g\rangle$ as above, what is $\langle x,\sin\pi x\rangle$?

$$\langle x,\sin\pi x\rangle = \int_0^1 x\sin\pi x\,dx = \left.\frac{-x\cos\pi x}{\pi}\right|_0^1 + \int_0^1 \frac{\cos\pi x}{\pi}\,dx$$
$$= \frac{-1}{\pi}[1(-1)-0(1)] + \frac{1}{\pi^2}[\sin\pi x]_0^1 = \frac{1}{\pi}.$$

■

These generalised inner products are crucial for many applications, *e.g.*, representing an arbitrary function as a weighted sum of basis functions (more on this in chapter 6).

5.2.3 Pythagoras' theorem

Pythagoras' theorem also holds, under the more general definition of orthogonality.

Theorem 5.2.1 — Pythagorean Theorem. If $\mathbf{v}$ and $\mathbf{w}$ are orthogonal, according to a given inner product (i.e. $\langle\mathbf{v},\mathbf{w}\rangle = 0$), then $\|\mathbf{v}+\mathbf{w}\|^2 = \|\mathbf{v}\|^2 + \|\mathbf{w}\|^2$.

Proof.

$$\begin{aligned}
\|\mathbf{v}+\mathbf{w}\|^2 &= \langle\mathbf{v}+\mathbf{w},\mathbf{v}+\mathbf{w}\rangle \\
&= \langle\mathbf{v},\mathbf{v}+\mathbf{w}\rangle + \langle\mathbf{w},\mathbf{v}+\mathbf{w}\rangle \text{ by 5.2.1 (3)} \\
&= \langle\mathbf{v}+\mathbf{w},\mathbf{v}\rangle + \langle\mathbf{v}+\mathbf{w},\mathbf{w}\rangle \text{ by 5.2.1 (2)} \\
&= \langle\mathbf{v},\mathbf{v}\rangle + 2\langle\mathbf{v},\mathbf{w}\rangle + \langle\mathbf{w},\mathbf{w}\rangle \text{ by 5.2.1 (2) and (3)} \\
&= \|\mathbf{v}\|^2 + 2\langle\mathbf{v},\mathbf{w}\rangle + \|\mathbf{w}\|^2 \text{ by definition} \\
&= \|\mathbf{v}\|^2 + \|\mathbf{w}\|^2 \text{ since } \mathbf{v},\mathbf{w} \text{ are orthogonal.}
\end{aligned}$$

■

5.3 How close are two vectors?

In two or three dimensions, we can talk meaningfully about the angle θ between two vectors, calculated via the dot product as

$$\cos\theta = \frac{\mathbf{x}^{\mathrm{T}}\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}, \quad 0 \le \theta \le \pi.$$

The angle is useful as a measure how closely aligned two vectors are: the smaller the angle, the closer they match. The dot product itself is also a good measure of this: the closer the dot product is to $\|\mathbf{x}\|\|\mathbf{y}\|$, they more aligned they are, while at the other extreme ($\mathbf{x}^{\mathrm{T}}\mathbf{y} = -\|\mathbf{x}\|\|\mathbf{y}\|$) they are pointing in opposite directions ($\cos\theta = 180°$).

Being able to measure and compare the difference in the orientation of two vectors is very useful. However, if we are using a different inner product, can we find an analogous definition for the angle between them? It turns out that we can, thanks to the Cauchy-Schwarz inequality.

Theorem 5.3.1 — Cauchy–Schwarz Inequality. Let $\mathbf{x}$ and $\mathbf{y}$ be arbitrary vectors in a real inner product space. Then

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

with equality holding if and only if one of the vectors is $\mathbf{0}$ or one vector is a multiple of the other.

The above inequality can be rewritten as (since $\langle \mathbf{x}, \mathbf{y} \rangle$ is a real number)

$$-\|\mathbf{x}\| \|\mathbf{y}\| \leq |\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

and therefore, provided $\|\mathbf{x}\| \|\mathbf{y}\| \neq 0$,

$$-1 \leq \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|}{\|\mathbf{x}\| \|\mathbf{y}\|} \leq 1$$

Thus each pair of non-zero vectors in a real inner product space determines a unique angle θ given by

$$\cos\theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}, \quad 0 \leq \theta \leq \pi.$$

In practice, the value of the angle itself is not particularly insightful[1], but the cosine of that angle, $\frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}$, gives a direct measure of how closely aligned two vectors are. The closer the cosine is to 1, the more aligned the vectors are, while they will point in opposite directions as it approaches -1. When the value is 0, the vectors are orthogonal to one another.

5.3.1 Application: information retrieval

Consider searching a database for documents that contains the frequency of certain key words. If we want to find documents on a single topic, then we can search for documents that contain that word. What is the best match? Is it the one that contains that word the most? Or is it the one more focussed on that topic? How could I find the best match?

One common approach is to represent the documents in the database as vectors, whose elements correspond to the frequency of our key words. We then search for the vector most closely aligned with the vector representing our search criteria, using an inner product to calculate the cosine of the angle between our search vector and the database vectors.

Let's consider a website that has eight modules for learning linear algebra. A database showing the frequency of key words in each module's headings and subheadings is shown in the table below. In matrix Q, we normalise each column of the database matrix, so that they are each unit vectors. Now we can search for a match to our search criteria. For example, to search for the words *orthogonality, spaces, vector* we use the normalised search vector $\hat{\mathbf{x}} = (0,0,0,0,0,0,0.577,0.577,0,0,0.577)^{\mathrm{T}}$ since $0.577 \approx \frac{1}{\sqrt{3}}$.

[1] Outside of using the dot product in 2D or 3D, it doesn't correspond to an actual angle we can measure.

Key Words	Modules							
	M1	M2	M3	M4	M5	M6	M7	M8
determinants	0	6	3	0	1	0	1	1
eigenvalues	0	0	0	0	0	5	3	2
linear	5	4	4	5	4	0	3	3
matrices	6	5	3	3	4	4	3	2
numerical	0	0	0	0	3	0	4	3
orthogonality	0	0	0	0	4	6	0	2
spaces	0	0	5	2	3	3	0	1
systems	5	3	3	2	4	2	1	1
transformations	0	0	0	5	1	3	1	0
vector	0	4	4	3	4	1	0	3

The normalised version of the above table gives

$$Q = \begin{pmatrix} 0 & 0.594 & 0.327 & 0 & 0.1 & 0 & 0.147 & 0.154 \\ 0 & 0 & 0 & 0 & 0 & 0.5 & 0.442 & 0.309 \\ 0.539 & 0.396 & 0.436 & 0.574 & 0.4 & 0 & 0.442 & 0.463 \\ 0.647 & 0.495 & 0.327 & 0.344 & 0.4 & 0.4 & 0.442 & 0.309 \\ 0 & 0 & 0 & 0 & 0.3 & 0 & 0.59 & 0.463 \\ 0 & 0 & 0 & 0 & 0.4 & 0.6 & 0 & 0.309 \\ 0 & 0 & 0.546 & 0.229 & 0.3 & 0.3 & 0 & 0.154 \\ 0.539 & 0.297 & 0.327 & 0.229 & 0.4 & 0.2 & 0.147 & 0.154 \\ 0 & 0 & 0 & 0.574 & 0.1 & 0.3 & 0.147 & 0 \\ 0 & 0.396 & 0.436 & 0.344 & 0.4 & 0.1 & 0 & 0.463 \end{pmatrix}$$

Because $\mathbf{x}$ and each column of Q is normalised, $\mathbf{y} = Q^{\mathrm{T}}\hat{\mathbf{x}}$ gives the cosine of the angle between our search criteria and the database entries. Using the standard inner product, we obtain

$$\mathbf{y} = (0, 0.229, 0.567, 0.331, 0.635, 0.577, 0, 0.535)^{\mathrm{T}}$$

Since $y_5 = 0.635$ is the largest entry, module 5 is the best match for our search. The next best matches are module 6 ($y_6 = 0.577$) and module 3 ($y_3 = 0.567$).

Note that $\mathbf{y} = Q^{\mathrm{T}}\hat{\mathbf{x}} \Rightarrow y_i = \mathbf{q}_i^{\mathrm{T}}\hat{\mathbf{x}} = \|\mathbf{q}_i\|\|\hat{\mathbf{x}}\| \cos\theta_i = \cos\theta_i$, where θ_i is the angle between $\hat{\mathbf{x}}$ and $\mathbf{q}_i$. This is because the columns of $\hat{\mathbf{x}}$ and of Q are normalised and answers the above question of why the normalisation was done. This expression means that the largest entry of $\mathbf{y}$, y_5, has the largest value of $\cos\theta_i$ and hence the direction of the search vector $\hat{\mathbf{x}}$ is closest to the direction of $\mathbf{q}_5$ and is therefore its 'best match'.

Using the standard inner product here is equivalent to assuming that each search term is independent and of equal weight. What if this was not the case? Let's consider two alternative cases here.

If the search term *orthogonality* was more important, we might use an inner product that re-weighted this term. For example, if it were twice as important, then we would define the inner product as

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + x_2y_2 + \cdots + x_5y_5 + \underline{2x_6y_6} + x_7y_7 + \cdots + x_{10}y_{10}.$$

This creates a new matrix of normalised columns

$$Q_2 = \begin{pmatrix} 0 & 0.594 & 0.327 & 0 & 0.093 & 0 & 0.147 & 0.147 \\ 0 & 0 & 0 & 0 & 0 & 0.428 & 0.442 & 0.295 \\ 0.539 & 0.396 & 0.436 & 0.574 & 0.371 & 0 & 0.442 & 0.442 \\ 0.647 & 0.495 & 0.327 & 0.344 & 0.371 & 0.343 & 0.442 & 0.295 \\ 0 & 0 & 0 & 0 & 0.279 & 0 & 0.590 & 0.442 \\ 0 & 0 & 0 & 0 & 0.371 & 0.515 & 0 & 0.295 \\ 0 & 0 & 0.546 & 0.229 & 0.279 & 0.257 & 0 & 0.147 \\ 0.539 & 0.297 & 0.327 & 0.229 & 0.371 & 0.175 & 0.147 & 0.147 \\ 0 & 0 & 0 & 0.574 & 0.093 & 0.257 & 0.147 & 0 \\ 0 & 0.396 & 0.436 & 0.344 & 0.371 & 0.086 & 0 & 0.442 \end{pmatrix}$$

and

$$\mathbf{y}_2 = (0, 0.198, 0.491, 0.287, 0.697, 0.686, 0, 0.590)$$

Now the matches with modules 5 and 6 are stronger, and the match with module 3 weaker, which arises because modules 5 and 6 have stronger references to orthogonality (module 3 has none).

We could have achieved a similar result simply by changing the search vector so that the *orthogonality* search term had double the weight: $\hat{\mathbf{x}} = (0,0,0,0,0,0.817,0.409,0,0,0.409)^{\mathrm{T}}$ since $0.409 \approx \frac{1}{\sqrt{6}}$. However, the two approaches have slightly different interpretations. The first says “make any reference to *orthogonality* twice as important as any other”, while the second says “search for references that contain twice as much on *orthogonality* as they do on *spaces* and *vectors*. The first fundamentally changes the value of reference to orthogonality, while the second changes the emphasis of the search.

5.3.2 Application: collaborative filtering

This approach to determining closeness of matches is used by social media platforms and streaming services to recommend content that users might like. Let us consider a similar set of data to the previous problem, but now with the values representing *ratings* (stars out of 5, for example) of 10 songs that 8 users have provided. A ‘0’ indicates that a user hasn’t provided a rating.

Song	User ratings							
	U1	U2	U3	U4	U5	U6	U7	U8
Song 1	0	5	3	0	1	0	1	1
Song 2	0	0	0	0	0	5	3	2
Song 3	5	4	4	5	4	0	3	3
Song 4	5	5	3	3	4	4	3	2
Song 5	0	0	0	0	3	0	4	3
Song 6	0	0	0	0	4	5	0	2
Song 7	0	0	5	2	3	3	0	1
Song 8	5	3	3	2	4	2	1	1
Song 9	0	0	0	5	1	3	1	0
Song 10	0	4	4	3	4	1	0	3

I sign up as a new user, and after listening to a few songs, have the rating vector

$$\mathbf{x} = (0,0,5,0,0,0,3,3,0,0)^{\mathrm{T}}.$$

What recommendation could the platform make?

First, it will calculate the inner product of my ratings with other users' ratings, to identify users with similar expressed tastes to my own. Again, we normalise each user's ratings, and my own, to produce

$$Q = \begin{pmatrix} 0 & 0.524 & 0.327 & 0 & 0.100 & 0 & 0.147 & 0.154 \\ 0 & 0 & 0 & 0 & 0 & 0.530 & 0.442 & 0.309 \\ 0.577 & 0.419 & 0.436 & 0.574 & 0.400 & 0 & 0.442 & 0.463 \\ 0.577 & 0.524 & 0.327 & 0.344 & 0.400 & 0.424 & 0.442 & 0.309 \\ 0 & 0 & 0 & 0 & 0.300 & 0 & 0.590 & 0.463 \\ 0 & 0 & 0 & 0 & 0.400 & 0.530 & 0 & 0.309 \\ 0 & 0 & 0.546 & 0.229 & 0.300 & 0.318 & 0 & 0.154 \\ 0.577 & 0.314 & 0.327 & 0.229 & 0.400 & 0.212 & 0.147 & 0.154 \\ 0 & 0 & 0 & 0.574 & 0.100 & 0.318 & 0.147 & 0 \\ 0 & 0.419 & 0.436 & 0.344 & 0.400 & 0.106 & 0 & 0.463 \end{pmatrix}$$

and

$$\hat{\mathbf{x}} = (0,0,0.762,0,0,0,0.457,0.457,0,0)^{\mathrm{T}}.$$

We can then estimate the closeness of a match with other users by taking the inner product, to obtain

$$\mathbf{y} = Q^{\mathrm{T}}\hat{\mathbf{x}} = (0.704,0.464,0.732,0.647,0.625,0.242,0.405,0.494)^{\mathrm{T}}.$$

The users with the closest matches—those with the largest values of y_i—are considered my 'nearest neighbours', in the jargon of collaborative filtering. The ratings of songs that I haven't already rated can then be used to suggest recommendations to me.

For a real social media or streaming platform, with vast numbers of both users and things to rate, these recommendations may be based only on ratings from my nearest N neighbours, for some constant N dependent on the size of the platform. Here we will use all 8 users.

We will first consider a simplified method that can be expressed neatly using matrix multiplication. If we put the unnormalised ratings in matrix R,

$$R = \begin{pmatrix} 0 & 5 & 3 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 5 & 3 & 2 \\ 5 & 4 & 4 & 5 & 4 & 0 & 3 & 3 \\ 5 & 5 & 3 & 3 & 4 & 4 & 3 & 2 \\ 0 & 0 & 0 & 0 & 3 & 0 & 4 & 3 \\ 0 & 0 & 0 & 0 & 4 & 5 & 0 & 2 \\ 0 & 0 & 5 & 2 & 3 & 3 & 0 & 1 \\ 5 & 3 & 3 & 2 & 4 & 2 & 1 & 1 \\ 0 & 0 & 0 & 5 & 1 & 3 & 1 & 0 \\ 0 & 4 & 4 & 3 & 4 & 1 & 0 & 3 \end{pmatrix},$$

then the product $R\mathbf{y}$ gives a weighted total of the ratings for each song, where ratings from users who are more similar to me receive a larger weight.

To put these values back on approximately the same scale as the original ratings, we divide by the total similarity weight. This gives the simplified recommendation score

$$\frac{R\mathbf{y}}{\sum_{i=1}^{8} y_i} = (1.400, 0.792, 3.880, 3.628, 1.154, 1.090, 1.867, 2.849, 1.158, 2.538)^{\mathrm{T}}.$$

Removing the songs I have already rated, just for clarity, leaves

$$(1.400, 0.792, -, 3.628, 1.154, 1.090, -, -, 1.158, 2.538)^{\mathrm{T}}.$$

Under this simplified system, the strongest recommendation is Song 4, with a score of 3.628, followed by Song 10, with a score of 2.538.

There is, however, a limitation to this calculation. A ‘0’ in the ratings matrix means that a user has *not rated* a song, rather than that they have given it a rating of zero. In the simplified calculation above, every user’s similarity value is included in the denominator, even if that user has not rated the particular song being considered. As a result, songs that have been rated by only a small number of users can receive artificially low recommendation scores.

We can improve the system by calculating a separate weighted average for each song, using only the users who have actually rated that song. For Song j, let p_j denote its predicted rating. We calculate

$$p_j = \frac{\sum\limits_{\substack{i=1 \\ R_{ji} \neq 0}}^{8} y_i R_{ji}}{\sum\limits_{\substack{i=1 \\ R_{ji} \neq 0}}^{8} y_i}.$$

The numerator is the similarity-weighted total of the ratings for Song j, while the denominator is the total similarity weight of only those users who have rated that song.

Applying this calculation to all 10 songs gives

$$\mathbf{p} = (2.220, 2.992, 4.111, 3.628, 3.266, 3.452, 2.938, 2.849, 2.601, 3.417)^{\mathrm{T}}.$$

Again removing the songs I have already rated gives

$$\mathbf{p} = (2.220, 2.992, -, 3.628, 3.266, 3.452, -, -, 2.601, 3.417)^{\mathrm{T}}.$$

Using this refined system, Song 4 is still the strongest recommendation, with a predicted rating of 3.628. The next strongest recommendation is Song 6, with a predicted rating of 3.452, followed closely by Song 10, with a predicted rating of 3.417.

Based on my rating of these songs, the streaming service could update my nearest neighbours and make further recommendations.

5.4 Orthogonal and orthonormal bases

Orthogonality is a very useful property in the context of developing bases.

Definition 5.4.1

1. The vectors $\mathbf{u}_1, \mathbf{u}_2, \dots \mathbf{u}_p$ form a set of orthogonal vectors if they are pairwise orthogonal; that is
$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \text{ if } i \neq j$$
2. An **orthonormal** set of vectors is an orthogonal set of unit vectors (*i.e.*, orthogonal vectors $\mathbf{v}_i$ also satisfy $\|\mathbf{v}_i\| = 1$ for all i).

Definition 5.4.2

A basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ for an inner product space is called

1. an **orthogonal** basis if $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ whenever $i \neq j$
2. an **orthonormal** basis if $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij}, \quad i, j = 1, 2, \dots, n$, where
$$\delta_{ij} = \begin{cases} 0 \text{ if } i \neq j \\ 1 \text{ if } i = j \end{cases} \quad \text{(Kronecker delta)}$$

5.4.1 Examples of orthogonal and orthonormal sets

■ **Example 5.3**

Consider $\mathbf{v}_1 = (1,1,1)^{\mathrm{T}}, \mathbf{v}_2 = (2,1,-3)^{\mathrm{T}}, \mathbf{v}_3 = (4,-5,1)^{\mathrm{T}}$. The set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal set in $\mathbb{R}^3$ since

$$\begin{aligned} \langle \mathbf{v}_1, \mathbf{v}_2 \rangle &= (1,1,1) \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} = 2(1) + 1(1) + 1(-3) = 0 \\ \langle \mathbf{v}_1, \mathbf{v}_3 \rangle &= 1(4) + 1(-5) + 1(1) = 0 \\ \langle \mathbf{v}_2, \mathbf{v}_3 \rangle &= 2(4) + 1(-5) + (-3)(1) = 0 \end{aligned}$$

and to form an orthonormal set let

$$\begin{aligned} \mathbf{w}_1 &= \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{1^2+1^2+1^2}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{3}}(1,1,1)^{\mathrm{T}} \\ \mathbf{w}_2 &= \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{2^2+1^2+(-3)^2}} \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} = \frac{1}{\sqrt{14}}(2,1,-3)^{\mathrm{T}} \\ \mathbf{w}_3 &= \frac{1}{\sqrt{42}}(4,-5,1)^{\mathrm{T}} \end{aligned}$$

■

■ **Example 5.4 — Orthonormal basis sets in $\mathbb{R}^2$.**

1.
$$\{\mathbf{e}_1, \mathbf{e}_2\} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

2.

$$\left\{\frac{1}{\sqrt{2}}(\mathbf{e}_1+\mathbf{e}_2), \frac{1}{\sqrt{2}}(\mathbf{e}_1-\mathbf{e}_2)\right\} = \left\{\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}\right\}$$

3. In $\mathbb{R}^2$, all orthonormal bases are simply rotated versions of the standard basis.

$$\{(\cos\theta\mathbf{e}_1+\sin\theta\mathbf{e}_2), (-\sin\theta\mathbf{e}_1+\cos\theta\mathbf{e}_2)\} = \left\{\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}, \begin{pmatrix}-\sin\theta\\ \cos\theta\end{pmatrix}\right\}$$

■

■ **Example 5.5 — Orthonormal basis sets in $\mathbb{R}^3$.**

1.

$$\{\mathbf{e}_1,\mathbf{e}_2,\mathbf{e}_3\} = \left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$$

2.

$$\left\{\frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\0\end{pmatrix}, \frac{1}{\sqrt{6}}\begin{pmatrix}1\\1\\-2\end{pmatrix}\right\}$$

■

5.4.2 Important results regarding orthogonal vectors

Theorem 5.4.1 If $\{\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_n\}$ is an orthogonal set of non–zero vectors, then $\mathbf{v}_1$,$\mathbf{v}_2$, $\ldots$, $\mathbf{v}_n$ are linearly independent.

Proof. We test for linear dependence or independence by considering for $c_i \in \mathbb{R}$

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots c_n\mathbf{v}_n = \mathbf{0}.$$

Taking the inner product of $\mathbf{v}_j$, for some j, $1 \le j \le n$, with both sides of this equation gives

$$c_1\langle\mathbf{v}_j,\mathbf{v}_1\rangle + c_2\langle\mathbf{v}_j,\mathbf{v}_2\rangle + \ldots + c_n\langle\mathbf{v}_j,\mathbf{v}_n\rangle = 0.$$

Since the vectors are pairwise orthogonal $\langle\mathbf{v}_j,\mathbf{v}_i\rangle = 0$, where $i \neq j$, and
$\therefore c_j\langle\mathbf{v}_j,\mathbf{v}_j\rangle = 0 \Rightarrow c_j\|\mathbf{v}_j\|^2 = 0 \Rightarrow c_j = 0$ and this is true for all $j = 1,2\ldots n$.
So $\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_n$ are linearly independent. ■

Orthonormal basis sets are very useful in applications as they are very simple to work with. We have seen previously that, for a vector space V, any vector $\mathbf{v} \in V$ can be written uniquely as a linear combination of the basis vectors, $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots c_n\mathbf{v}_n$, where the c_i values are the components of $\mathbf{v}$ relative to the given basis. It is easier to determine the c_i values when the basis is orthogonal, since

$$\begin{aligned}\langle\mathbf{v},\mathbf{v}_i\rangle &= \langle c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots c_n\mathbf{v}_n, \mathbf{v}_i\rangle\\ &= c_1\langle\mathbf{v}_1,\mathbf{v}_i\rangle + c_2\langle\mathbf{v}_2,\mathbf{v}_i\rangle + \ldots + c_n\langle\mathbf{v}_n,\mathbf{v}_i\rangle\\ &= c_i\langle\mathbf{v}_i,\mathbf{v}_i\rangle.\end{aligned}$$

We have now proved the following Theorem.

Theorem 5.4.2 If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthogonal basis for $\mathbb{R}^n$, then any vector in $\mathbb{R}^n$ can be expanded in the form

$$\mathbf{v} = \frac{\langle \mathbf{v}, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle}\mathbf{v}_1 + \frac{\langle \mathbf{v}, \mathbf{v}_2 \rangle}{\langle \mathbf{v}_2, \mathbf{v}_2 \rangle}\mathbf{v}_2 + \ldots + \frac{\langle \mathbf{v}, \mathbf{v}_n \rangle}{\langle \mathbf{v}_n, \mathbf{v}_n \rangle}\mathbf{v}_n$$

For an orthonormal basis set this becomes

$$\mathbf{v} = \langle \mathbf{v}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{v}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \ldots + \langle \mathbf{v}, \mathbf{v}_n \rangle \mathbf{v}_n$$

■ **Example 5.6** Consider the basis for $\mathbb{R}^2$ given by $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$. To find the coefficients of the vector $\mathbf{v} = (-1, 7)^{\mathrm{T}}$ relative to this basis, we must solve $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ for c_1 and c_2. So

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 7 \end{pmatrix} \Rightarrow c_1 = 3, c_2 = -4.$$

Alternatively, using Theorem 5.4.2,

$$c_1 = \frac{\langle \mathbf{v}, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle}\mathbf{v}_1 = \frac{(-1\ 7)\begin{pmatrix} 1 \\ 1 \end{pmatrix}}{\left\| \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\|^2} = \frac{6}{2} = 3 \text{ and } c_2 = \frac{\langle \mathbf{v}, \mathbf{v}_2 \rangle}{\langle \mathbf{v}_2, \mathbf{v}_2 \rangle}\mathbf{v}_2 = \frac{(-1\ 7)\begin{pmatrix} 1 \\ -1 \end{pmatrix}}{\left\| \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\|^2} = -\frac{8}{2} = -4.$$

This calculation would be even easier if the basis were orthonormal and the denominator terms were 1. ■

It is possible to take any basis and use projections to construct an orthonormal basis from it using the **Gram–Schmidt orthogonalisation** process. We will cover this in section 7.3.

5.5 Orthogonal matrices

Orthogonal matrices are of particular importance in the diagonalisation of a matrix, which we considered earlier.

Definition 5.5.1 An $n \times n$ matrix Q is said to be an **orthogonal matrix** if the column vectors of Q form an orthonormal set in $\mathbb{R}^n$ (with respect to the standard inner product).

Theorem 5.5.1 An $n \times n$ matrix is orthogonal iff $Q^{\mathrm{T}}Q = I$.

Proof. By definition, Q is orthogonal iff $\mathbf{q}_i^{\mathrm{T}}\mathbf{q}_j = \delta_{ij}$, where $\mathbf{q}_k$ is the kth column vector of Q. However, $\mathbf{q}_i^{\mathrm{T}}\mathbf{q}_j$ is the (i, j) entry of the matrix $Q^{\mathrm{T}}Q$. Therefore Q is orthogonal iff $Q^{\mathrm{T}}Q = I$. ■

Corollary 5.5.2 If Q is an orthogonal matrix then Q is invertible and $Q^{-1} = Q^{\mathrm{T}}$.

This is an extremely useful result in the case that Q also corresponds to the eigenvectors of a matrix, as we will see in section 5.6.

■ **Example 5.7** For any fixed angle θ, the matrix $Q = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is orthogonal.
You can prove this by showing $QQ^{\mathrm{T}} = Q^{\mathrm{T}}Q = I$ or by showing that the inner product of the columns is 0, as well as the norm of each column is 1.
Then you can find Q^{-1} easily using corollary 5.5.2:

$$Q^{-1} = Q^{\mathrm{T}} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

■

Note:
Recall that Q in the above example can be thought of as a linear transformation $\mathbb{R}^2 \to \mathbb{R}^2$ that rotates each vector through an angle of θ, but leaves its length unchanged. Similarly, Q^{-1} can be thought of as a rotation through the angle $-\theta$. It is possible to prove in general that multiplication by an orthogonal matrix preserves the length of vectors (see below).

5.5.1 Properties of orthogonal matrices

If Q is an $n \times n$ orthogonal matrix, then

1. The column vectors of Q form an orthonormal basis for $\mathbb{R}^n$
2. $Q^{\mathrm{T}}Q = I$
3. $Q^{\mathrm{T}} = Q^{-1}$
4. $\langle Q\mathbf{x}, Q\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$
5. $||Q\mathbf{x}|| = ||\mathbf{x}||$ (where $||\ ||$ is the Euclidean norm).
6. $\det(Q) = \pm 1$

- Proof of 4:

$$\langle Q\mathbf{x}, Q\mathbf{y}\rangle = (Q\mathbf{x})^{\mathrm{T}} Q\mathbf{y} = \mathbf{x}^{\mathrm{T}} Q^{\mathrm{T}} Q\mathbf{y} = \mathbf{x}^{\mathrm{T}}\mathbf{y} = \langle \mathbf{x}, \mathbf{y}\rangle$$

- Proof of 5:

$$\therefore \text{ if } \mathbf{x} = \mathbf{y},\ ||Q\mathbf{x}||^2 = \langle Q\mathbf{x}, Q\mathbf{x}\rangle = \langle \mathbf{x}, \mathbf{x}\rangle = ||\mathbf{x}||^2 \quad \Rightarrow ||Q\mathbf{x}|| = ||\mathbf{x}||.$$

- Proof of 6:

$$Q^{\mathrm{T}}Q = I \Rightarrow \det(Q^{\mathrm{T}}Q) = 1 \Rightarrow \det(Q)\det(Q^{\mathrm{T}}) = 1 \Rightarrow (\det(Q))^2 = 1 \Rightarrow \det(Q) = \pm 1$$

5.6 Real symmetric matrices

Symmetric matrices occur surprisingly often (*e.g.*, in classical and quantum mechanics, statistics, probability theory). Their eigenvalues and eigenvectors have special (and useful) properties.

Theorem 5.6.1 If A is a real symmetric matrix (ie. $A^{\mathrm{T}} = A$), then

1. all the eigenvalues of A are real; and
2. the eigenvectors corresponding to *distinct* eigenvalues of A are orthogonal.

We will not present a proof of 1, as it requires some generalisations to complex matrices that we are not going to investigate in this course.

Proof of 2. Let λ_1 and λ_2 be two distinct eigenvalues of A and let $\mathbf{v_1}$ and $\mathbf{v_2}$ be their corresponding eigenvectors. First, we observe that

$$\mathbf{v_2}^{\mathrm{T}} A\mathbf{v_1} = \mathbf{v_2}^{\mathrm{T}}(\lambda_1 \mathbf{v_1}) = \lambda_1 \mathbf{v_1}^{\mathrm{T}}\mathbf{v_2}.$$

But this is just a real number, so it must be its own transpose. Using this and the fact that $A^{\mathrm{T}} = A$, we see that

$$\mathbf{v_2}^{\mathrm{T}} A\mathbf{v_1} = (\mathbf{v_2}^{\mathrm{T}} A\mathbf{v_1})^{\mathrm{T}} = \mathbf{v_1}^{\mathrm{T}} A^{\mathrm{T}}\mathbf{v_2} = \mathbf{v_1}^{\mathrm{T}} A\mathbf{v_2} = \mathbf{v_1}^{\mathrm{T}}(\lambda_2\mathbf{v_2}) = \lambda_2\mathbf{v_1}^{\mathrm{T}}\mathbf{v_2}$$

Since $\lambda_1 \neq \lambda_2$, we see that

$$0 = (\lambda_2 - \lambda_1)\mathbf{v_1}^{\mathrm{T}}\mathbf{v_2} \implies \mathbf{v_1}^{\mathrm{T}}\mathbf{v_2} = 0 \implies \langle \mathbf{v_1}, \mathbf{v_2}\rangle = 0.$$

■

■ **Example 5.8** The matrix $A = \begin{pmatrix} -1 & 2 & 5 \\ 2 & 2 & 2 \\ 5 & 2 & -1 \end{pmatrix}$ is symmetric and its eigenvalues and vectors are

$$\lambda_1 = 6,\ \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \lambda_2 = 0,\ \mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \quad \lambda_3 = -6,\ \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

So in this case, all the eigenvalues are real numbers and the eigenvectors are orthogonal (you must check that $\langle \mathbf{v}_1, \mathbf{v}_2\rangle = (1,1,1)\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 0$ and similarly for the other two pairs of vectors). ■

We can now put together a number of previous results to come up with a theorem for symmetric matrices. Recall from Theorem 4.4.1 that a matrix with linearly independent eigenvectors can be rewritten as $A = VDV^{-1}$, where D is the diagonal matrix of eigenvalues and V is the matrix with eigenvectors corresponding to these eigenvalues as columns. So from Theorem 4.4.1, for a symmetric matrix, these eigenvectors are orthogonal and if we normalise them V is an orthogonal matrix and Theorem 5.4.2 tells us that for an orthogonal matrix $V^{\mathrm{T}}V = I$ ie. $V^{-1} = V^{\mathrm{T}}$. This means that we can write $A = VDV^{-1} = VDV^{\mathrm{T}}$.

Theorem 5.6.2 — Spectral Theorem. For every symmetric real matrix A there exists a real orthogonal matrix V and a diagonal matrix D such that $A = VDV^{\mathrm{T}}$ (or $D = V^{\mathrm{T}}AV$). Here D is the diagonal matrix of eigenvalues of A and V has the corresponding normalised eigenvectors as columns.

This may seem like a rather abstract mathematical statement but in fact it was the starting point of all modern ways to calculate the eigenvalues of symmetric matrices. The method was first discovered by the astronomer Carl Gustav Jacobi in 1846 when he was trying to calculate the stability of planetary orbits in the solar system. He needed to get the eigenvalues of a 7×7 matrix and had to develop a better method than the determinant method.

The basic idea of the Jacobi method is to try to find a matrix V so that $V^{\mathrm{T}}AV$ is a purely diagonal matrix D and then we would have all the eigenvalues at once. The columns in V would hold the eigenvectors of A and the diagonal in D would give the eigenvalues of A. The method is iterative and works by moving toward V in small steps using simple rotation matrices to reduce the size of successive off–diagonal elements.

5.7 Positive definite matrices

In the previous section we have seen that symmetric matrices have some special and useful properties compared with other matrices. The final category of matrices that we will consider (positive definite matrices) have even more special properties. These types of matrices occur in a wide variety of applications, for example, in solutions of initial value problems by finite difference and finite element methods.

Definition 5.7.1 The expression $\mathbf{x}^{\mathrm{T}}A\mathbf{x}$ is known as a **quadratic form**. It is always possible to construct the matrix A such that it is symmetric $\left(A = A^{\mathrm{T}}\right)$.

For $n = 2$ the expression $\mathbf{x}^{\mathrm{T}}A\mathbf{x} = (x\ y)\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + 2bxy + cy^2$. Notice the form of the matrix with only 3 unique entries because it is symmetric.

Quadratic forms occur in many applications (*e.g.*, in mechanics as the energy of vibrating multi–particle systems in quantum mechanics as approximations for the energy of complex molecules, and in statistics for random quantities that have a Gaussian distribution).

Definition 5.7.2 A real symmetric $n \times n$ matrix A is said to be **positive definite** if $\mathbf{x}^{\mathrm{T}}A\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0} \in \mathbb{R}^n$.

■ **Example 5.9** Consider the symmetric matrix $A = \begin{pmatrix} 5 & 2 \\ 2 & 3 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. Then

$$\begin{aligned} \mathbf{x}^{\mathrm{T}}A\mathbf{x} = (x_1\ x_2)\begin{pmatrix} 5 & 2 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= (x_1\ x_2)\begin{pmatrix} 5x_1 + 2x_2 \\ 2x_1 + 3 \end{pmatrix} \\ &= 5x_1^2 + 4x_1x_2 + 3x_2^2 \\ &= (2x_1 + x_2)^2 + x_1^2 + 2x_2^2 > 0 \end{aligned}$$

(here we complete the square). So A is positive definite. ■

Theorem 5.7.1 Let A be a real symmetric $n \times n$ matrix. Then A is positive definite if and only if all its eigenvalues are positive.

Proof that a positive definite matrix has only positive eigenvalues. We have a real $n \times n$ symmetric matrix A that is positive definite, so

$$\mathbf{x}^{\mathrm{T}} A \mathbf{x} > 0 \,\forall\, \mathbf{x} \neq \mathbf{0}.$$

By the spectral theorem, A can be decomposed as $A = VDV^{\mathrm{T}}$. Thus,

$$\mathbf{x}^{\mathrm{T}} VDV^{\mathrm{T}} \mathbf{x} > 0 \,\forall\, \mathbf{x} \neq \mathbf{0}.$$

Note that we can now define $\mathbf{y} = V^{\mathrm{T}}\mathbf{x}$ and rewrite the condition as

$$\mathbf{y}^{\mathrm{T}} D \mathbf{y} > 0 \,\forall\, \mathbf{y} \neq \mathbf{0}.$$

This follows from the relationship $(AB)^{\mathrm{T}} = B^{\mathrm{T}} A^{\mathrm{T}}$ and the fact that the columns of V form a basis for $\mathbb{R}^n$. Our result must hold for the standard vectors $\mathbf{y} = \mathbf{e}_i$, so we see $d_{ii} > 0 \,\forall\, i$. The diagonal elements of D are the eigenvalues of A. Thus, positive definite symmetric matrices have positive eigenvalues.
We must also prove that positive eigenvalues of a real symmetric matrix implies positive definiteness. This is straightforward, starting from the observation that

$$\mathbf{u}^{\mathrm{T}} D \mathbf{u} > 0 \,\forall\, \mathbf{u} \neq \mathbf{0}$$

if D only contains positive diagonal elements. ■

Once we know that a matrix is positive definite then many other useful properties can be established about the matrix that can be used in the solution of practical problems.

5.7.1 Application – finding a global minimum

Consider the graph of the function of two variables $f(x,y) = 2x^2 - 4xy + 5y^2$. It is not clear *a priori* whether the stationary point at $(0,0)$ is a global minimum or a saddle point. We can use information about positive definiteness to decide the issue.

Recall that for $f(x,y)$ we set the partial derivatives to zero to find the stationary point. Now

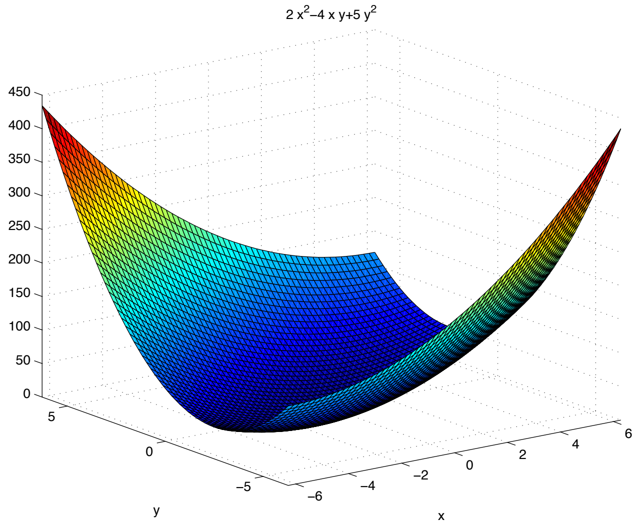
$$\frac{\partial f}{\partial x} = 4x - 4y = 0 \text{ and } \frac{\partial f}{\partial y} = -4x + 10y = 0 \Rightarrow (x,y) = (0,0).$$

We can write $f(x,y) = 2x^2 - 4xy + 5y^2$ in quadratic form

$$f(x,y) = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

and since $f(0,0) = 0$, then if $f(x,y) > 0$, (0,0) will be a global minimum.

Now $f(x,y) = \mathbf{x}^{\mathrm{T}} A \mathbf{x}$ so we just need to show this is positive for all x,y; that is, we need to see if it is positive definite. The eigenvalues for $A = \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}$ are $\lambda_1 = 6$ and $\lambda_2 = 1$ $(\mathrm{trace}(A) = 7 = \lambda_1 + \lambda_2$ and $\det(A) = 6 = \lambda_1\lambda_2)$. The eigenvalues are both positive and so by Theorem 5.7.1 the matrix is positive definite and therefore (0,0) is a global minimum.



We can extend this idea to the less trivial case where the critical point is not at the origin, but at some other point $\mathbf{x}_0$. In this case, the corresponding quadratic form becomes

$$\begin{aligned}(\mathbf{x}-\mathbf{x}_0)^{\mathrm{T}}A(\mathbf{x}-\mathbf{x}_0)^{\mathrm{T}} &= \mathbf{x}^{\mathrm{T}}A\mathbf{x} - 2\mathbf{x}_0^{\mathrm{T}}A\mathbf{x} + \mathbf{x}_0A\mathbf{x}_0 \\ &= ax^2 + 2bxy + cy^2 + dx + ey + g.\end{aligned}$$

We've used g for the constant offset to avoid confusion with the function $f(x,y)$.

To find the location and nature of the critical point when presented with an example of this more general quadratic form, we first re-write it in the form

$$ax^2 + 2bxy + cy^2 + dx + ey + g = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} d & e \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + g = \mathbf{x}^{\mathrm{T}}A\mathbf{x} + B\mathbf{x} + g.$$

Note that $B \in \mathbb{R}^{(1\times 2)}$. The nature of the critical point (minimum, maximum, or saddle point) is determined by the eigenvalues of matrix A, in exactly the same way as above, and the location of the critical point is determined by comparing the two vector forms of the equation (remembering $A = A^{\mathrm{T}}$):

$$2\mathbf{x}_0^{\mathrm{T}}A = B \implies \mathbf{x}_0 = \frac{1}{2}A^{-1}B^{\mathrm{T}}$$

Notice that g shifts the value of the function, but does not affect the location or nature of its critical points.

So, if the quadratic form was $f(x,y) = 2x^2 - 4xy + 5y^2 + 10x + 8y + 2$, then we can rewrite this as

$$f(x,y) = \mathbf{x}^{\mathrm{T}} \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 10 & 8 \end{pmatrix} \mathbf{x} + 2$$

in which case the critical point is at

$$\mathbf{x}_0 = \frac{1}{2} \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 10 \\ 8 \end{pmatrix} = \begin{pmatrix} \frac{11}{2} \\ 3 \end{pmatrix}$$

and is a local minimum (as per the first example).

5.7.2 Application – least square fitting

A very similar problem of much more practical interest is finding the line of best fit through a set of data points.

Let's begin by considering a straight line of best fit $y = mx + c$ through data points (x_1, y_1), (x_2, y_2), ..., (x_n, y_n). If the data points were all collinear, then we would have that

$$\begin{aligned} y_1 &= mx_1 + c \\ y_2 &= mx_2 + c \\ &\vdots \\ y_n &= mx_n + c \end{aligned} \implies \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix} \begin{pmatrix} m \\ c \end{pmatrix} = X\mathbf{k} \implies \mathbf{y} - X\mathbf{k} = \mathbf{0}$$

However, for typical data, the y_i values do not all fall on the line, so instead we find that $\mathbf{y} - X\mathbf{k}$ is not zero, but represents the discrepancy – or *error* – between what the y_i values actually are, and what they would be if they sat on the line $y = mx + c$.

If we want a measure of this error, the norm $\|\mathbf{y} - X\mathbf{k}\|$ is an excellent choice – we know that it must be a positive value, unless there is no error (when it will be zero). It cannot be negative. Smaller values of the discrepancy correspond to better fits, and we can find the line of best fit by minimising the error by our choices for m and c – that is, by our choice of $\mathbf{k}$. We also note that minimising a positive quantity is equivalent to minimising its square, which in this case makes the calculations a little easier[2]. Therefore, the optimal choice of m and c, represented by $\mathbf{k}_0$, is the choice that minimises

$$\begin{aligned} \|\mathbf{y} - X\mathbf{k}\|^2 &= (\mathbf{y}^{\mathrm{T}} - \mathbf{k}^{\mathrm{T}} X^{\mathrm{T}})(\mathbf{y} - X\mathbf{k}) \\ &= \mathbf{k}^{\mathrm{T}} X^{\mathrm{T}} X\mathbf{k} - 2\mathbf{y}^{\mathrm{T}} X\mathbf{k} + \mathbf{y}^{\mathrm{T}}\mathbf{y}. \end{aligned}$$

Fortunately, this is just a quadratic form, as we have seen in the previous section, and we now know how to find its minimum:

$$A = X^{\mathrm{T}} X, B = 2\mathbf{y}^{\mathrm{T}} X \implies \mathbf{k}_0 = \frac{1}{2} A^{-1} B^{\mathrm{T}} = \frac{1}{2} (X^{\mathrm{T}} X)^{-1} (2X^{\mathrm{T}}\mathbf{y}) = (X^{\mathrm{T}} X)^{-1} X^{\mathrm{T}}\mathbf{y}$$

The beauty of this approach is its generality. We can easily extend it to different choices of inner product – typically we would want to do this if some data had larger errors associated with them than others, because it allows us to provide a closer fit for better data points. It can also be extended to fitting any function of the form

$$y = a\,f(x) + b\,g(x) + c\,h(x) + \cdots,$$

that is, when y is a linear combination of given functions of x, and we just need to determine their coefficients $a, b, c, \ldots$. For example, to fit data to a cubic $y = ax^3 + bx^2 + cx + d$, the solution is given by

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1^3 & x_1^2 & x_1 & 1 \\ x_2^3 & x_2^2 & x_2 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ x_n^3 & x_n^2 & x_n & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = X\mathbf{k} \implies \mathbf{k}_0 = (X^{\mathrm{T}} X)^{-1} X^{\mathrm{T}}\mathbf{y}$$

[2] This boils down to the continuous differentiability of x^2, whereas $|x|$ is not differentiable at $x = 0$.

As an example, let us try to fit three different functions through the same data set, shown in the table and figure below: a linear fit, a quadratic fit, and a fit using two exponential functions (a 'bi-exponential' fit).

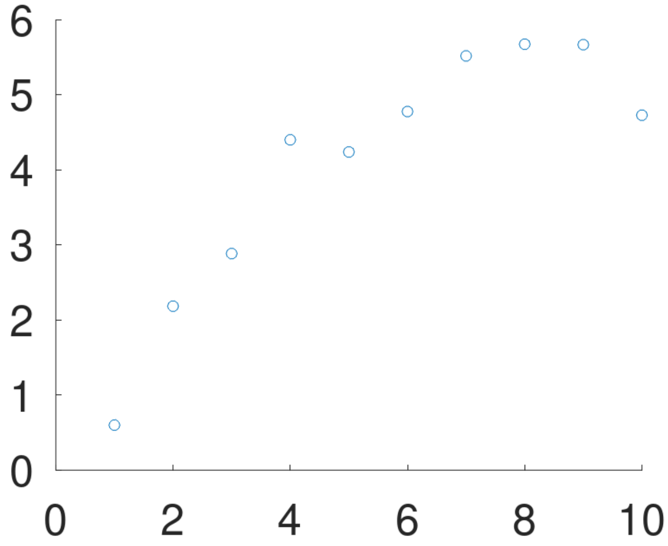


Figure 5.1: Data that we are attempting to fit using the least-squares method.

x_i	1	2	3	4	5	6	7	8	9	10
y_i	0.6	2.1837	2.8844	4.3985	4.2369	4.7755	5.5160	5.6721	5.6654	4.7268

For the least-squares fit to the linear equation $y = mx + c$, we have

$$\mathbf{y} = \begin{pmatrix} 0.60007 \\ 2.18372 \\ \vdots \\ 4.72679 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ \vdots & \vdots \\ 10 & 1 \end{pmatrix} \begin{pmatrix} m \\ c \end{pmatrix} = X\mathbf{k} \qquad \Longrightarrow \qquad \mathbf{k}_0 = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}\mathbf{y}$$

where the columns of X are the values of x_i and 1. Now,

$$X^{\mathrm{T}}X = \begin{pmatrix} 1 & 2 & \dots & 10 \\ 1 & 1 & \dots & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ \vdots & \vdots \\ 10 & 1 \end{pmatrix} = \begin{pmatrix} 385 & 55 \\ 55 & 10 \end{pmatrix} \Longrightarrow (X^{\mathrm{T}}X)^{-1} = \frac{1}{825} \begin{pmatrix} 10 & -55 \\ -55 & 385 \end{pmatrix}$$

So

$$\begin{pmatrix} m \\ c \end{pmatrix} = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}\mathbf{y} = \frac{1}{825} \begin{pmatrix} 10 & -55 \\ -55 & 385 \end{pmatrix} \begin{pmatrix} 1 & 2 & \dots & 10 \\ 1 & 1 & \dots & 1 \end{pmatrix} \begin{pmatrix} 0.60007 \\ 2.18372 \\ \vdots \\ 4.72679 \end{pmatrix} \approx \begin{pmatrix} 0.48 \\ 1.42 \end{pmatrix}$$

For the least-squares fit to the quadratic equation $y = ax^2 + bx + c$, we have

$$\mathbf{y} = \begin{pmatrix} 0.60007 \\ 2.18372 \\ \vdots \\ 4.72679 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ \vdots & \vdots & \vdots \\ 100 & 10 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = X\mathbf{k} \qquad \Longrightarrow \qquad \mathbf{k}_0 = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}\mathbf{y}$$

where the columns of X are the values of x_i^2, x_i and 1. Now,

$$X^{\mathrm{T}}X = \begin{pmatrix} 1 & 4 & \dots & 100 \\ 1 & 2 & \dots & 10 \\ 1 & 1 & \dots & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ \vdots & \vdots & \vdots \\ 100 & 10 & 1 \end{pmatrix} = \begin{pmatrix} 25333 & 3025 & 385 \\ 3025 & 385 & 55 \\ 385 & 55 & 10 \end{pmatrix}$$

$$\Longrightarrow \; (X^{\mathrm{T}}X)^{-1} = \frac{1}{2640} \begin{pmatrix} 5 & -55 & 110 \\ -55 & 637 & -1386 \\ 110 & -1386 & 3652 \end{pmatrix}$$

So

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}\mathbf{y}$$

$$= \frac{1}{2640} \begin{pmatrix} 5 & -55 & 110 \\ -55 & 637 & -1386 \\ 110 & -1386 & 3652 \end{pmatrix} \begin{pmatrix} 1 & 4 & \dots & 100 \\ 1 & 2 & \dots & 10 \\ 1 & 1 & \dots & 1 \end{pmatrix} \begin{pmatrix} 0.60007 \\ 2.18372 \\ \vdots \\ 4.72679 \end{pmatrix} \approx \begin{pmatrix} -0.10 \\ 1.59 \\ -0.80 \end{pmatrix}$$

Finally, imagine we had some reason for believing that the data should be well modelled by a function of the form $y = ae^{-x} + be^{-x/2} + c$. Then we can repeat the above process:

$$\mathbf{y} = \begin{pmatrix} 0.60007 \\ 2.18372 \\ \vdots \\ 4.72679 \end{pmatrix} = \begin{pmatrix} 0.367879 & 0.606531 & 1 \\ 0.135335 & 0.367879 & 1 \\ \vdots & \vdots & \vdots \\ 0.000045 & 0.006738 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = X\mathbf{k} \qquad \Longrightarrow \qquad \mathbf{k}_0 = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}\mathbf{y}$$

where the columns of X are the values of e^{-x_i}, $e^{-x_i/2}$ and 1. Now

$$X^{\mathrm{T}}X = \begin{pmatrix} 0.367879 & 0.135335 & \dots & 0.000045 \\ 0.606531 & 0.367879 & \dots & 0.006738 \\ 1 & 1 & \dots & 1 \end{pmatrix} \begin{pmatrix} 0.367879 & 0.606531 & 1 \\ 0.135335 & 0.367879 & 1 \\ \vdots & \vdots & \vdots \\ 0.000045 & 0.006738 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.15652 & 0.28722 & 0.58195 \\ 0.28722 & 0.58195 & 1.53111 \\ 0.58195 & 1.53111 & 10 \end{pmatrix}$$

$$\Longrightarrow \; (X^{\mathrm{T}}X)^{-1} = \begin{pmatrix} 102.97874 & -58.70585 & 2.99565 \\ -58.70585 & 36.34439 & -2.14833 \\ 2.99565 & -2.14833 & 0.25460 \end{pmatrix}$$

Thus,

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}\mathbf{y}$$
$$\approx \begin{pmatrix} 5.9457 \\ -11.6435 \\ 5.5027 \end{pmatrix}.$$

In the figure below, we show the three least-square lines of best fit that have been calculated.

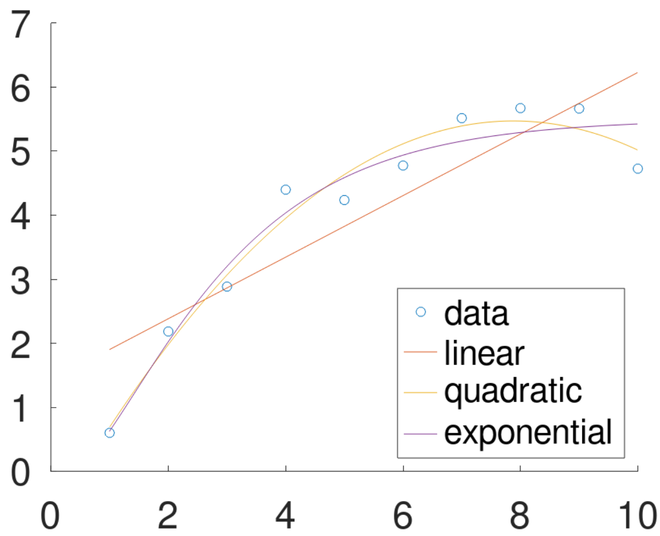


Figure 5.2: Linear, quadratic and bi-exponential least-squares lines of best fit for the tabulated data set.

Errors in the fit can be estimated from $\|\mathbf{y} - X\mathbf{k}\|$ in each case, which shows that the quadratic fit is a little better than the exponential fit, and much better than the linear fit (in agreement with what we observe from the figure).

Another beautiful aspect of this approach is that it gives us some insight into the nature of the solution. We see that the best-fit parameters satisfy the relation

$$\mathbf{k}_0 = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}\mathbf{y} \implies X^{\mathrm{T}}X\mathbf{k}_0 = X^{\mathrm{T}}\mathbf{y} \implies X^{\mathrm{T}}(\mathbf{y} - X\mathbf{k}_0) = \mathbf{0}$$

This result tells us that the *minimum error in our estimate*, $\mathbf{y} - X\mathbf{k}_0$, is orthogonal to each column vector in X. This turns out to be a unique property to the best estimate, and can be considered as an alternative criterion for finding the optimal set of parameters.

6. Vector spaces

Up to this point, we have focussed our attention on $\mathbb{R}^n$, with only passing reference to how the ideas we have covered might also apply to other types of mathematical objects, such as functions. Now we will formalise this extension to other mathematical objects. First, we will discuss what types of mathematical objects we can extend these ideas to, and come up with the abstract idea of a *vector space*. Then we will upgrade some of the definitions we introduced earlier in the context of point spaces $\mathbb{R}^m$, so that they apply in the more abstract setting. We will then finish by looking at some useful applications of these ideas beyond those we have already seen.

6.1 What are vector spaces?

6.1.1 Why vector spaces?

As we have seen, a linear map $\mathscr{L}$ is any map with the property

$$\mathscr{L}(\alpha\mathbf{x}+\beta\mathbf{y}) = \alpha\mathscr{L}(\mathbf{x})+\beta\mathscr{L}(\mathbf{y})$$

We saw that this is a property of scalar and matrix multiplication:

$$a\times(\alpha x+\beta y) = \alpha ax+\beta ay \quad \text{and} \quad A(\alpha\mathbf{x}+\beta\mathbf{y}) = \alpha A\mathbf{x}+\beta A\mathbf{y}$$

but it is also a property of differentiation or integration

$$\frac{\mathrm{d}}{\mathrm{d}x}(\alpha f(x)+\beta g(x)) = \alpha\frac{\mathrm{d}f}{\mathrm{d}x}+\beta\frac{\mathrm{d}g}{\mathrm{d}x}$$
$$\int[\alpha f(x)+\beta g(x)]\ \mathrm{d}x = \alpha\int f(x)\ \mathrm{d}x+\beta\int g(x)\ \mathrm{d}x$$

So the objects that linear maps can be applied to might be scalars, matrices, or functions, or indeed other mathematical objects that we haven't yet considered. These objects seem very different in nature. If we want to develop a single, robust, comprehensive theory of linear maps, we need to somehow get beyond the *differences* of these objects that we can apply such maps to, and focus on their *similarities*. What is it that is common to the way linear maps are applied to scalars, matrices, functions, *etc.*?

A first observation is that, whatever type of mathematical objects $\mathbf{x}$ and $\mathbf{y}$ are, the object $\alpha\mathbf{x}+\beta\mathbf{y}$ must exist and be the same type of object, for any real numbers α and β[1]. If this isn't the case, then we can't evaluate $\mathscr{L}(\alpha\mathbf{x}+\beta\mathbf{y})$, so the rule defining our linear map wouldn't hold! We also want all the usual algebraic laws of addition and scalar multiplication (commutativity, associativity, *etc.*) to hold in the same way they have held up to now, while we've considered $\mathbf{x},\mathbf{y} \in \mathbb{R}^n$ and linear maps as matrices $\mathbf{y} = A\mathbf{x}$. Without these rules holding, we can't extend some of the key results we've developed up to this point to more general problems.

6.1.2 Why 'vectors'?

We therefore introduce the concept of a **vector space**, as a set of mathematical objects (like matrices or functions) that obey all of these rules. Once we establish that a set of objects obeys the rules for a vector space, the objects in the vector spaces are called **vectors**. Calling these numbers, matrices or functions 'vectors' might seem confusing at first, but there is a very good reason for doing so. Just as we have done in $\mathbb{R}^n$, we will choose a **basis** for our vector space — a collection $\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_n$ of members of the vector space such that any vector $\mathbf{v}$ in the space can be written as

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$$

for appropriate (and unique) coefficients $c_1,c_2,\ldots,c_n$. The list of coefficients $(c_1,c_2,\ldots,c_n)$ that represents each member of our vector space can be thought of as a vector, in the usual sense — even if $\mathbf{v}$ itself is a matrix or a function[2]. This abstraction is a major strength of linear algebra — in being able to represent quite different types of problems using the common framework of a vector space, we can apply results derived, for example, in solving matrix problems to help solve completely different types of problems (such as ordinary differential equations).

6.1.3 Definition of a vector space

In mathematics, the words 'set' and 'space' have somewhat different connotations. A 'set' is simply a collection of objects, while a 'space' is a collection of objects with some sort of mathematical structure or rules for how the objects interact. There are many different types of 'spaces', defined by the specific mathematical structure imposed on the objects in the set.

The days of the week constitute a set, but we don't think of them as a space because we don't have any meaningful definitions for mathematical operations that take two days and give a single day in return. On the other hand, we have well defined notions of adding and multiplying real numbers, so we can think of the real numbers as a set (when we don't care about these operations) or as a space (when we do).

[1] The usual requirement is that α and β be the same type of number as the scalars, matrices or functions acted upon by the map – everything we shall consider will be for real numbers or subsets thereof.

[2] As an example, we can think of the coefficients of a polynomial ax^2+bx+c being represented as the vector (a,b,c).

A **vector space** is a collection of objects (called vectors) with a notion of addition and a notion of scalar multiplication. Vector spaces are **closed**: adding two vectors in the space produces another vector in the space; and rescaling a vector in the space produces another vector in the space. Also, addition and scalar multiplication as defined for the vector space must obey the 'usual' rules of associativity, commutativity, distributivity, identity[3], and (additive) inverse.

Definition 6.1.1 — Vector Space. A *vector space* is a set V whose elements obey the following rules for any $\mathbf{x},\mathbf{y},\mathbf{z} \in V$ and any scalar $\alpha \in \mathbb{R}$:

1. $\mathbf{x}+\mathbf{y}$ is always in V (closure under addition)
2. $\alpha\mathbf{x}$ is always in V (closure under scalar multiplication)
3. $\mathbf{x}+\mathbf{y} = \mathbf{y}+\mathbf{x}$ (commutativity)
4. $(\mathbf{x}+\mathbf{y})+\mathbf{z} = \mathbf{x}+(\mathbf{y}+\mathbf{z})$ (associativity)
5. There is an element $\mathbf{0}$ such that $\mathbf{x}+\mathbf{0} = \mathbf{x}$ (additive identity)
6. Each element $\mathbf{x}$ has an element $-\mathbf{x}$ such that $\mathbf{x}+(-\mathbf{x}) = \mathbf{0}$ (additive inverse)
7. $\alpha(\mathbf{x}+\mathbf{y}) = \alpha\mathbf{x}+\alpha\mathbf{y}$ (distributivity over vector addition)
8. $(\alpha+\beta)\mathbf{x} = \alpha\mathbf{x}+\beta\mathbf{x}$ (distributivity over scalar addition)
9. $(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x})$ (associativity of scalar multiplication)
10. $1\mathbf{x} = \mathbf{x}$ (multiplicative identity)

6.1.4 Common examples

In order to prove that a $\mathbb{R}^n$ is a vector space (under the usual definition of addition and scalar multiplication), we would have to establish that all of the above 10 rules hold. The proofs are not difficult but they are tedious, because each of them depends on the fact that these results all hold for the $\mathbb{R}$, and so therefore must hold for each component of the vectors in $\mathbb{R}^n$. In the same way, they must hold for matrices of a given size, $\mathbb{R}^{m\times n}$, and for the set of all functions defined on a given domain (again, under the usual definition of addition and scalar multiplication).

Theorem 6.1.1 The following spaces are all vector spaces, under the usual definition of addition and scalar multiplication in each case:

- $\mathbb{R}^n$ for $n \in \mathbb{N}$
- $\mathbb{R}^{m\times n}$ for $m,n \in \mathbb{N}$
- the set of *all* functions defined on a given domain, *e.g.*, the set of all functions $f : [0,1] \mapsto \mathbb{R}$

6.2 Subspaces

Often we will be working not with an entire vector space V, but with a subset[4] $S \subset V$. Examples might include the subset of 3×3 matrices that are symmetric, or the subset of functions defined on $[0,1]$ that are continuous. In these cases, we often want to show that

[3] Note that all the conditions in definition 6.1.1 only depend on vector addition/subtraction and multiplication of a vector by a scalar; we don't have any requirements on the products of vectors with other vectors. This means that the additive inverse and additive identity have to be vectors, but the multiplicative identity only needs to be the scalar 1. We can come up with pathological examples that break requirement 10, but they're outside the scope of this course.

[4] $S \subset V$ means that S is a subset of V, *i.e.*, that $x \in S \Rightarrow x \in V$.

this subset S forms a vector space *in its own right* (when using the algebraic operations as defined for V), in which case we call S a **(vector)**[5] **subspace** of V.

To show that S is a subspace, we have to establish that the 10 rules from Definition 6.1.1 hold. However, V is a vector space, so the algebraic rules 3–10 (which hold for V) are all guaranteed for $S \subset V$. The only way in which S can fail to be a subspace is if the closure rules don't hold: if the subset is not closed under addition (rule 1) or scalar multiplication (rule 2). For a subspace, these are the only rules that we need check, which hugely reduces our workload.

Definition 6.2.1 — Subspace. If S is a nonempty subset of a vector space V, where
1. $\alpha\mathbf{x} \in S$ wherever $\mathbf{x} \in S$ and $\alpha \in \mathbb{R}$; and
2. $\mathbf{x}+\mathbf{y} \in S$ whenever $\mathbf{x} \in S$ and $\mathbf{y} \in S$

then S is a *subspace* of V.

Note that a subspace must contain *at least one member*, and that a subspace is a vector space in its own right.

6.2.1 Examples of subspaces

■ Example 6.1 — The set S of all 3D vectors $\mathbf{x}$ with $x_3 = 0$ is a subspace of $\mathbb{R}^3$. We only need to establish closure. The sum of any two vectors $\mathbf{x},\mathbf{y} \in S$ is

$$\mathbf{x}+\mathbf{y} = (x_1,x_2,0)+(y_1,y_2,0) = (x_1+y_1,x_2+y_2,0) \in S$$

and a scalar multiple of any vector $\mathbf{x} \in S$ is

$$\alpha\mathbf{x} = \alpha(x_1,x_2,0) = (\alpha x_1,\alpha x_2,0) \in S$$

so we conclude that the set of all 3D vectors $\mathbf{x}$ with $x_3 = 0$ *is a subspace* of $\mathbb{R}^3$ ■

■ Example 6.2 — The set S of all 3D vectors $\mathbf{x}$ with $x_3 = 1$ is *not* a subspace of $\mathbb{R}^3$. We only need to establish closure. The sum of any two vectors $\mathbf{x},\mathbf{y} \in S$ is

$$\mathbf{x}+\mathbf{y} = (x_1,x_2,1)+(y_1,y_2,1) = (x_1+y_1,x_2+y_2,2) \notin S$$

and a scalar multiple of any vector $\mathbf{x} \in S$ is

$$\alpha\mathbf{x} = \alpha(x_1,x_2,1) = (\alpha x_1,\alpha x_2,\alpha) \notin S$$

so we conclude that the set of all 3D vectors $\mathbf{x}$ with $x_3 = 1$ *is not a subspace* of $\mathbb{R}^3$ ■

■ Example 6.3 — The set S of all 2×2 matrices where $a_{21} = -a_{12}$ is a subspace of $\mathbb{R}^{2\times 2}$. We only need to establish closure. The sum of any two vectors $\mathbf{x},\mathbf{y} \in S$ is

$$\mathbf{x}+\mathbf{y} = \begin{pmatrix} x_{11} & x_{12} \\ -x_{12} & x_{22} \end{pmatrix} \begin{pmatrix} y_{11} & y_{12} \\ -y_{12} & y_{22} \end{pmatrix} = \begin{pmatrix} x_{11}+y_{11} & x_{12}+y_{12} \\ -(x_{12}+y_{12}) & x_{22}+y_{22} \end{pmatrix} \in S$$

and a scalar multiple of any vector $\mathbf{x} \in S$ is

$$\alpha\mathbf{x} = \alpha \begin{pmatrix} x_{11} & x_{12} \\ -x_{12} & x_{22} \end{pmatrix} \begin{pmatrix} \alpha x_{11} & \alpha x_{12} \\ \alpha(-x_{12}) & \alpha x_{22} \end{pmatrix} \in S$$

so we conclude that the set of all 2×2 matrices where $a_{21} = -a_{12}$ *is a subspace* of $\mathbb{R}^{2\times 2}$.
■

[5] for problems involving different types of spaces, we would keep the descriptor 'vector', but we usually drop it if there is no ambiguity

■ **Example 6.4 — The set of all polynomials less than a given order.** Under the usual rules of addition and scalar multiplication for functions $\mathbf{f},\mathbf{g}$ defined on $\mathbb{R}$

$$\boxed{\mathbf{f}+\mathbf{g} = \sum_{k=0}^{n-1} f_k x^k + \sum_{k=0}^{n-1} g_k x^k = \sum_{k=0}^{n-1} (f_k+g_k)x^k \in V} \text{ and } \boxed{\alpha\mathbf{f} = \alpha\sum_{k=0}^{n-1} f_k x^k = \sum_{k=0}^{n-1} (\alpha f_k)x^k \in V}$$

so that the sum of any polynomials of order less than n is always a polynomial of order less than n (Rule 1), and a rescaled polynomial of order less than n is always a polynomial of order less than n (Rule 2). Thus, the set of all functions defined on $\mathbb{R}$ is a vector space and the set of all polynomials of order less than n is a vector subspace of it. ■

■ **Example 6.5 — The set of all continuous functions on a given interval.** This is an important result, but unfortunately establishing the closure rules here is beyond the scope of this course. We need to prove that the sum of any two continuous functions is also continuous, and that the scalar multiple of a continuous function is also continuous. These results seem reasonable, but to prove them requires more formal definitions of continuity than you have seen to this point. Armed with those definitions, we can show that the closure rules hold, and that the continuous functions form a subspace of the space of (real) functions. ■

6.2.2 A special subspace: the null space

Definition 6.2.2 — The Null Space of a Matrix. The null space of a matrix $A \in \mathbb{R}^{m\times n}$, denoted $N(A)$, is the set of all solutions to the homogeneous system $A\mathbf{x} = 0$, *i.e.*,

$$N(A) = \{\mathbf{x} \in \mathbb{R}^n : \ A\mathbf{x} = \mathbf{0}\}.$$

Proposition 6.2.1 $N(A)$ is a subspace of $\mathbb{R}^n$

Proof.

1. The zero solution always exists, so $N(A)$ is not empty
2. If $\mathbf{x} \in N(A)$ and $\mathbf{y} \in N(A)$, then $\mathbf{x}+\mathbf{y} \in N(A)$, since

$$A(\mathbf{x}+\mathbf{y}) = A\mathbf{x}+A\mathbf{y} = \mathbf{0}+\mathbf{0} = \mathbf{0}$$

3. If $\mathbf{x} \in N(A)$, then $\alpha\mathbf{x} \in N(A)$, since

$$A(\alpha\mathbf{x}) = \alpha A\mathbf{x} = \alpha\cdot\mathbf{0} = \mathbf{0}$$

■

To find the null space of a matrix, we solve the augmented system $(A|\mathbf{0})$.

■ **Example 6.6 — Finding the null space of a matrix.** To find the null space of $A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}$, we need to find the set of $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -2 & 1 & 0 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 \end{array}\right)$$

x_3 and x_4 are free variables, leading to

$$x_1 = x_3 - x_4$$
$$x_2 = -2x_3 + x_4$$

If we set $x_3 = \alpha$ and $x_4 = \beta$, then the general solution is

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} \alpha - \beta \\ -2\alpha + \beta \\ \alpha \\ \beta \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

■

Let's confirm that the null space of A we found above is indeed a subspace of $\mathbb{R}^4$. If we denote

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

then we can express $N(A) = \{\alpha\mathbf{v}_1 + \beta\mathbf{v}_2,\ \alpha, \beta \in \mathbb{R}\}$ in terms of basis vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Therefore $\mathbf{x}, \mathbf{y} \in N(A) \Rightarrow \mathbf{x} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2$ and $\mathbf{y} = y_1\mathbf{v}_1 + y_2\mathbf{v}_2$, so

$$\mathbf{x} + \mathbf{y} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \mathbf{v}_1 + y_2\mathbf{v}_2 = (x_1 + y_1)\mathbf{v}_1 + (x_2 + y_2)\mathbf{v}_2 \in N(A)$$

and

$$\alpha\mathbf{x} = \alpha(x_1\mathbf{v}_1 + x_2\mathbf{v}_2) = (\alpha x_1)\mathbf{v}_1 + (\alpha x_2)\mathbf{v}_2 \in N(A)$$

Therefore $N(A)$ is indeed a subspace, since it obeys the closure rules.

The null space plays an important role in linear problems, since it provides a way of finding *all* the solutions to the problem.

Proposition 6.2.2 If $\mathbf{y}$ is a single solution to the linear problem $A\mathbf{x} = \mathbf{b}$, then *the set of all solutions* is precisely the set $\{\mathbf{y} + \mathbf{x} : \mathbf{x} \in N(A)\}$.

Proof. To prove the equivalence of these statements, we first prove that if $\mathbf{y} + \mathbf{x}$, $\mathbf{x} \in N(A)$, then $\mathbf{y} + \mathbf{x}$ is indeed a solution. Then we prove that any solution must have this form.

First, since $\mathbf{y}$ is a solution, it follows that $A\mathbf{y} = \mathbf{b}$. Since $\mathbf{x} \in N(A)$, we know that $A\mathbf{x} = \mathbf{0}$. Therefore

$$A(\mathbf{y} + \mathbf{x}) = A\mathbf{y} + A\mathbf{x} = \mathbf{b} + \mathbf{0} = \mathbf{b}$$

so $\mathbf{y} + \mathbf{x}$ is a solution.

Second, if $\mathbf{z}$ is some other solution, then $A\mathbf{y} = A\mathbf{z} = \mathbf{b}$, and therefore

$$A(\mathbf{z} - \mathbf{y}) = A\mathbf{z} = A\mathbf{y} = \mathbf{b} - \mathbf{b} = \mathbf{0}$$

From the definition of the null space, this means that $(\mathbf{z} - \mathbf{y}) \in N(A)$. In other words, $\mathbf{z} - \mathbf{y} = \mathbf{x}$ for some $\mathbf{x} \in N(A)$, and therefore $\mathbf{z} = \mathbf{y} + \mathbf{x}$ for some $\mathbf{x} \in N(A)$. But the choice of $\mathbf{z}$ was arbitrary, meaning that this result must be true *for any solution* $\mathbf{z}$. ■

We can use this result to find the general solution to a linear problem: once we have the null space, we can generate all the solutions to a problem by finding one solution, and then adding on the null space of solutions. Let's look at some examples.

■ **Example 6.7 — Solving a linear problem using the null space** $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$**.** Notice that we can easily find a specific solution, since

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \Longrightarrow \quad \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

From Example 6.6 we can write the elements of the null space as

$$\alpha \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad \alpha, \beta \in \mathbb{R}$$

The full solution is therefore the sum of the specific solution and the elements of the null space:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad \alpha, \beta \in \mathbb{R}$$

■

■ **Example 6.8 — The set of all solutions of a linear, homogeneous ODE is a null space.** This is a more sophisticated example, also of great importance in finding the solutions to ODEs. Define V as the set of solutions: a subset of the space of functions. We can write a linear, homogeneous ODE in the form $L[f] = 0$, where L is a map of f that returns some linear combination of f and its derivatives (for example, $L[f] = f'' + 3f' + 2f$). As we have seen earlier, differentiation is a linear transformation, so $L[f]$ is a linear map of f. Any function that satisfies $L[f] = 0$ must be a solution to the ODE.

Suppose that $\mathbf{f}$ and $\mathbf{g}$ are known solutions to the ODE, so that $\mathbf{f} \in V$ and $\mathbf{g} \in V$. It follows that their sum must also be a solution, i.e. that $[\mathbf{f} + \mathbf{g}] \in V$, because

$$L[\mathbf{f} + \mathbf{g}] = L[\mathbf{f}] + L[\mathbf{g}] = 0 + 0 = 0$$

A scalar multiple of any solution must also be a solution — $[\alpha \mathbf{f}] \in V$ for any $\mathbf{f} \in V$ — because

$$L[\alpha \mathbf{f}] = \alpha L[\mathbf{f}] = \alpha \cdot 0 = 0$$

so we conclude that the set of all solutions of a linear, homogeneous ODE is a vector space in its own right, as a subspace of the space of functions.

How do we use this result in practice? To find the solution of $f'' + 3f' + f = g$ for some function g, we first find the solution to the homogenous equation, which gives us the null space $f_h \in N(L)$. Then we find a particular solution to the original problem f_p. Finally, we combine these to give the general solution $f = f_p + f_h$ for any $f_h \in N(L)$. You will have learnt about (and will continue to encounter) this procedure in other courses: the reasoning why it gives all possible solutions is tied up in the linear algebra theory of the null space. ■

6.3 Bases

We spent some time in Chapter 4 developing the idea of a basis of $\mathbb{R}^n$ — a set of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ for which every $\mathbf{x} \in \mathbb{R}^n$ can be represented by a unique linear combination of the $\mathbf{v}_i$. This idea is incredible useful — it allows us to represent any element in $\mathbb{R}^n$ as a list[6] of the coefficients $(c_1, \ldots, c_n)$ of the basis vectors $\mathbf{v}_i$. Even if we are dealing with a space of matrices or functions, we can still identify each member of the space by a list or vector of coefficients, which means we can consider any problem involving these objects as a problem of vectors in $\mathbb{R}^n$, and use all the ideas we've developed to this point.

But how do we define a basis for spaces involving other types of objects, like functions? In Chapter 4 we could use the dimension of $\mathbb{R}^n$ and our knowledge of the solution of linear problems $A\mathbf{x} = \mathbf{b}$ to characterise the requirements for basis vectors. It is perhaps not too difficult to see how we can extend this idea to $\mathbb{R}^{m\times n}$, but less clear how to proceed more generally (such as for subspaces of $\mathbb{R}^n$, or for function spaces). To get around this, we take a slightly different approach to the one we took in Chapter 4, but one that is closely connected with our earlier work.

6.3.1 Spanning

For a given list of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$, it is useful to be able to characterise the range of vectors that can be 'reached' through their linear combinations. We call this range the **span** of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Note that we can find the span of any list of vectors in a space — there are no special requirements (such as linear independence) for the vectors in the list.

Definition 6.3.1 — The span of a set of vectors. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ be vectors in a vector space V. The set of all *linear combinations*

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n, \qquad \alpha_i \in \mathbb{R}$$

is called the **span** of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, denoted $\text{Span}\,(\mathbf{v}_1, \ldots, \mathbf{v}_n)$

■ **Example 6.9** The null space of $A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}$ is

$$N(A) = \text{Span}\left(\begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}\right)$$

■

■ **Example 6.10** If we define

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

then

$$\mathbb{R}^3 = \text{Span}\,(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$$

■

[6] Or vector, in the traditional sense.

■ **Example 6.11** The set $\mathscr{P}_4$ of all polynomials of degree less than 4 is given be

$$\mathscr{P}_4 = \text{Span}\left(1, x, x^2, x^3\right)$$

since any polynomial in $\mathscr{P}_4$ is a linear combination $ax^3 + bx^2 + cx + d$. Note that not all of these polynomials are cubics: for example, if $a = 0$ but $b \neq 0$, we will have a quadratic but not a cubic. ■

■ **Example 6.12** The solutions of the ODE $f'' + 3f' + 2f = 0$ are

$$\text{Span}\left(e^{-x}, e^{-2x}\right)$$

■

Proposition 6.3.1 The span of a set of vectors $\mathbf{v}_i$ is a vector subspace

Proof. We won't give full details here, but the proof is straightforward. Each element of the span can be written as a linear combination of the $\mathbf{v}_i$, and so therefore can the sum of two such elements. Since the sum has this form, it is in the span as well, so addition is closed. A similar argument shows that scalar multiplication must be closed too. ■

Sometimes, rather than wanting to know what set can be 'reached' using a set of vectors, we want to know whether a set of vectors 'reaches' every point in a given set. We thus introduce the idea of a **spanning set** of vectors for the subset of a vector space.

Definition 6.3.2 — A spanning set of vectors. The set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ of vectors in V is a **spanning set** for V if and only if *every* vector in V can be written as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, *i.e.*,

$$\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n \text{ for every } \mathbf{v} \in V.$$

In this case, we can say that the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ *span* V.

6.3.2 Examples

■ **Example 6.13**

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

are a spanning set of the null space of $A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 0 \end{pmatrix}$ ■

■ **Example 6.14**

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

are a spanning set of $\mathbb{R}^3$ ■

■ **Example 6.15**

$$f_1(x) = e^{-x} \text{ and } f_2(x) = e^{-2x}$$

are a spanning set for the solutions of the ODE $f'' + 3f' + 2f = 0$. ■

How do we work out whether a set of vectors span a space V? We need a way of describing each vector in V, so that we can convert the question into a matrix problem to solve using the techniques we have developed so far in the course.

■ **Example 6.16 — Spanning set in $\mathbb{R}^3$.** Do the vectors

$$\mathbf{v}_1 = \begin{pmatrix}1\\2\\4\end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix}2\\1\\3\end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix}0\\0\\1\end{pmatrix},$$

span $\mathbb{R}^3$? What we are asking is, do $\alpha_1, \alpha_2, \alpha_3$ exist such that

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 = \begin{pmatrix}x\\y\\z\end{pmatrix} \quad \text{for any } x, y, z \in \mathbb{R} \text{ ?}$$

Now, this is

$$\begin{aligned}\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 &= \begin{pmatrix}| & | & |\\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3\\ | & | & |\end{pmatrix}\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix}\\ &= \begin{pmatrix}1 & 2 & 0\\2 & 1 & 0\\4 & 3 & 1\end{pmatrix}\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix} = A\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix} = \begin{pmatrix}x\\y\\z\end{pmatrix}\end{aligned}$$

If A^{-1} exists, there will be a (unique) $\alpha_1, \alpha_2, \alpha_3$ for any x, y, z; but if A^{-1} doesn't exist, the equations will be inconsistent for some $(x, y, z)^T$.

$$\det A = 1 \cdot (1-4) = -3 \neq 0 \quad \Rightarrow \quad A^{-1} \text{ exists}$$

and we conclude that the vectors $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ span $\mathbb{R}^3$. ■

■ **Example 6.17 — Spanning set in $\mathscr{P}_3$.** Do the vectors $\mathbf{v}_1 = 1 - x^2, \mathbf{v}_2 = x + 2, \mathbf{v}_3 = x^2$ span $\mathscr{P}_3$, the space of all polynomials of order less than 3? What we are asking is, do $\alpha_1, \alpha_2, \alpha_3$ exist such that $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 = ax^2 + bx + c$ for any $a, b, c \in \mathbb{R}$? Now,

$$\begin{aligned}\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 &= \alpha_1(1-x^2) + \alpha_2(x+2) + \alpha_3 x^2\\ &= (\alpha_3 - \alpha_1)x^2 + (\alpha_2)x + (\alpha_1 + 2\alpha_2)\end{aligned}$$

Therefore

$$\begin{pmatrix}-1 & 0 & 1\\0 & 1 & 0\\1 & 2 & 0\end{pmatrix}\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix} = A\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix} = \begin{pmatrix}a\\b\\c\end{pmatrix}$$

If A^{-1} exists, there will be a (unique) $\alpha_1, \alpha_2, \alpha_3$ for any x, y, z; but if A^{-1} doesn't exist, the equations will be inconsistent for some x, y, z.

$$\det A = 1 \cdot (1-0) = 1 \neq 0 \quad \Rightarrow \quad A^{-1} \text{ exists}$$

and we conclude that the vectors $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ (or equivalently, that the functions $1 - x^2$, $x + 2$, and x^2) span $\mathscr{P}_3$. ■

6.3.3 Definition of a basis

We are now able to give a definition for a basis that is equivalent to our previous one, but that is more practically useful for general vector spaces. Rather than stipulating a specific number of basis vectors, we require that our basis vectors span the space. In order to recover a unique representation for each member of the vector space in terms of the basis, we retain the condition of linear independence. This has the effect of producing a spanning set that doesn't contain any redundant members.

Definition 6.3.3 — Basis. The vectors $\mathbf{v}_1,\dots,\mathbf{v}_n$ in vector space V form a *basis* for V if and only if

1. $\mathbf{v}_1,\dots,\mathbf{v}_n$ are linearly independent
2. $\mathbf{v}_1,\dots,\mathbf{v}_n$ span V

■ **Example 6.18** The *standard basis* for $\mathbb{R}^3$ is

$$\left\{\mathbf{e}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix}0\\1\\0\end{pmatrix}, \mathbf{e}_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$$

but we could also choose $\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}1\\1\\0\end{pmatrix}, \begin{pmatrix}1\\1\\1\end{pmatrix}\right\}$ or $\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}, \begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}2\\0\\1\end{pmatrix}\right\}$.
Two vectors, *e.g.*, $\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}, \begin{pmatrix}1\\0\\1\end{pmatrix}\right\}$ will not be enough to span $\mathbb{R}^3$, and four vectors, *e.g.*, $\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}, \begin{pmatrix}1\\0\\1\end{pmatrix}\right\}$ cannot be linearly independent. ■

■ **Example 6.19** The linearly independent vectors

$$\left\{\begin{pmatrix}1\\-2\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\1\\0\\1\end{pmatrix}\right\}$$

form a basis for the null space of $\begin{pmatrix}1&1&1&0\\2&1&0&1\end{pmatrix}$. ■

■ **Example 6.20** The matrices

$$\left\{E_{11} = \begin{pmatrix}1&0\\0&0\end{pmatrix}, E_{12} = \begin{pmatrix}0&1\\0&0\end{pmatrix}, E_{21} = \begin{pmatrix}0&0\\1&0\end{pmatrix}, E_{22} = \begin{pmatrix}0&0\\0&1\end{pmatrix}\right\}$$

form *the standard basis* for $\mathbb{R}^{2\times 2}$. ■

■ **Example 6.21** The polynomial terms

$$\{1, x, x^2, \dots, x^{n-1}\}$$

form *the standard basis* for the space $\mathscr{P}_n$ of all polynomials of order $< n$. ■

6.3.4 Important results relating to bases

Proposition 6.3.2 If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ spans V, then any set $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ with more than n vectors is linearly dependent.

Proof. Express each $\mathbf{u}_j$ as a linear combination of the $\mathbf{v}_i$; solve the system of equations for a linear combination to be zero; there are n equations and $> n$ unknowns; the system is homogeneous, so a family of solutions exists, so the larger set must be linearly dependent. ■

Proposition 6.3.3 If both $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ are bases for V, then $n = m$.

Proof. The proof follows from the above: if $m > n$ the $\mathbf{u}_i$ must be linearly dependent, and if $n > m$ the $\mathbf{v}_i$ must be linearly independent. Neither is the case, leaving only $m = n$. ■

We used these two results for point spaces $\mathbb{R}^n$ when we developed our definition for a basis in Chapter 4, from which we concluded that we needed n linearly independent vectors to form a basis.

Proposition 6.3.4 If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for V, each $\mathbf{v} \in V$ is a unique linear combination of $\mathbf{v}_i$.

Proof. If it isn't true, take the difference of the two representations, which must be a linear combination of the $\mathbf{v}_i$ summing to zero. Since the $\mathbf{v}_i$ form a basis, the difference in the corresponding coefficients must be zero, meaning the two representations must be the same. ■

6.3.5 Dimension

In Chapter 4 we started using the term 'dimension' in relation to the point space $\mathbb{R}^n$, as a way of referring to the number n of components in the vector. This number-of-components definition is fine for entire point spaces, but needs refining for subspaces, and for function spaces which you might not intuitively think of as occupying a space of some dimension. Here we present the formal definition of the dimension of a vector space, which (unsurprisingly) is consistent with our intuitive understanding of the dimension of $\mathbb{R}^n$.

The key idea that we use to define the dimension of a (sub)space is the result of Proposition 6.3.3 — any basis for a vector space must contain the same number of vectors. This number is independent of the particular choice of basis, so represents a characteristic value for the vector space.

> **Definition 6.3.4 — Dimension of a vector space.** The **dimension** of a vector space is *the number of vectors in any of its bases.*

■ **Example 6.22** What is the dimension of $\mathbb{R}^3$? It has a basis

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\},$$

so it has dimension 3. This is consistent with our intuition that the dimension of $\mathbb{R}^n$ is the number of components of the vector, since each component has its own basis vector $\mathbf{e}_i$. ■

■ **Example 6.23** What is the dimension of $\mathbb{R}^{2\times 2}$? It has a basis

$$\left\{\begin{pmatrix}1 & 0\\0 & 0\end{pmatrix},\begin{pmatrix}0 & 1\\0 & 0\end{pmatrix},\begin{pmatrix}0 & 0\\1 & 0\end{pmatrix},\begin{pmatrix}0 & 0\\0 & 1\end{pmatrix}\right\},$$

so it has dimension 4. ■

■ **Example 6.24** What is the dimension of $\mathscr{P}_4$ (polynomials of degree ≤ 3)? It has a basis $\{1,x,x^2,x^3\}$, so it has dimension 4. ■

■ **Example 6.25** What is the dimension of $\mathscr{C}[0,1]$ (continuous functions defined for $0\leq x\leq 1$)? It has a basis $\{1,x,x^2,\ldots,x^k,\ldots\}$, so it has infinite dimension. ■

Proposition 6.3.5 If V is a vector with dimension $n>0$, then

1. any set of n linearly independent vectors spans V,
2. any n vectors that span V are linearly independent,
3. no fewer than n vectors can span V ,
4. any set of $<n$ vectors can be extended to form a basis for V,
5. any spanning set containing $>n$ vectors can be reduced to form a basis for V.

6.3.6 Changing basis

Examples 6.16 and 6.17 also shows us how we can change basis. We have seen this already for changing bases in the point space $\mathbb{R}^n$ — the same approach applies for other spaces such as $\mathscr{P}_n$.

Building on Example 6.17, how can we write x^2+2 in terms of the basis

$$\mathbf{v}_1=1-x^2,\mathbf{v}_2=x+2,\mathbf{v}_3=x^2?$$

From that exercise we know that a solution must exist, and that it must be unique (both because of the non-zero determinant in that exercise). So how do we find c_1,c_2,c_3 such that $c_1\mathbf{v}_1+c_2\mathbf{v}_2+c_3\mathbf{v}_3=x^2+2$?

Using the working from that problem, we collect terms to obtain

$$\begin{aligned}c_1\mathbf{v}_1+c_2\mathbf{v}_2+c_3\mathbf{v}_3&=c_1(1-x^2)+c_2(x+2)+c_3x^2\\&=(c_3-c_1)x^2+(c_2)x+(c_1+2c_2)=x^2+2\end{aligned}$$

Equating the coefficients of each term gives us the three equations we need to determine the c_i. For more complicated problems, we can turn to Gaussian elimination or matrix inversion, but here we can solve directly:

- equating the linear (x) term coefficients gives us $c_2=0$
- equating the constant term coefficients gives us $c_1+2c_2=2 \implies c_1=2$
- finally, equating the quadratic term coefficients gives us $c_3=c_1=1 \implies c_3=3$.

We therefore deduce that

$$x^2+2=2\mathbf{v}_1+3\mathbf{v}_3,$$

which we can quickly confirm:

$$2\mathbf{v}_1+3\mathbf{v}_3=2(1-x^2)+3(x^2)=2+x^2. \qquad ✔$$

6.4 Column and row spaces

Armed with our new terminology and understanding, let's return to the matrix problem $A\mathbf{x} = \mathbf{b}$, and see how we can characterise the solutions of this problem.

6.4.1 Definition of the column and row spaces

First, we will introduce two spaces associated with the columns and rows of a matrix A, whose properties are closely connected with the properties of the solutions $A\mathbf{x} = \mathbf{b}$.

Definition 6.4.1 — Column and Row Spaces. The **column space** of a matrix $A \in \mathbb{R}^{m\times n}$ is the subspace of $\mathbb{R}^m$ spanned by the columns of A, and the **row space** of a matrix $A \in \mathbb{R}^{m\times n}$ is the subspace of $\mathbb{R}^n$ spanned by the rows of A.

Equivalently, we can define the column space as the set of all vectors obtainable when *post*-multiplying by a column vector:

$$\text{column space of } A = \{\mathbf{y} : \mathbf{y} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\},$$

and the row space as the set of all vectors obtainable when *pre*-multiplying by a row vector:

$$\text{row space of } A = \{\vec{\mathbf{y}} : \vec{\mathbf{y}} = \vec{\mathbf{x}}A \text{ for some } \vec{\mathbf{x}} \in \mathbb{R}^m\}.$$

■ **Example 6.26** For the matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$, the column space is the set of all vectors of the form

$$\alpha_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha_1 + \alpha_3 \\ \alpha_2 + \alpha_3 \end{pmatrix}$$

and the row space is the set of all vectors of the form

$$\alpha_1 \begin{pmatrix} 1 & 0 & 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_1 + \alpha_2 \end{pmatrix}.$$

■

Now let's consider some important results related to column spaces:

Proposition 6.4.1 A linear system $A\mathbf{x} = \mathbf{b}$ is consistent iff $\mathbf{b}$ is in the column space of A, since

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

and therefore $\mathbf{b}$ can only be reached if it is a linear combination of the $\mathbf{a}_i$. We saw this result already back in proposition 2.3.1.

Proposition 6.4.2 A linear system $A\mathbf{x} = \mathbf{b}$ ($A \in \mathbb{R}^{m\times n}$) is consistent for any $\mathbf{b}$ iff the column space of A is $\mathbb{R}^m$. This follows immediately from the previous point, where now the linear combination of the $\vec{\mathbf{a}}_i$ must reach all of $\mathbb{R}^m$.

Proposition 6.4.3 An $n \times n$ matrix is invertible iff the column vectors of A form a basis in $\mathbb{R}^n$. This follows because A being invertible is equivalent to there being a unique solution to $A\mathbf{x} = \mathbf{b}\ \forall\ \mathbf{b}$. Following the above, the $\vec{\mathbf{a}}_i$ must reach all of $\mathbb{R}^n$.

6.4.2 Orthogonality of spaces

We can generalise the concept orthogonality to vector spaces in a straightforward manner – a pair of vector spaces are orthogonal if any element of one space is orthogonal to any element of the second space, *i.e.*, spaces V and W are orthogonal if

$$\mathbf{x}\cdot\mathbf{y} = 0\ \forall \mathbf{x} \in V \text{ and } \mathbf{y} \in W.$$

From this definition, we see that *the null space and row space of a matrix are orthogonal.* This is because

$$A\mathbf{x} = \mathbf{0} \Rightarrow \begin{pmatrix} \text{——} & \vec{\mathbf{a}}_1 & \text{——} \\ \text{——} & \vec{\mathbf{a}}_1 & \text{——} \\ & \vdots & \end{pmatrix} \mathbf{x} = \begin{pmatrix} \vec{\mathbf{a}}_1\mathbf{x} \\ \vec{\mathbf{a}}_2\mathbf{x} \\ \vdots \end{pmatrix} = \mathbf{0} \Rightarrow \vec{\mathbf{a}}_j\mathbf{x} = 0\ \forall j.$$

Similarly, *the column space of A is orthogonal to the null space of A^{T}.*

6.4.3 Rank and nullity

As we saw above, the column space of A is strongly connected with the nature of solutions to the problem $A\mathbf{x} = \mathbf{b}$. We can make quite specific claims about whether we will obtain unique solutions, families of solutions, or no solutions, directly from the dimensions of these spaces.

The two key quantities we consider are the **rank** and the **nullity**.

Definition 6.4.2 — Rank. The **rank** of a matrix
- is the dimension of its row space.
- is, equivalently, the dimension of its column space.
- equals the number of non-zero rows in its reduced-row echelon form
- equals the number of lead variables in its reduced-row echelon form

Definition 6.4.3 — Nullity. The **nullity** of a matrix is
- is the dimension of its null space.
- equals the number of free variables in its reduced-row echelon form

The rank and nullity are related by the following theorem:

Theorem 6.4.4 — Rank–nullity theorem. If A is an $m \times n$ matrix, then

$$\text{the rank of } A \ + \ \text{the nullity of } A \ = \ n.$$

Proof. The number of columns in the row-reduced matrix is n. They are either free variables (nullity) or lead variables (rank). ■

6.4.4 Characterising solutions using column and row spaces

■ **Example 6.27** Find bases for the row space, column space, and null space of the matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}$$

and give their dimensions. Does $A\mathbf{x} = \mathbf{b}$ have a solution for any $\mathbf{b}$?

1. First, row reduce: $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \to \cdots \to \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \end{pmatrix}$

2. *Use lead variable columns from A* for the column space:

$$\text{Column space has basis } \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}, \text{ has dimension } 2, \therefore \text{ rank} = 2$$

3. *Use all non-zero rows from reduced A* for the row space (check dimension=rank):

$$\text{Row space has basis } \left\{ \begin{pmatrix} 1 & 0 & -1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 2 & -1 \end{pmatrix} \right\}, \text{ has dimension } 2$$

4. Calculate the null space as shown earlier (check rank+nullity = #columns of A) :

$$\text{Null space has basis } \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\}, \therefore \text{ has dimension } 2.$$

5. Does $A\mathbf{x} = \mathbf{b}$ have a solution for any $\mathbf{b}$? From the problem, $\mathbf{b} \in \mathbb{R}^2$. The column space has dimension 2 (rank is 2), so the range of values of $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has a solution has dimension 2. This means that any $\mathbf{b} \in \mathbb{R}^2$ will have a solution. Furthermore, because the nullity is 2, there will be a (two-dimensional) family of solutions for any $\mathbf{b} \in \mathbb{R}^2$.

■

■ **Example 6.28** Find bases for the row space, column space, and null space of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

and give their dimensions. Does $A\mathbf{x} = \mathbf{b}$ have a solution for any $\mathbf{b}$?

1. First, row reduce: $\begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \cdots \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

2. *Use lead variable columns from A* for the column space:

$$\text{Column space has basis } \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right\}, \text{ has dimension } 2, \therefore \text{ rank} = 2$$

3. *Use all non-zero rows from reduced A* for the row space (check dimension=rank):

$$\text{Row space has basis } \left\{ \begin{pmatrix} 1 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 1 \end{pmatrix} \right\}, \text{ has dimension } 2$$

4. Calculate the null space as shown earlier (check rank+nullity = #columns of A) :

$$\text{Null space has basis } \left\{ \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}, \text{ has dimension } 1$$

5. Does $A\mathbf{x} = \mathbf{b}$ have a solution for any $\mathbf{b}$? From the problem, $\mathbf{b} \in \mathbb{R}^3$. The column space has dimension 2 (rank is 2), so the range of values of $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has a solution has dimension 2. This means that some $\mathbf{b} \in \mathbb{R}^3$ must lie outside of the column space of A, and will *not* have a solution. Because the nullity is 1, there will be a (one-dimensional) family of solutions for any $\mathbf{b} \in \mathbb{R}^3$ in the column space of A.

■

There is often some confusion regarding how to construct the basis for the row and column vectors. For the column space, any of the columns of A could be in the basis, but we need a systematic way to identify a linearly independent subset. Choosing columns corresponding to lead variables provides such a system[7]. We *shouldn't* choose these columns from the row-reduced matrix, since the row-reduction process adds and re-orders rows. If the rank is the number of rows then the column space will be all of $\mathbb{R}^n$, in which case this mistake won't have any consequence. If the rank is less than this, choosing the columns from the row-reduced matrix will only be correct by chance!

For the row space, the row reduction process effectively generates a basis for us. Any row of zeros arises because we have created a zero linear combination of rows, and in the process we remove one of the rows that was a linear combination of the others. If there are no non-zero rows, then all the rows are linearly independent, so the original rows can be used. If there *are* non-zero rows, then the original rows are not linearly independent so do not form a basis for the row space. We also cannot choose the rows with lead variables from the original matrix, again because the row-reduction process re-orders the rows.

For these reasons, we must choose the column space from lead variable columns in the **original** *matrix, but we must choose the row space from the non-zero rows of the* **reduced** *matrix.*

6.5 Application: introduction to coding theory

The null space may seem like an obscure property of a matrix, but it plays a valuable role in some quite important applications of linear algebra. Recall that the null space $N(A)$ of matrix $A \in \mathbb{R}^{m\times n}$ is defined as

$$N(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

The null space helps us find all solutions to a given linear problem; if we have any single particular solution, all the other solutions are obtained by adding elements from the null space.

- For solutions to matrix problems, *e.g.*, $A\mathbf{x} = \mathbf{b}$:
  1. find the null space (set of homogeneous solutions) $N(A)$;
  2. find a particular solution $\mathbf{x}_p$;
  3. then $\mathbf{x}_p + \mathbf{x}_h$ is a solution for any $\mathbf{x}_h \in N(A)$.
- For solutions to differential equations, *e.g.*, $\mathscr{L}[f] = f'' - 3f' + 2f = \sin(t)$
  1. find the null space (set of homogeneous solutions) $N(\mathscr{L})$;
  2. find a particular solution $f_p(t)$;
  3. then $f_p(t) + f_h(t)$ is a solution for any $f_h(t) \in N(\mathscr{L})$.

Beyond this, null spaces play a key role in *coding theory*. Computers store and transmit data in 1s and 0s. We will consider such data as *binary numbers* – our data will be elements of a vector space of $\{0,1\}$ rather than $\mathbb{R}$. We use standard algebraic rules for $\{0,1\}$, viz

$$0+0=0,\ 0+1=1,\ 1+0=1,\ 1+1=0 \qquad \text{[XOR]}$$

[7]The rationale behind this is that we lock in a specific solution for a given $\mathbf{b}$ by first choosing the free variables, which determines the lead variables. We are therefore allowed to set the free variables to 0, which tells us that the lead-variable columns provide a basis for the column space.

$$0 \times 0 = 0,\ 0 \times 1 = 0,\ 1 \times 0 = 0,\ 1 \times 1 = 1 \qquad \text{[AND]},$$

which we may think of as our usual operations modulo 2.

Sometimes, our data will become corrupted. In this example, we consider the simple case where one binary digit ('bit') gets flipped ($0 \to 1$ or $1 \to 0$). How could we detect and correct such errors?

Let's think about a four-bit message being sent between computers. This could encode any number between 0 and 127, inclusive, *inter alia*. We are going to extend our message to seven bits, giving us three bits to use for error detection purposes.

Now consider the matrix

$$H = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

whose columns $\mathbf{c}_i$ are the $2^3 - 1 = 7$ non-zero 3D binary vectors[8]. H acts on the set of binary vectors with 7 elements ($\{0,1\}^7$), which contains seven basis vectors denoted $\mathbf{e}_i$. Now, we note the following useful properties of H.

1. If $H\mathbf{v} = \mathbf{0}$, then $H(\mathbf{v} + \mathbf{e}_i) = \mathbf{c}_i \neq \mathbf{0}$ for any of the basis vectors (this is because none of the columns of H are all zeroes);
2. $H(\mathbf{e}_i - \mathbf{e}_j) = \mathbf{c}_i - \mathbf{c}_j \neq \mathbf{0}$ for any of the basis vectors ($i \neq j$);
3. therefore, if $H\mathbf{v} = \mathbf{c}_j$, then $H(\mathbf{v} + \mathbf{e}_i) = \mathbf{0}$ only if $i = j$.

Alternatively, we can write these conditions as

$$\mathbf{v} \in N(H) \Rightarrow \mathbf{v} + \mathbf{e}_i \notin N(H); \qquad \mathbf{e}_i - \mathbf{e}_j \notin N(H)\ (i \neq j); \qquad H\mathbf{v} = \mathbf{c}_j \Rightarrow \mathbf{v} + \mathbf{e}_j \in N(H).$$

In short, if we encode our message so that it lies in the nullspace of H then we can always detect any bit-flips, which result in a $\mathbf{v}$ that lies outside $N(H)$!

Firstly, we find the null space of H; solving $H\mathbf{x} = \mathbf{0}$ *via* Gaussian elimination yields

$$N(H) = \operatorname{Span}\left(\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right).$$

Note that the nullity is 4, the size of the message we want to encode. For convenience, we write these basis vectors as rows of a matrix G:

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

[8] Elements of $\{0,1\}^3$, *i.e.*, the set of all vectors with three elements with values of either 0 or 1. Reading down each column, we see that the first column is the binary representation of one, the second is two in binary, third is three, ..., and the nth column is n in binary.

The matrix G gives us our means of encoding our message such that it is guaranteed to be in $N(H)$. If $\vec{\mathbf{a}} = [a_1a_2a_3a_4]$ is our four-bit message, then our seven-bit encoding is

$$\mathbf{v} = (\vec{\mathbf{a}}G)^{\mathrm{T}} = \left\{ [a_1a_2a_3a_4] \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix} \right\}^{\mathrm{T}} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_2+a_3+a_4 \\ a_1+a_3+a_4 \\ a_1+a_2+a_4 \end{pmatrix}.$$

We observe that the first four digits of $\mathbf{v}$ should be $\vec{\mathbf{a}}$, and by construction, $\mathbf{v}$ should be in the null space of $N(H)$. Therefore, if any single digit in the message becomes incorrect, $H\mathbf{v} = c_j$, identifying the incorrect jth digit! In this instance (considering only single-bit errors) it is then simple to rectify the issue – just add 1 to the jth digit.

If we receive $\mathbf{v} = 0101010$, then

$$H\mathbf{v} = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

so the message is *correct*, and $\vec{\mathbf{a}} = 0101$. If we receive $\mathbf{v} = 0111010$, then

$$H\mathbf{v} = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

so the 011 (third) digit is *incorrect*, and $\vec{\mathbf{a}} = 0101$.

We can make this system more efficient with more rows in H. For example, 4 rows gives $2^4 - 1 = 15$ columns, so 11-digit messages (because $15 - 4 = 11$). But this system only guarantees catching single errors, so we need to extend to more sophisticated error-detecting/correcting methods for more troublesome errors (such as higher error rates). Things become even more complicated when trying to detect and correct errors in quantum bits[9].

One application of this null-space method is the *Luhn algorithm*, used for credit-card numbers (and some bank account numbers). This algorithm will pick up single-digit transcription/neighbour swapping/some doubling errors. The Luhn algorithm is applied as follows:

[9]Rather than being 0 or 1, a qubit (quantum bit, pronounced ‘Q bit’) can take on any value between 0 and 1, and also has an associated ‘phase’ between 0 and 2π. Both quantities must be error checked. This is further exacerbated by the properties of quantum measurement and entanglement – more on this if you study quantum mechanics!

1. Take the first 15 digits $d_1 d_2 d_3 d_4 \dots d_{14} d_{15}$
2. Calculate $s_1 = 2d_1, s_2 = d_2, s_3 = 2d_3, s_4 = d_4, \dots, s_{14} = d_{14}, s_{15} = 2d_{15}$
3. Calculate S, the sum all the digits in s_1 to s_{15} [= sum base 9]
4. The final credit card digit $d_{16} = 10 -$ [the final digit in S]

This algorithm can be used to validate Australian credit cards and Go-cards *inter alia*.

7. Factorisations

So far, we have already looked at a number of processes that involve **matrix factorisation**, also called **matrix decomposition**. Some of these have been explicit factorisations, such as matrix diagonalisation – others have been implicit, such as row reduction. Much like with real numbers, factorisation is an important part of many applications involving matrices. Perhaps differently from factorisation with real numbers, factorisation of matrices can involve a number of substantively different approaches, each with their own particular advantages and applications. In this chapter, we will motivate the benefits of factorisation, and then explore a few key approaches to matrix factorisation that we will rely on as we turn out attention to various linear algebra applications.

7.1 Why factorisation?

We use factorisation commonly when performing calculations with real numbers:

- When we add fractions, we need to find the lowest common multiple of all their denominators, to avoid unnecessary multiplication and then cancellation at the end. An efficient way to find the lowest common multiple of a set of numbers is to write out their **prime factorisation**, where we write each number as a product of powers of prime numbers only. The lowest common multiple will be the product of the highest power of each prime that appears.
- The process of cancelling fractions can be achieved by identifying the highest common factor of the numerator and denominator, and then dividing each by this highest common factor. Again, an efficient way to find the highest common factor among a set of numbers starts by writing out their prime factorisations. The highest common factor will be the product of the lowest power of each prime that appears (so the contribution will be 1 if the number doesn't appear in each number).
- This process can allow us to break a more complicated multiplication or division into easier stages. For example, try to calculate $1554 \div 42$ in your head – not so

easy! But once you remember that $42 = 6 \times 7 = 2 \times 3 \times 7$, we can first calculate $1554 \div 2 = 777$, then find $777 \div 7 = 111$, and finally $111 \div 3 = 37$. This step-by-step approach is a common process we follow in practice, when cancelling fractions.

Can we perform each of these processes without prime factorisation? Yes, we can, using long-hand procedures like long division, or a computer. However, prime factorisation allows us to do these processes efficiently, converts a complex task into a sequence of easier, more manageable steps, and allows more insights into what is going on along the way. They are often what lies behind the computational approach to solving the problem. Note also that we might not always use factorisation into primes: for example, we wouldn't calculate $2000000 \div 10000$ using prime factorisation – here we would work with factors of 10 (or, equivalently, consider the factorisation required to express a number in scientific notation) to quickly get to the answer.

This is the same situation with matrix factorisation. There are a range of different factorisations, each suited to different problems. These factorisations convert a particular problems into a sequence of more manageable, easily performed steps. Many of those problems can nowadays be solved by turning to software, but the software that solves the problem typically uses the factorisations we will be looking at[1]. Furthermore, these factorisations can have properties that lead to applications beyond the first one that led to their introduction, and so can lead to a number of other uses in a range of applications – as we shall see in coming weeks.

7.1.1 Which factorisations will we consider?

So far, we have already considered some factorisations, whether we've explored them explicitly as factorisations or not. Some of these factorisations have more general forms – for example, applying beyond the limit of only square matrices – which make them much more broadly applicable approaches. Below, we will describe the main factorisations we will consider in this chapter.

These factorisations involve certain types of matrices, which are usually given standard symbols (often appearing in the name of the factorisation approach). These include:

- The matrix to be factorised, usually called A. Any special properties will be noted in the description of the factorisation (such as being square or positive-definite).
- **Upper triangular matrices**, denoted U. This is any matrix where the only non-zero values appear on or above the main diagonal, *e.g.*,

$$\begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & 3 & 2 \\ 0 & 0 & 4 \\ 0 & 0 & -2 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & 3 & 2 & 1 \\ 0 & 0 & 4 & 3 \\ 0 & 0 & -1 & -2 \end{pmatrix}.$$

  It's important to realise that U might not be a square matrix (it depends on context).
- **Lower triangular matrices**, denoted L. This is a matrix where the only non-zero values appear on or below the main diagonal, *e.g.*,

$$\begin{pmatrix} 4 & 0 \\ -1 & 2 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & 0 & 0 \\ -1 & 3 & 0 \\ 2 & 1 & -1 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & 0 & 0 \\ -1 & 3 & 0 \\ 2 & 1 & -1 \\ 5 & 2 & 1 \end{pmatrix}.$$

[1] Or variations based on them.

As with the upper triangular matrices, L might not be a square matrix, depending on context.

- **Diagonal matrices**, denoted D. This is a matrix whose only non-zero values appear on the main diagonal, *e.g.*,

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix}.$$

  Once again, D does not have to be a square matrix.
- **Orthogonal matrices**, denoted Q. Recall, these are square matrices whose columns are a set of orthonormal vectors, so $Q^{\mathrm{T}}Q = I$.

There are others which we'll define as we come to them, but these are the main ones. Let's now list the main factorisations we will consider at in this chapter.

- **LU**: A factorisation of the form $A = LU$. This factorisation is closely associated with the processes Gaussian elimination. In general, a matrix might not have an LU-decomposition, so we will have to consider some variations on this theme to guarantee an associated factorisation, and to guarantee that it is unique.
- **QU/QR**: A factorisation of the form[2] $A = QU$. This factorisation represents the columns of matrix A as linear combinations of the orthonormal basis vectors forming the columns of Q. It has a close connection with the least-squares problem (finding the line of best fit through data)
- **Cholesky**[3]: A special case of the above arises for positive-definite symmetric matrices, with a factorisation of the form $A = U^{\mathrm{T}}U = LL^{\mathrm{T}}$ since the transpose of an upper triangular matrix must be lower triangular.

Diagonalisation is an incredibly useful factorisation, because it allows us to extend the use the Taylor series from functions of real numbers to functions of matrices. We can then extend our use of these real numbers to analogous problems involving matrices, such as in finding solutions to differential equations. Because of the connection with **spectral** properties of a matrix – its eigenvalues and eigenvectors – diagonalisation provides an important starting point for approximating matrices, as we will see. However, diagonalisation in the form we have seen required square matrices that have as many eigenvectors as they have rows or columns. There are two factorisations that allow us to move beyond these constraints in generating diagonalised representations: the first is quite closely related to the approach we have already seen; the second has some similarities, but also some fundamental differences:

- **JCF**: The Jordan canonical form (JCF) (also known as Jordan normal form) extends diagonalisation to square matrices that do not have as many eigenvectors as they have rows or columns. The JCF produces a representation $A = VJV^{-1}$, where the columns of V comprise all the eigenvectors of A, and its *pseudo-eigenvectors*, and J is a matrix with a block-diagonal form. This completes the diagonalisation approach we started in chapter 4, for any square matrix. However, the Jordan canonical form

[2] R is an common alternative symbol used to denote upper-triangular matrices, so this is often called QR factorisation.

[3] Pronounced 'shaLESkie'.

is numerically unstable, because a small change in any matrix element will typically have a significant impact on the number of eigenvectors that matrix has. It is a very useful construct for understanding certain problems, however.

- **SVD**: The singular value decomposition (SVD) allows us to extend the benefits of diagonalisation to arbitrary matrices, with the representation $A = Q_1 \Sigma Q_2{}^{\mathrm{T}}$ where the Q_i are appropriately-sized orthogonal matrices, and Σ is a diagonal matrix[4]. This representation is related to the spectral properties of $A^{\mathrm{T}}A$ and AA^{T}, which must be real symmetric matrices.

7.2 LU factorisation

Back at the start of the course, we were interested in solving systems of equations with matrix form $A\mathbf{x} = \mathbf{b}$. We quickly came to realise that we needed a systematic approach that could accommodate complicated systems of equations, where there might be a family of solutions, or no solution.

The systematic approach we adopted was row reduction.

1. We wanted to perform a series of operations on the rows of the augmented matrix $[A|\mathbf{b}]$ that had the same solution(s) $\mathbf{x}$.
2. Each operation was represented by pre-multiplication by an elementary matrix E_i. After k steps, the augmented matrix became
$$[E_k \dots E_1 A | E_k \dots E_1 \mathbf{b}]$$
3. Our objective was to convert A in this fashion into an upper triangular matrix U, so that the augmented matrix became $[U|\mathbf{r}]$.

Why did we target a row-reduced form with upper triangular matrix U? Because, from this point, obtaining the solution to the original problem is simple. For systems with a unique solution, this leads to the equivalent problem $U\mathbf{x} = \mathbf{r}$, which we could easily solve using back-substitution.

Example 7.1 — Back-substitution with a unique solution (from Example 1.6).

$$\begin{aligned} 3x_1 + 2x_2 - x_3 &= -2 \\ 3x_1 + x_2 - x_3 &= -5 \\ 3x_1 + 2x_2 + x_3 &= 2 \end{aligned} \quad \Rightarrow \quad \left(\begin{array}{ccc|c} 3 & 2 & -1 & -2 \\ 3 & 1 & -1 & -5 \\ 3 & 2 & 1 & 2 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 3 & 2 & -1 & -2 \\ 0 & -1 & 0 & -3 \\ 0 & 0 & 2 & 4 \end{array}\right)$$

Back-substitution gives

$$\begin{aligned} 2x_3 = 4 &\quad \Rightarrow \quad x_3 = 2 \\ -x_2 = -3 &\quad \Rightarrow \quad x_2 = 3 \\ 3x_1 + 2x_2 - x_3 = 3x_1 + 6 - 2 = -2 &\quad \Rightarrow \quad x_1 = -2 \end{aligned}$$

■

For systems with a family of solutions, we could stipulate free variables – the problem also reduced to straight-forward back-substitution, with unique values emerging if values for any free variables were provided.

[4]We could use the symbol D here, but traditionally, and for reasons we will see, the symbol Σ is used instead.

■ Example 7.2 — Back-substitution with free variables.

$$\left.\begin{array}{rl} 3x_1+2x_2-x_3+2x_4 & =-2 \\ 3x_1+x_2-x_3+x_4 & =-5 \\ 3x_1+2x_2+x_3+2x_4 & =2 \end{array}\right\} \Rightarrow \left(\begin{array}{cccc|c} 3 & 2 & -1 & 2 & -2 \\ 3 & 1 & -1 & 1 & -5 \\ 3 & 2 & 1 & 2 & 2 \end{array}\right)$$

$$\Rightarrow \left(\begin{array}{cccc|c} 3 & 2 & -1 & 2 & -2 \\ 0 & -1 & 0 & -1 & -3 \\ 0 & 0 & 2 & 0 & 4 \end{array}\right)$$

Back-substitution gives

$$\begin{aligned} 2x_3 = 4 \quad &\Rightarrow \quad x_3 = 2 \\ -x_2 - x_4 = -3 \quad &\Rightarrow \quad x_2 = 3 - x_4 \\ 3x_1+2x_2-x_3+2x_4 = 3x_1+6-2 = -2 \quad &\Rightarrow \quad x_1 = -2 \end{aligned}$$

■

And, as we saw in Chapter 1, we could quickly identify systems that would have no solution using this approach.

If it takes m steps to reduce the augmented matrix to row-echelon form, the augmented matrix becomes

$$\begin{aligned} [E_m \dots E_1 A | E_m \dots E_1 \mathbf{b}] = [U|\mathbf{r}] &\implies [EA|E\mathbf{b}] = [U|\mathbf{r}] \\ &\implies EA = U \\ &\implies A = E^{-1}U \end{aligned}$$

where we have defined $E = E_m \dots E_1$, and where we know E is invertible because of the properties of the elementary matrices E_i (which are themselves all invertible). If E^{-1} was guaranteed to be lower-triangular, then we have shown that A is guaranteed to have an LU factorisation. Unfortunately, however, *this is not the case*!

To understand why, let's consider what the equation $EA = U$ represents, by writing it in the block form.

$$U = EA = E \begin{pmatrix} \text{——} & \vec{\mathbf{a}}_1 & \text{——} \\ \text{——} & \vec{\mathbf{a}}_2 & \text{——} \\ & \vdots & \\ \text{——} & \vec{\mathbf{a}}_n & \text{——} \end{pmatrix} = \begin{pmatrix} E_{11}\vec{\mathbf{a}}_1 + E_{12}\vec{\mathbf{a}}_2 + \cdots + E_{1n}\vec{\mathbf{a}}_n \\ E_{21}\vec{\mathbf{a}}_1 + E_{22}\vec{\mathbf{a}}_2 + \cdots + E_{2n}\vec{\mathbf{a}}_n \\ \vdots \\ E_{n1}\vec{\mathbf{a}}_1 + E_{n2}\vec{\mathbf{a}}_2 + \cdots + E_{nn}\vec{\mathbf{a}}_n \end{pmatrix}.$$

That is, the elements of E are the coefficients describing how the original rows of A must be combined in order to obtain the row-reduced form U. Now, if $A = E^{-1}U$ produces an LU factorisation, then E^{-1} must be lower-triangular, in which case E must be lower-triangular as well. So if E is lower-triangular, then $E_{ij} = 0$ if $j > i$, meaning that

$$U = EA = \begin{pmatrix} E_{11}\vec{\mathbf{a}}_1 \\ E_{21}\vec{\mathbf{a}}_1 + E_{22}\vec{\mathbf{a}}_2 \\ \vdots \\ E_{n1}\vec{\mathbf{a}}_1 + E_{n2}\vec{\mathbf{a}}_2 + \cdots + E_{nn}\vec{\mathbf{a}}_n \end{pmatrix}$$

This can only happen if the row-reduction process doesn't require us to swap any rows along the way. In that case, the row-reduction steps look like the following

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ E_{21} & 1 & 0 & 0 \\ E_{31} & 0 & 1 & 0 \\ E_{41} & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & b_{32} & b_{33} & b_{34} \\ 0 & b_{42} & b_{43} & b_{44} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ E_{21} & 1 & 0 & 0 \\ E_{31} & E_{32} & 1 & 0 \\ E_{41} & E_{42} & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & c_{43} & c_{44} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ E_{21} & 1 & 0 & 0 \\ E_{31} & E_{32} & 1 & 0 \\ E_{41} & E_{42} & E_{43} & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & 0 & d_{44} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

So the values $E_{11} = E_{22} = \cdots = E_{nn} = 1$ and E is a lower-triangular matrix. This brings us to an important observation

Theorem 7.2.1 A matrix A can be written as the product $A = LU$, where

- L is a lower-triangular matrix; and
- U is an upper-triangular matrix

provided that A can be reduced to row-echelon form without elementary steps involving the swapping of rows.

7.2.1 PLU variation for any matrix

When do we need to perform row-swapping during row reduction? During each step of row reduction, we use a *pivot row*. This is the row that we use to set all other entries in that column to 0[5]. Typically, we proceed with row reduction by using the top row as the pivot row for the first column, then the (new) second row as the pivot row for the second column, then the (new) third row as the pivot row for the third column, and so on. Our theorem above tells us that, in this circumstance, we can reach an LU factorisation. However, if after the first pivot we ended up with a 0 in the second column of the second row, we cannot use that row as the next pivot row – we would have to swap it with another row. The same would happen if we obtained a 0 in the third column of the third row after the second round of pivoting, *etc.*

[5]If we want to reach the unique reduced row echelon form.

In practice, we can always get around this problem by swapping rows until we have moved a suitable row into the pivot row position. What implication does this have for our LU factorisation? It means that, for any matrix, we can have an LU factorisation, but we may need to include an extra matrix out the front of this factorisation, representing the effect of swapping rows around. Therefore, we end up with a 'PLU' factorisation instead, where the symbol P represents the inclusion of **permutation matrix**. Permutation matrices are a subset of the elementary matrices and have the following properties:

- Permutation matrices are square matrices
- Each row and each column of the permutation matrix contains precisely one 1, with 0s otherwise.
- Permutation matrices have determinant ± 1.
- There must be some integer m for which $P^m = I$, for permutation matrix P.

While permutations are not uncommon in row reduction, we still call this the LU factorisation, even when a permutation matrix is required.

Theorem 7.2.2 Any matrix A can be written as the product $A = PLU$, where

- P is a permutation matrix;
- L is a lower-triangular matrix; and
- U is an upper-triangular matrix.

Now we have a result that holds for *any* matrix A.

7.2.2 Uniqueness – the PLU and PLDU factorisations

An important question is whether the PLU factorisation is unique. A matrix can have several PLU factorisations, particularly when different row permutations are possible. However, if A is an invertible square matrix, the permutation matrix P is fixed, and L is required to be unit lower-triangular, then the factors L and U are unique.

The matrix U is the row-echelon form obtained by Gaussian elimination; it is not generally the reduced row-echelon form. The requirement that the diagonal entries of L equal 1 removes the freedom to transfer scaling factors between L and U.

The diagonal entries of U can be separated into a diagonal matrix D. Writing

$$U = D\widehat{U},$$

where $\widehat{U}$ is unit upper-triangular, gives

$$A = PLD\widehat{U}.$$

For a fixed P, this PLDU factorisation is unique.

Theorem 7.2.3 Any matrix A has a factorisation of the form

$$A = PLU,$$

where

- P is a permutation matrix;
- L is a unit lower-triangular matrix; and
- U is an upper-triangular or upper-trapezoidal matrix in row-echelon form.

This factorisation is not necessarily unique. If A is an invertible square matrix and P is fixed, then L and U are unique.

Furthermore, an invertible square matrix has a factorisation of the form

$$A = PLD\widehat{U},$$

where
- P and L are the same as in the PLU factorisation;
- D is an invertible diagonal matrix; and
- $\widehat{U}$ is a unit upper-triangular matrix.

The factors are related by

$$U = D\widehat{U}.$$

7.2.3 Calculating the PLU factorisation

To obtain the matrices P, L and U, we perform Gaussian elimination until the matrix is in row-echelon form:
- U is the resulting upper-triangular or upper-trapezoidal row-echelon matrix;
- L records the row-elimination multipliers used during Gaussian elimination; and
- P records the row swaps performed during the elimination process.

Keeping track of the interaction between the row eliminations and row swaps can be complicated, so in practice we often use software to calculate the factors.

■ Example 7.3 — Calculation of PLU factorisation using Octave/MATLAB.

```
> A=[4 5 0; 2 5 2; 0 5 4];
> [L,U]=lu(A)
L =

   1.0000        0        0
   0.5000   0.5000   1.0000
        0   1.0000        0

U =

   4   5   0
   0   5   4
   0   0   0
```

Notice that, without calling for P, the matrix L generated by the lu command is *not lower-triangular*. Including P among the return arguments of the call rectifies this:

```
> [L,U,P]=lu(A)
L =

   1.0000        0        0
        0   1.0000        0
   0.5000   0.5000   1.0000

U =

   4   5   0
   0   5   4
   0   0   0
```

```
P =

Permutation Matrix

   1    0    0
   0    0    1
   0    1    0
```

■

■ **Example 7.4 — Calculation of PLU factorisation using R.** R creates a data structure that we need to pull each of the relevant matrices out of using the `expand` command. Note that R also automatically outputs P as a 'sparse' matrix – sparse matrices are almost entirely populated by zeroes, with only occasional non-zero entries (this allows computers to efficiently handle memory). Permutation matrices are always sparse because only one element per row is non-zero.

```
> library(Matrix)
> A <- matrix(c(4,5,0,2,5,2,0,5,4),3,3,byrow=TRUE)
> pludecomposition <- lu(A)
> P <- expand(pludecomposition)$P
> L <- expand(pludecomposition)$L
> U <- expand(pludecomposition)$U
> P
3 x 3 sparse Matrix of class "pMatrix"

[1,] | . .
[2,] . . |
[3,] . | .
> L
3 x 3 Matrix of class "dtrMatrix" (unitriangular)
     [,1] [,2] [,3]
[1,]  1.0    .    .
[2,]  0.0  1.0    .
[3,]  0.5  0.5  1.0
> U
3 x 3 Matrix of class "dtrMatrix"
     [,1] [,2] [,3]
[1,]    4    5    0
[2,]    .    5    4
[3,]    .    .    0
```

■

■ **Example 7.5 — Calculation of PLU factorisation using Python.**

```
> import scipy
> A = np.array([[4,5,0],[2,5,2],[0,5,4]])
> P,L,U = scipy.linalg.lu(A)
> P
    array([[1., 0., 0.],
           [0., 0., 1.],
           [0., 1., 0.]])
> L
    array([[1. , 0. , 0. ],
           [0. , 1. , 0. ],
```

```
            [0.5 ,  0.5 ,  1.  ]])
> U
      array([[4. ,  5. ,  0.],
            [0. ,  5. ,  4.],
            [0. ,  0. ,  0.]])
```

■

7.3 QR factorisation

We have seen how the solution to the problem $A\mathbf{x} = \mathbf{b}$ can be considered in the context of the column space of matrix A;

- any solution $\mathbf{x}$ represents a column vector of coefficients x_i such that

$$\sum_i x_i \mathbf{a}_i = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b},$$

  where $\mathbf{a}_j$ is a column vector of A;
- if $\mathbf{b}$ is not in the column space of A, there will be no solution;
- if $\mathbf{b}$ is in the column space of A, then there will be at least one solution; and
- if the null space of A is larger than just the zero vector, there will be a family of solutions.

If the columns of A were an orthonormal set of vectors, finding these x_j coefficients would be very simple. To see why this is the case, consider

$$\begin{aligned}\mathbf{a}_j \cdot \mathbf{b} &= \mathbf{a}_j \cdot (x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n) \\ &= x_1\mathbf{a}_j \cdot \mathbf{a}_1 + x_2\mathbf{a}_j \cdot \mathbf{a}_2 + \cdots + x_n\mathbf{a}_j \cdot \mathbf{a}_n.\end{aligned}$$

Now, because the columns of A are all orthonormal, $\mathbf{a}_j \cdot \mathbf{a}_i = \delta_{ij}$, which is 0 if $i \neq j$, and 1 if $i = j$. Therefore, there is only one non-zero contribution in the sum, and

$$\mathbf{a}_j \cdot \mathbf{b} = x_j.$$

An alternative way to see this result is to recall that, if the columns of A were an orthonormal set of vectors, then $A^{\mathrm{T}}A = I$, so

$$A\mathbf{x} = \mathbf{b} \implies \mathbf{x} = I\mathbf{x} = A^{\mathrm{T}}A\mathbf{x} = A^{\mathrm{T}}\mathbf{b} \implies x_j = \mathbf{a}_j \cdot \mathbf{b}$$

But what if the column vectors of A are not orthonormal? Well, we know that they span a vector space in their own right, so we must be able to construct a basis for that space. What if we could find an *orthonormal* basis for them?

And indeed, we can. There is a process for constructing an orthonormal basis that has the same span as a given set of vectors, known as **Gram-Schmidt orthogonalisation**. The recipe to construct a set of orthonormal basis vectors $\mathbf{v}_i$ with the same span as the given set of vectors $\mathbf{a}_i$ is as follows[6]:

[6] There is nothing special about the fact that we are considering the column vectors of a matrix, since there is no restriction on what the column vectors can be!)

Definition 7.3.1 — Gram-Schmidt orthogonalisation. To construct a set of orthonormal basis vectors $\hat{\mathbf{v}}_i$ with the same span as the given set of vectors $\mathbf{a}_i$, complete these steps.

1. Since we are free to choose any (non-zero) vector as our starting point, choose a vector pointing in the same direction as the $\mathbf{a}_1$, *i.e.*, set $\mathbf{v}_1 = \mathbf{a}_1$.
2. Since we want ortho*normal* vectors, set its size to 1[a]: $\hat{\mathbf{v}}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$.
3. For the second vector, start with a vector pointing in the direction of $\mathbf{a}_2$, and then remove the component pointing in the direction $\mathbf{v}_1$ so that the resulting vector is orthogonal to $\mathbf{v}_1$: $\mathbf{v}_2 = \mathbf{a}_2 - (\mathbf{a}_2 \cdot \hat{\mathbf{v}}_1)\hat{\mathbf{v}}_1$.
4. Again, set the size of this vector to 1: $\hat{\mathbf{v}}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|}$.
5. For the third vector, start with a vector pointing in the direction of $\mathbf{a}_3$, and then remove the components pointing in the directions of $\mathbf{v}_1$ and $\mathbf{v}_2$. This ensures that the resulting vector is orthogonal to both $\mathbf{v}_1$ and $\mathbf{v}_2$: $\mathbf{v}_3 = \mathbf{a}_3 - (\mathbf{a}_3 \cdot \hat{\mathbf{v}}_1)\hat{\mathbf{v}}_1 - (\mathbf{a}_3 \cdot \hat{\mathbf{v}}_2)\hat{\mathbf{v}}_2$.
6. Again, set the size of this vector to 1: $\hat{\mathbf{v}}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|}$.
7. Continue in this fashion, with

$$\mathbf{v}_m = \mathbf{a}_m - \sum_{i=1}^{m-1} (\mathbf{a}_m \cdot \hat{\mathbf{v}}_i)\hat{\mathbf{v}}_i \quad \text{and} \quad \hat{\mathbf{v}}_m = \frac{\mathbf{v}_m}{\|\mathbf{v}_m\|},$$

until all elements of the set of $\mathbf{a}_i$ have been considered (or until the number of $\hat{\mathbf{v}}$ vectors equals the rank of the matrix with columns given by $\mathbf{a}_i$).

[a] At every stage, if $\mathbf{v}_j$ turns out to be the zero vector, ignore it and move on to the next vector. This eliminates any linear dependence that might be present in the set of $\mathbf{a}_i$ so that we obtain a truly orthonormal *basis*.

Notice that, in this process, we produce a basis such that

- $\mathbf{v}_1$ is proportional to $\mathbf{a}_1$;
- $\mathbf{v}_2$ is a linear combination of $\mathbf{a}_2$ and $\mathbf{v}_1$, so therefore also a linear combination of $\mathbf{a}_2$ and $\mathbf{a}_1$;
- $\mathbf{v}_3$ is a linear combination $\mathbf{a}_3$, $\mathbf{v}_2$ and $\mathbf{v}_1$, so therefore also a linear combination of $\mathbf{a}_3$, $\mathbf{a}_2$ and $\mathbf{a}_1$.
- It follows that $\mathbf{v}_m$ will be a linear combination of $\mathbf{a}_m$, $\mathbf{a}_{m-1}, \ldots, \mathbf{a}_2$ and $\mathbf{a}_1$.

Therefore, if Q is the matrix whose columns are the orthonormal vectors $\mathbf{v}_i$, it follows that

$$Q = \begin{pmatrix} | & | & | & \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \cdots \\ | & | & | & \end{pmatrix} = \begin{pmatrix} | & | & | & \\ c_{11}\mathbf{a}_1 & c_{12}\mathbf{a}_1 + c_{22}\mathbf{a}_2 & c_{13}\mathbf{a}_1 + c_{23}\mathbf{a}_2 + c_{33}\mathbf{a}_3 & \cdots \\ | & | & | & \end{pmatrix} = AC$$

where A is our original matrix (whose columns are the $\mathbf{a}_i$), and C is the upper-triangular matrix whose elements are the coefficients c_{ij}. Finally, because the inverse of an upper-triangular matrix is itself upper-triangular, it follows that $A = QU$, where $U = C^{-1}$ is an upper-triangular matrix.

Theorem 7.3.1 Any matrix A has a factorisation of the form $A = QU$, where

- Q is an orthogonal matrix; and

- U is an upper-triangular matrix.

The QR factorisation *is not unique* – it is straightforward to construct many different bases that all span the same space. For example, the vectors

$$\begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix}, \begin{pmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

form an orthonormal basis for $\mathbb{R}^3$ *for any value of* $\theta \in \mathbb{R}$. Applying the Gram-Schmidt algorithm to the columns of A, in arbitrary order, will likely produce different bases, all spanning precisely the same column space.

7.3.1 Calculating the QR factorisation

The method outlined above can be applied to determine the QR factorisation for a given matrix. In practice, we generally use software to perform these calculations.

Example 7.6 — Calculation of QR factorisation using Octave/MATLAB.

```
> A=[4 5 0; 2 5 2; 0 5 4]; % same A as in earlier example
> [Q,U]=qr(A)
Q =
   -0.8944    0.1826    0.4082
   -0.4472   -0.3651   -0.8165
         0   -0.9129    0.4082

U =
   -4.4721   -6.7082   -0.8944
         0   -5.4772   -4.3818
         0         0   -0.0000
```

■

Example 7.7 — Calculation of QR factorisation using R.

```
> library(Matrix)
> A <- matrix(c(4,5,0,2,5,2,0,5,4),3,byrow=TRUE);
> qrobj <- qr(A)
> Q <- qr.Q(qrobj)
> R <- qr.R(qrobj)
> Q
           [,1]       [,2]       [,3]
[1,] -0.8944272  0.1825742  0.4082483
[2,] -0.4472136 -0.3651484 -0.8164966
[3,]  0.0000000 -0.9128709  0.4082483
> R
          [,1]      [,2]          [,3]
[1,] -4.472136 -6.708204 -8.944272e-01
[2,]  0.000000 -5.477226 -4.381780e+00
[3,]  0.000000  0.000000  4.440892e-16
```

Note that R produces an object `qrobj` that contains a whole bunch of information about the QR factorisation, and then we need to extract the components that we're interested in by feeding the object into some associated functions `qr.R`, *etc.* ■

Example 7.8 — Calculation of QR factorisation using Python.

```
> A = np.array([[4,5,0], [2,5,2], [0,5,4]])
> Q, R = np.linalg.qr(A)
> Q
    array([[-0.89442719,  0.18257419,  0.40824829],
           [-0.4472136 , -0.36514837, -0.81649658],
           [-0.        , -0.91287093,  0.40824829]])
> R
    array([[-4.47213595e+00, -6.70820393e+00, -8.94427191e-01],
           [ 0.00000000e+00, -5.47722558e+00, -4.38178046e+00],
           [ 0.00000000e+00,  0.00000000e+00, -1.77635684e-15]])
```

■

7.3.2 The *thin* QR factorisation

Let's consider the QR factorisation for a non-square matrix.

■ Example 7.9 — Calculation of non-square QR factorisation using Octave/MATLAB.

```
    > A=[3 3 3; 2 1 2; -1 -1 1; 2 -3 1];
> [Q,U]=qr(A)
Q =
   -0.707107    0.471405   -0.208758    0.483940
   -0.471405    0.078567   -0.205595   -0.854011
    0.235702   -0.157135   -0.955228    0.085401
   -0.471405   -0.864242    0.041119    0.170802
U =
   -4.2426   -1.4142   -3.2998
         0    4.2426    0.5500
         0         0   -1.9516
         0         0         0
```

■

Notice that the upper-triangular matrix U has the same shape as A. Because A has more rows than columns, the bottom rows of U must all be zeros. For an $n \times m$ matrix A where $n > m$, the top m rows of U will be a square upper-triangular matrix, and the bottom $n-m$ rows will all be zeros (here $n = 4$ and $m = 3$, and indeed we have 1 row of zeros).

This means, if $n > m$, we can consider U as a block matrix with square upper-triangular matrix U_{mm} sitting on top of the $(n-m)$-by-m 0 matrix. If we also split the *columns* of the n-by-n orthogonal Q into two blocks – Q_{mn} comprising the first m columns, and Q_{rest} comprising the remaining rows – then we see that

$$A = QU = \left[Q_{mn} \mid Q_{\text{rest}}\right] \begin{bmatrix} U_{mm} \\ 0 \end{bmatrix} = Q_{mn}U_{mm}.$$

As a consequence, if $n > m$, we can drop all the zero rows from the bottom rows of U, and drop a matching number of columns from the right-hand side of Q, and still have a QR factorisation for A. This factorisation has the added advantage that we can talk sensibly about the inverse of U, because it is a square matrix.

This QR factorisation where we strip away any zero rows of U, and the corresponding redundant columns of Q, is known as the **thin QR factorisation** of A. It only works when $n > m$ – otherwise there will not be any zero rows in U (although in practice we can extend this approach to the A^{T}, when it proves useful to do so).

In practical applications, it is not uncommon to be working with matrices for which the thin factorisation is very useful. We consider one such application in the next section.

7.3.3 The least-squares problem revisited

Recall the least-squares problem from Section 5.7.2: for a given matrix X and vector $\mathbf{y}$, we want to find the vector $\mathbf{k}$ that minimises

$$\|\mathbf{y} - X\mathbf{k}\|.$$

If X has full column rank, then $X^\mathsf{T}X$ is invertible and the unique least-squares solution is

$$\mathbf{k}_0 = (X^\mathsf{T}X)^{-1}X^\mathsf{T}\mathbf{y}.$$

Express X in terms of its thin QR factorisation,

$$X = QU,$$

where Q has orthonormal columns and U is an invertible upper-triangular matrix. We use the identities

- $Q^\mathsf{T}Q = I$;
- $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$; and
- $(AB)^{-1} = B^{-1}A^{-1}$, provided that A and B are invertible.

It follows that

$$\begin{aligned}
\mathbf{k}_0 &= (X^\mathsf{T}X)^{-1}X^\mathsf{T}\mathbf{y} \\
&= (U^\mathsf{T}Q^\mathsf{T}QU)^{-1}U^\mathsf{T}Q^\mathsf{T}\mathbf{y} \\
&= (U^\mathsf{T}U)^{-1}U^\mathsf{T}Q^\mathsf{T}\mathbf{y} \\
&= U^{-1}(U^\mathsf{T})^{-1}U^\mathsf{T}Q^\mathsf{T}\mathbf{y} \\
&= U^{-1}Q^\mathsf{T}\mathbf{y}.
\end{aligned}$$

Therefore, the least-squares solution satisfies

$$U\mathbf{k}_0 = Q^\mathsf{T}\mathbf{y}.$$

Since U is upper triangular, $\mathbf{k}_0$ can be calculated efficiently using back-substitution, without explicitly computing U^{-1}.

To see this in practice, let us revisit the least-squares problems we solved in section 5.7.2, where we fit a linear and quadratic function through through the data set shown in the table below. For the linear fit to $y = mx + c$, we obtained optimal parameters $m \approx 0.48$ and $c \approx 1.42$, while for the quadratic fit to $y = ax^2 + bx + c$, we obtained optimal parameters $a \approx -0.10$, $b \approx 1.59$ and $c \approx -0.80$.

x_i	1	2	3	4	5	6	7	8	9	10
y_i	0.6	2.1837	2.8844	4.3985	4.2369	4.7755	5.5160	5.6721	5.6654	4.7268

For the least-squares fit to the linear equation $y = mx + c$, we have

$$X = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ \vdots & \vdots \\ 10 & 1 \end{pmatrix}$$

which has thin QR factorisation $X = QU$, where Q is a 10×2 matrix (shown below) and

$$U = \begin{pmatrix} -19.6214 & -2.8031 \\ 0 & -1.4639 \end{pmatrix}$$

Calculating $U^{-1}Q^{\mathrm{T}}$ gives

$$U^{-1}Q^{\mathrm{T}} = \frac{1}{165}\begin{pmatrix} -9 & -7 & -5 & -3 & -1 & 1 & 3 & 5 & 7 & 9 \\ 66 & 55 & 44 & 33 & 22 & 11 & 0 & -11 & -22 & -33 \end{pmatrix} = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}.$$

Using this,

$$\begin{pmatrix} m \\ c \end{pmatrix} = U^{-1}Q^{\mathrm{T}}\mathbf{y} = \frac{1}{165}\begin{pmatrix} -9 & -7 & -5 & -3 & -1 & 1 & 3 & 5 & 7 & 9 \\ 66 & 55 & 44 & 33 & 22 & 11 & 0 & -11 & -22 & -33 \end{pmatrix}\begin{pmatrix} 0.60007 \\ 2.18372 \\ \vdots \\ 4.72679 \end{pmatrix}$$

$$\approx \begin{pmatrix} 0.48 \\ 1.42 \end{pmatrix},$$

which matches what we found in section 5.7.2.

For the least-squares fit to the quadratic equation $y = ax^2 + bx + c$, we have

$$X = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ \vdots & \vdots & \vdots \\ 100 & 10 & 1 \end{pmatrix}$$

which has QR factorisation $X = QU$, where Q is a 10×3 matrix (shown below) and

$$U = \begin{pmatrix} -159.1634 & -19.0056 & -2.418 \\ 0 & -4.8771 & -1.851 \\ 0 & 0 & 0.8502 \end{pmatrix}.$$

Calculating $U^{-1}Q^{\mathrm{T}}$ gives

$$U^{-1}Q^{\mathrm{T}} = \frac{1}{1320}\begin{pmatrix} 30 & 10 & -5 & -15 & -20 & -20 & -15 & -5 & 10 & 30 \\ -402 & -166 & 15 & 141 & 212 & 228 & 189 & 95 & -54 & -258 \\ 1188 & 660 & 242 & -66 & -264 & -352 & -330 & -198 & 44 & 396 \end{pmatrix},$$

so

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = U^{-1}Q^{\mathrm{T}}\mathbf{y}$$

$$= \frac{1}{1320}\begin{pmatrix} 30 & 10 & -5 & -15 & -20 & -20 & -15 & -5 & 10 & 30 \\ -402 & -166 & 15 & 141 & 212 & 228 & 189 & 95 & -54 & -258 \\ 1188 & 660 & 242 & -66 & -264 & -352 & -330 & -198 & 44 & 396 \end{pmatrix}\begin{pmatrix} 0.60007 \\ 2.18372 \\ \vdots \\ 4.72679 \end{pmatrix}$$

$$\approx \begin{pmatrix} -0.10 \\ 1.59 \\ -0.80 \end{pmatrix}$$

This demonstrates an important characteristic of the QR factorisation; because the QR factorisation is closely related to the column space of the matrix being factorised – it provides an orthonormal basis for the column space – it can be a useful factorisation for problems related to that space. The least-squares problem can be interpreted as the determination of the point closest to $\mathbf{y}$ that lies within the column space of X; a problem that is easy to solve when an orthonormal basis for the column space is available.

Another advantage of the QR factorisation is that it tends to involve numerically more stable calculations than the LU factorisation, meaning that calculations using the QR factorisation are generally more straightforward to implement. This is only usually a problem if you are writing your own software to perform these calculations, however; the majority of software distributions (*e.g.*, all those used in this course) have robust algorithms to accommodate all but the most pathological of numerical problems.

7.3.4 Cholesky factorisation

If the matrix A is a real, symmetric, positive-definite matrix, then the QR factorisation leads to a special case:

Theorem 7.3.2 Any real, symmetric, positive-definite matrix A has a unique factorisation of the form $A = U^{\mathrm{T}}\,U$, or equivalently, $A = L\,L^{\mathrm{T}}$, where

- U is an upper-triangular matrix; and
- L is a lower-triangular matrix

To see that this is the case, consider the diagonalisation of $A = VDV^{-1}$.

- Since A is real and symmetric, we know that its eigenvectors are orthogonal, and therefore V is an orthogonal matrix, so $A = VDV^{\mathrm{T}}$
- Since A is positive-definite, its eigenvalues are all positive. Therefore we can define $S = D^{1/2}$ as the diagonal matrix whose elements are the (positive) square roots of the elements of D, so $S^2 = D$
- Since S is diagonal, $S = S^{\mathrm{T}}$, and therefore $D = S^2 = SS^{\mathrm{T}}$
- Putting these together gives $A = VDV^{\mathrm{T}} = VSS^{\mathrm{T}}V^{\mathrm{T}} = (VS)(VS)^{\mathrm{T}}$
- The matrix $(VS)^{\mathrm{T}}$ must have its own QR factorisation, $(VS)^{\mathrm{T}} = QU$, for orthogonal matrix Q and upper-triangular matrix U
- This means that $A = (QU)^{\mathrm{T}}(QU) = U^{\mathrm{T}}Q^{\mathrm{T}}QU = U^{\mathrm{T}}U$ since $Q^{\mathrm{T}}Q = I$ for orthogonal matrices.

This final step proves the existence of the Cholesky factorisation $U^{\mathrm{T}}U$. Defining the lower-triangular matrix $L = U^{\mathrm{T}}$ proves the existence of the alternative factorisation LL^{T}.

The method outlined above can be applied to determine the Cholesky factorisation for a given matrix. In practice, we generally use software to perform these calculations.

■ Example 7.10 — Calculation of Cholesky factorisation using Octave/MATLAB.

```
> C = [4 3 2; 3 6 3; 2 3 4];
> U = chol(C)
U =
    2.0000    1.5000    1.0000
         0    1.9365    0.7746
         0         0    1.5492

> U'*U  % should recover C
ans =
```

```
4   3   2
3   6   3
2   3   4
```

■

■ **Example 7.11 — Calculation of Cholesky factorisation using R.**

```
> library(Matrix)
> C <- matrix(c(4,3,2,3,6,3,2,3,4),3,byrow=TRUE);
> chol(C)
     [,1]     [,2]      [,3]
[1,]    2 1.500000 1.0000000
[2,]    0 1.936492 0.7745967
[3,]    0 0.000000 1.5491933
```

■

■ **Example 7.12 — Calculation of Cholesky factorisation using Python.**

```
> import numpy as np
> C = np.array([[4,3,2],[3,6,3],[2,3,4]])
> np.linalg.cholesky(C)
    array([[2.        , 0.        , 0.        ],
           [1.5       , 1.93649167, 0.        ],
           [1.        , 0.77459667, 1.54919334]])
```

■

7.4 Singular value decomposition

The final factorisation we introduce in this chapter is the **singular value decomposition**, commonly abbreviated to SVD:

Theorem 7.4.1 Any matrix $A \in \mathbb{R}^{m\times n}$ can be written as a product of the form

$$A = Q_m \, \Sigma \, Q_n^{\mathrm{T}},$$

where

- Q_m is an $m \times m$ orthogonal matrix;
- Σ is an $m \times n$ diagonal matrix, with all non-negative diagonal elements; and
- Q_n is an $n \times n$ orthogonal matrix.

This decomposition is very commonly written as $A = U\Sigma V^{\mathrm{T}}$, but this notation is inconsistent with our convention of using U for upper-triangular matrices. *It is important to bear in mind, if you see this expansion, that U and V represent the orthogonal matrices in the SVD expansion, which are usually **not** upper triangular.* In fact, the only way for a matrix to be orthogonal and upper triangular is to be diagonal.

Recall that the eigen-decomposition theorem (4.4.1) states that a square matrix A can be factorised using its eigenvalues and eigenvectors, $A = VDV^{-1}$, with V composed of the eigenvectors. This did not apply to non-square matrices, but we did note that $A^{\mathrm{T}}A$ and AA^{T} will always be square matrices. Thus, it is reasonable to expect that the SVD will be related to the spectral properties of these two matrices. Let's see this with a numerical example.

■ **Example 7.13 — Calculation of SVD factorisation using MATLAB.** Let's first consider the diagonalisation of the matrix

$$A = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}.$$

```
> [W,D]=eig(A) %Use W so as not confused with V that will be used in
    the SVD
W =
    0.8944    0.7071
    0.4472    0.7071
D =
Diagonal Matrix
    3    0
    0    2
```

Now use MATLAB to find the SVD of A.

```
> [U,S,V]=svd(A) %Use S as shorthand for Sigma
U =
   -0.9940   -0.1091
   -0.1091    0.9940
S =
Diagonal Matrix

    4.4966         0
         0    1.3343
V =
   -0.9085    0.4179
    0.4179    0.9085
```

We can see that there's no obvious connection between the spectral properties of A and its SVD. However, let's consider the spectral properties of $A^{\mathrm{T}}A$:

```
> [W2,D2]=eig(A'*A)
W2 =
   -0.4179   -0.9085
   -0.9085    0.4179
D2 =
Diagonal Matrix

    1.7805          0
         0    20.2195
> sqrt(D2)
ans =
Diagonal Matrix
    1.3343         0
         0    4.4966
```

So we have verified that the elements in the diagonal matrix in the SVD of A (in the matrix S) are the square roots of the eigenvalues of $A^{\mathrm{T}}A$. Also, we notice that the eigenvectors of $A^{\mathrm{T}}A$ are the columns of the orthogonal matrix V in the SVD; orthogonality is guaranteed because they are eigenvectors corresponding to distinct eigenvalues.

How might we obtain the orthogonal matrix U? Consider AA^{T}:

```
> [W3,D3]=eig(A*A')
```

```
W3 =
    0.1091   -0.9940
   -0.9940   -0.1091
D3 =
Diagonal Matrix
     1.7805          0
          0    20.2195
```

It seems we can obtain the orthogonal matrix `U` by considering the eigenvectors of the matrix $AA^{\mathrm{T}} = (A^{\mathrm{T}}A)^{\mathrm{T}}$. This connection makes more sense when we consider the SVD of a non-square matrix. ■

■ **Example 7.14 — Calculation of non-square SVD factorisation using MATLAB.** Consider the SVD of

$$A = \begin{pmatrix} 1 & -3 & 2 \\ 2 & 0 & 3 \end{pmatrix}.$$

```
> [U,S,V]=svd(A)
U =
   -0.7288   -0.6847
   -0.6847    0.7288
S =
Diagonal Matrix
    4.6385         0         0
         0    2.3419         0
V =
   -0.452350    0.330059   -0.828517
    0.471378    0.877114    0.092057
   -0.757088    0.348902    0.552345
```

Since A is not square, we cannot obtain a diagonalisation for it. Instead, let's consider the spectral properties of $A^{\mathrm{T}}A$:

```
> [W,D]=eig(A'*A)
W =
   -0.828517    0.330059    0.452350
    0.092057    0.877114   -0.471378
    0.552345    0.348902    0.757088
D =
Diagonal Matrix
    2.5208e-15            0            0
             0       5.4844            0
             0            0      21.5156
> sqrt(D)
ans =
Diagonal Matrix
    5.0207e-08            0            0
             0       2.3419            0
             0            0       4.6385
```

So again we have verified that the singular values of A (in the matrix `S`) are the square roots of the eigenvalues of $A^{\mathrm{T}}A$, and that the eigenvectors of $A^{\mathrm{T}}A$ are the columns of the orthogonal matrix `V` in the SVD. Notice that there must be three eigenvalues of $A^{\mathrm{T}}A$, but also that one of them must be zero. Finally, to obtain the orthogonal matrix `U` we evaluate the eigenvectors of AA^{T},

```
> [W2,D2]=eig(A*A')
W2 =
    0.6847   -0.7288
   -0.7288   -0.6847
D2 =
Diagonal Matrix
     5.4844         0
          0   21.5156
```

■

As we've seen in these examples, comparisons between the diagonalised form obtained through eigenvector decomposition and the factorisation obtained *via* SVD should be treated with care. Certainly, SVD results in a diagonalised form. However, we have noted some important distinctions:

- The diagonal elements of the matrix Σ are *not* the eigenvalues of A. They are called the **singular values** of the matrix A, and are equivalent (up to a square root) to the set of eigenvalues of AA^{T}. In general, there is no simple relation between the eigenvalues and singular values of a matrix, although in special circumstances simple relations do emerge (for example, if A is square and symmetric, the singular values of A are given by the absolute values of its eigenvalues). To see why these eigenvalues arise in the SVD, consider that for a square matrix A

$$
\begin{aligned}
A\,A^{\mathrm{T}} &= U\Sigma\,V^{\mathrm{T}}\left(U\Sigma\,V^{\mathrm{T}}\right)^{\mathrm{T}} \\
&= U\Sigma\,V^{\mathrm{T}}V\Sigma\,U^{\mathrm{T}} \\
&= U\Sigma^2\,U^{\mathrm{T}},
\end{aligned}
$$

  which has the same form as the eigen-decomposition of $A\,A^{\mathrm{T}}$ (U formed from eigenvectors of $A\,A^{\mathrm{T}}$ and Σ^2 formed from the corresponding eigenvalues). A more general argument can be made for the case of non-square A.
- If A has rank r, then only r elements of Σ are non-zero. Traditionally, these are placed as the first elements along the diagonal of Σ, in descending order (see the output in the examples earlier).
- Because the SVD can easily be calculated for non-square matrices, U and V are not the same size in general. Their columns are *not* the eigenvectors of A. Instead, they are the eigenvectors of $A^{\mathrm{T}}A$ and its transpose, AA^{T}, which must (both) be orthogonal sets of vectors, because $A^{\mathrm{T}}A$ and its transpose are real symmetric matrices.

Very helpfully, it can be shown that the column space of A is precisely the same as the column space of AA^{T}, and that the column space of A^{T}, equivalently the row space of A, is precisely the same as the column space of $A^{\mathrm{T}}A$.

The eigenvectors of AA^{T} corresponding to nonzero eigenvalues form an orthonormal basis for the column space of A, while those corresponding to zero eigenvalues form an orthonormal basis for the null space of A^{T}.

Similarly, the eigenvectors of $A^{\mathrm{T}}A$ corresponding to nonzero eigenvalues form an orthonormal basis for the row space of A, while those corresponding to zero eigenvalues form an orthonormal basis for the null space of A. This is shown in Fig. 7.1

To summarise all of this: for an $m \times n$ matrix A with rank r:

1. the first r columns of U are an orthonormal basis for the column space of A;
2. the first r columns of V are an orthonormal basis for the column space of A^{T};

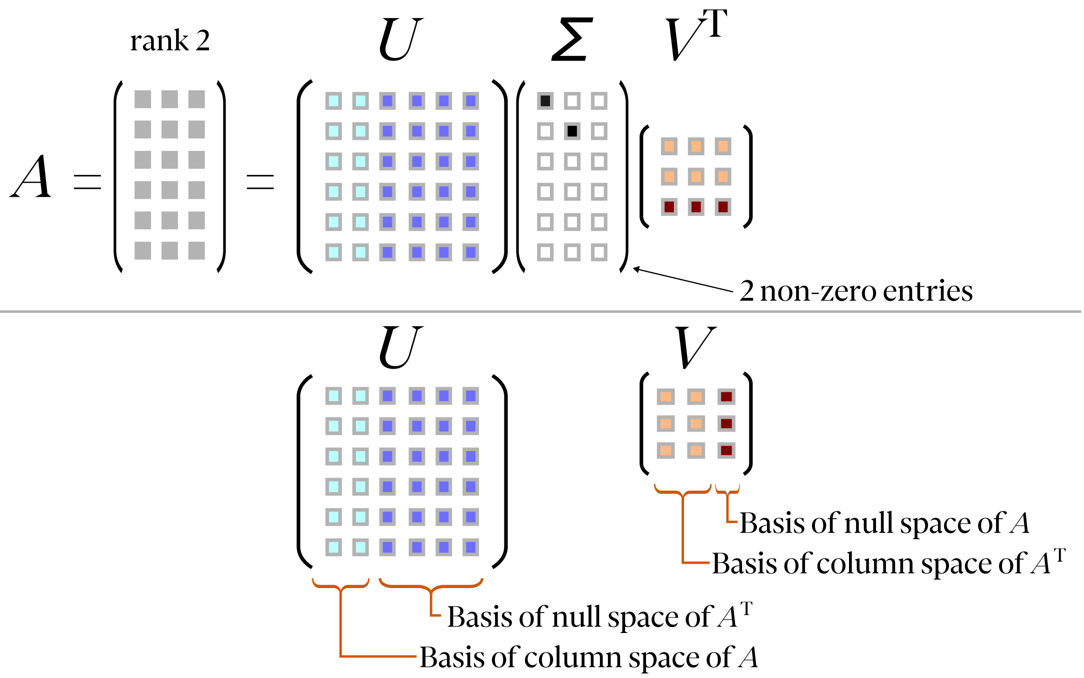


Figure 7.1: Schematic of the SVD expansion and its relationship to orthogonal bases of the column and null spaces of A and A^{T}. Here we assume that A is rank 2, so the Σ matrix contains two non-zero diagonal terms.

3. the final $m-r$ columns of U are an orthonormal basis for the null space of A^{T}; and
4. the final $n-r$ columns of V are an orthonormal basis for the null space of A.

Proof. Property 4. Suppose we have $A \in \mathbb{R}^{m\times n}$ with rank r. Using the SVD, we have $A = U\Sigma V^{\mathrm{T}}$. Let the columns of V be $\mathbf{v}_i$ and the columns of U be $\mathbf{u}_i$. Then,

$$
\begin{aligned}
A\mathbf{v}_i &= U\Sigma V^{\mathrm{T}}\mathbf{v}_i \\
&= U\Sigma \begin{pmatrix} | & | & | & \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \cdots \\ | & | & | & \end{pmatrix}^{\mathrm{T}} \mathbf{v}_i \\
&= U\Sigma\mathbf{e}_i.
\end{aligned}
$$

This arises due to the orthonormality of the columns of V. Next, note that only the first r diagonal elements of Σ ($\sigma_i = [\Sigma]_{ii}$) are non-zero. Thus,

$$
A\mathbf{v}_i = \begin{cases} U\sigma_i\mathbf{e}_i, & i \le r \\ \mathbf{0}, & i > r. \end{cases}
$$

Note that the second line implies $A\sum_{i>r}^{n}\alpha_i\mathbf{v}_i = \mathbf{0}\ \forall\ \alpha_i \in \mathbb{R}$, which means that *the last $n-r$ columns of V are an orthonormal basis for the null space of A.* Orthonormality is guaranteed by $V^{\mathrm{T}}V = I$, and we know they span the null space because the rank–nullity theorem demands the nullity be $n-r$. ■

Proof. Property 3. A^{T} can be written $A^{\mathrm{T}} = \left(U\Sigma V^{\mathrm{T}}\right)^{\mathrm{T}} = V\Sigma^{\mathrm{T}}U^{\mathrm{T}}$. Following the same line of reasoning as the previous proof (whilst noting that Σ is diagonal), we obtain

$$A^{\mathrm{T}}\mathbf{u}_i = \begin{cases} V\sigma_i\mathbf{e}_i, & i \le r \\ \mathbf{0}, & i > r. \end{cases}$$

Hence, *the last $m-r$ columns of U are an orthonormal basis for the null space of A^{T}.* ■

The proof of property 2 is left as an exercise.

Proof. Property 1. Any vector $A\mathbf{x}$ ($\mathbf{x} \in \mathbb{R}^n$) is an element of the column space of A. We can construct an arbitrary vector $\mathbf{x}$ by making a basis change on another arbitrary vector using an orthogonal matrix, $\mathbf{y}$, *viz.* $\mathbf{x} = Q\mathbf{y}$ for orthogonal $Q \in \mathbb{R}^{n\times n}$. Thus, choosing $Q = V$ shows that any vector $AV\mathbf{y}$ is an element of the column space of A. Using the SVD,

$$AV\mathbf{y} = U\Sigma V^{\mathrm{T}}V\mathbf{y} = U\Sigma\mathbf{y}.$$

The column space of A is thus spanned by the columns of $U\Sigma$. Consider now that there are only r non-zero elements of Σ, so

$$\begin{aligned} U\Sigma\mathbf{y} &= \begin{pmatrix} | & | & | & \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \cdots \\ | & | & | & \end{pmatrix} \left(\begin{array}{ccc|c} \sigma_1 & \cdots & 0 & \\ \vdots & \ddots & \vdots & \mathbf{0} \\ 0 & \cdots & \sigma_r & \\ \hline & \mathbf{0} & & \mathbf{0} \end{array}\right) \mathbf{y} \\ &= \begin{pmatrix} | & | & | & | & | \\ \mathbf{u}_1 & \cdots & \mathbf{u}_r & \mathbf{0} & \cdots \\ | & | & | & | & | \end{pmatrix} \mathbf{y} \\ &= \sigma_1 y_1 \mathbf{u}_1 + \sigma_2 y_2 \mathbf{u}_2 + \cdots + \sigma_r y_r \mathbf{u}_r. \end{aligned}$$

From this, we see that the span of columns of $U\Sigma$, which is equal to the span of columns of A, is also equal to the span of the first r columns of U. Hence, *the first r columns of U form an orthonormal basis for the column space of A.* ■

As with the QR factorisation, the presence of orthogonal vectors suggests a close connection with with the column and row spaces of A. The presence of non-negative diagonal elements of Σ also provides a sense of ordering by significance – this is explored further in chapter **??**.

Here endeth the notes – for now.

"You can't learn too much linear algebra."

Prof. Benedict Gross, Harvard University

II Exercises

i. Tutorial questions

Practice Questions – Ch. 1

1. For each of the following system of equations:
   - Write the corresponding augmented matrix.
   - Use elementary row operations to reduce the matrix to reduced row echelon form.
   - Use the reduced matrix to determine the solution to the system.

   (a) $\begin{aligned} x_1 - 2x_2 &= 5 \\ 3x_1 + x_2 &= 1 \end{aligned}$

   (b) $\begin{aligned} 3x_1 + 2x_2 + x_3 &= 0 \\ -2x_1 + x_2 - x_3 &= 2 \\ 2x_1 - x_2 + 2x_3 &= -1 \end{aligned}$

   (c) $\begin{aligned} x_1 + x_2 &= 1 \\ x_1 - x_2 &= 1 \\ -x_1 + 3x_2 &= 3 \end{aligned}$

   (d) $\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ x_1 - x_2 - x_3 &= 1 \end{aligned}$

2. For each of the following augmented matrices:
   - Use elementary row operations to reduce the matrix to reduced row echelon form.
   - Use the reduced matrix to determine the solution to the system.

   (a) $\left(\begin{array}{rrr|r} 5 & -2 & 1 & 3 \\ 2 & 3 & -4 & 0 \end{array}\right)$

   (b) $\left(\begin{array}{rr|r} 2 & 1 & 3 \\ 1 & -1 & 0 \\ 4 & -1 & 3 \end{array}\right)$

   (c) $\left(\begin{array}{rrr|r} 2 & 1 & 4 & 0 \\ 4 & 2 & 3 & 5 \\ 5 & 2 & 6 & 1 \end{array}\right)$

   (d) $\left(\begin{array}{rr|r} 3 & 2 & 8 \\ 1 & 5 & 7 \end{array}\right)$

3. Determine whether the following row echelon forms correspond to (i) inconsistent systems, (ii) systems with a unique solution, or (iii) system with a family of solutions.

   (a) $\left(\begin{array}{rr|r} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right)$

   (b) $\left(\begin{array}{rrr|r} 1 & -2 & 2 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right)$

   (c) $\left(\begin{array}{rr|r} 1 & 3 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right)$

   (d) $\left(\begin{array}{rrr|r} 1 & -2 & 4 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right)$

4. Balance the following chemical equations.
   (a) $x_1 C_4H_{10} + x_2 O_2 \ \rightarrow \ x_3 CO_2 + x_4 H_2O$
   (b) $x_1 Cu + x_2 HNO_3 \ \rightarrow \ x_3 Cu(NO_3)_2 + x_4 NO + x_5 H_2O$
   (c) $x_1 P_4O_{10} + x_2 H_2O \ \rightarrow \ x_3 H_3PO_4$

5. Port Ashfield has a local railroad supporting its coal mine and electricity power plant. Mining \$100 of coal costs \$25 in electricity and \$25 in rail access. Generating \$100 of electricity costs \$65 in coal, \$5 in electricity and \$5 in rail access. Finally, in charging \$100 for rail access, the railroad company spends \$55 on coal and \$10 in electricity.

Over a given period, the Port Ashfield receives purchase orders for \$2M of coal and \$1M of electricity. How much must each industry produce (measured in \$) so that it can meet all demands.

6. A medieval village has a dairy farmer whose farm produces 500 litres of milk each week, a corn farmer who produces 100 kg each week on average, and a cheese maker who produces 10 kg of cheese each week. They exchange goods with each other through the village market. Each week, the diary farmer trades 150 litres for 20kg of corn and 4 kg of cheese; the corn farmer trades 30 kg of corn for 3 kg of cheese and 100 litres of milk; and the cheese maker takes the remaining 50 litres of milk and 10 kg of corn in exchange for the 7kg of cheese provided.

   One day, a foreigner arrives at the village market, paying 100 coins for 1kg of cheese, 2 kg of corn, and 3 litres of milk. How should these coins be distributed among the diary farmer, corn farmer and cheese maker, so that each is happy with the amount they receive?

7. Below is a table of car service costs to a small business.

Month	# trips	total time	total distance	total bill
January	6	3 hr	60 km	\$161.70
February	3	2 hr	60 km	\$129.75
March	10	5 hr	200 km	\$394.50

   How much should the company allow in the April budget if it expects 20 trips, lasting 8 hours, covering a total distance of 300km?

Practice Questions – Ch. 2

1. Find AB and BA, where possible

(a) $A = \begin{pmatrix} 3 & 5 & 1 \\ -2 & 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \\ 4 & 1 \end{pmatrix}$

(b) $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \\ 4 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$

(c) $A = \begin{pmatrix} 4 & 6 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 1 & 5 \\ 4 & 1 & 6 \end{pmatrix}$

(d) $A = \begin{pmatrix} 1 & 4 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 2 \\ 1 & 1 \\ 4 & 5 \end{pmatrix}$

2. If $A = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 2 \\ -3 & 1 & 1 \end{pmatrix}$, evaluate the following expressions:

(a) A^2B

(b) AB and BA where possible

(c) $(A-I)B$

(d) A^2

(e) $B^{\mathrm{T}}B$

(f) $A^{\mathrm{T}}B$

(g) BB^{T}

(h) $B^{\mathrm{T}}A$

3. Use block multiplication to simplify the working in the following questions.

(a) Let I_2 be the 2×2 identity matrix, let 0 denote the 2×2 zero matrix, and let P, Q, R, S be arbitrary 2×2 matrices. Use block multiplication to evaluate the following.

$$\begin{pmatrix} I_2 & 0 \\ P & I_2 \end{pmatrix}\begin{pmatrix} Q & R \\ 0 & S \end{pmatrix}$$

(b) The matrices

$$A = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 4 \\ 3 & 2 \end{pmatrix}$$

can be partitioned as

$$A = \begin{pmatrix} I_3 & \mathbf{u} \\ \vec{\mathbf{0}} & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} C \\ \vec{\mathbf{r}} \end{pmatrix},$$

where $\mathbf{u} \in \mathbb{R}^{3\times 1}$, $C \in \mathbb{R}^{3\times 2}$ and $\vec{\mathbf{r}} \in \mathbb{R}^{1\times 2}$.
Use block multiplication to calculate AB.

(c) Let X be any $m \times n$ matrix. Use block multiplication to show that

$$\begin{pmatrix} I_m & X \\ 0 & I_n \end{pmatrix} \begin{pmatrix} I_m & -X \\ 0 & I_n \end{pmatrix} = I_{m+n}.$$

Hence write down the inverse of

$$\begin{pmatrix} I_m & X \\ 0 & I_n \end{pmatrix}.$$

4. Find the inverse of the following matrices using Gaussian elimination.

(a) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ -1 & -1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 2 & 2 \\ -1 & -2 & -1 \end{pmatrix}$

(d) $\begin{pmatrix} -1 & -1 & -1 \\ 0 & 1 & 1 \\ 2 & 2 & 1 \end{pmatrix}$

5. Using the results from the previous question, solve the following linear systems.

(a) $\begin{aligned} x + 2y &= 5, \\ y - z &= -1, \\ x + 2y + z &= 8. \end{aligned}$

(b) $\begin{aligned} x + y &= 1, \\ y + 2z &= 4, \\ -x - y + z &= 0. \end{aligned}$

(c) $\begin{aligned} y + z &= 2, \\ x + 2y + 2z &= 6, \\ -x - 2y - z &= -3. \end{aligned}$

(d) $\begin{aligned} -x - y - z &= -3, \\ y + z &= 2, \\ 2x + 2y + z &= 2. \end{aligned}$

6. Find the determinant of the following matrices

(a) $\begin{pmatrix} 3 & 1 & 2 \\ 2 & 4 & 5 \\ -1 & 3 & 3 \end{pmatrix}$

(b) $\begin{pmatrix} 4 & 3 & 0 \\ 3 & 1 & 2 \\ 5 & -1 & 4 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} 2 & -1 & 2 \\ 1 & 3 & 2 \\ 5 & 1 & 6 \end{pmatrix}$

7. Using the determinants calculated in the previous question, determine if the following systems of equations would have a unique solution. You are NOT required to solve the systems of equations.

(a) $\begin{pmatrix} 3 & 1 & 2 \\ 2 & 4 & 5 \\ -1 & 3 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 11 \\ 5 \end{pmatrix}$ (c) $\begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \\ 2 \end{pmatrix}$

(b) $\begin{pmatrix} 4 & 3 & 0 \\ 3 & 1 & 2 \\ 5 & -1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 10 \\ 11 \\ 15 \end{pmatrix}$ (d) $\begin{pmatrix} 2 & -1 & 2 \\ 1 & 3 & 2 \\ 5 & 1 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \\ 0 \end{pmatrix}$

8. Solve the systems of equations from Question 5 using Cramer's Rule.

9. Given the 2×2 matrices A, B, and C below, solve the equations by finding the 2×2 matrix X.

$$A = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \; B = \begin{pmatrix} 6 & 2 \\ 2 & 4 \end{pmatrix}, \text{ and } C = \begin{pmatrix} 4 & -2 \\ -6 & 3 \end{pmatrix}$$

(a) $AX + B = C$ (c) $AX + B = X$

(b) $XA + B = C$ (d) $XA + C = X$

Practice Questions – Ch. 3

1. For each mapping below, show that T is **not** a linear transformation.
   Give a specific counterexample that demonstrates at least one of the following:

$$T(\mathbf{u}+\mathbf{v}) \neq T(\mathbf{u})+T(\mathbf{v}), \qquad \text{or} \qquad T(c\mathbf{u}) \neq cT(\mathbf{u}).$$

   State clearly which linearity condition fails.

   (a) $T : \mathbb{R}^2 \to \mathbb{R}^2$, where

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1x_2 \\ 1 \end{pmatrix}.$$

   (b) $T : \mathbb{R}^2 \to \mathbb{R}^2$, where

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1x_2 \\ 2x_1 \end{pmatrix}.$$

   (c) $T : \mathbb{R}^2 \to \mathbb{R}^2$, where

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_2 \\ x_1 - 1 \end{pmatrix}.$$

   (d) $T : \mathbb{R}^2 \to \mathbb{R}^2$, where

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 + x_2 \\ 2 \end{pmatrix}.$$

   (e) $T : \mathbb{R}^2 \to \mathbb{R}^4$, where

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 \\ 1 \\ x_2 \\ 1 \end{pmatrix}.$$

2. For each mapping below:
   (i) show that $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation; and
   (ii) describe geometrically what T does to vectors in the plane.

   (a)

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}.$$

   (b)

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = -\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

(c)

$$T\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right)=\begin{pmatrix}-x_1\\x_2\end{pmatrix}.$$

(d)

$$T\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right)=\frac{1}{2}\begin{pmatrix}x_1\\x_2\end{pmatrix}.$$

(e)

$$T\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right)=\begin{pmatrix}x_2\\x_1\end{pmatrix}.$$

3. Show that each mapping below is a linear transformation.
   For arbitrary vectors $\mathbf{u},\mathbf{v}$ and scalar c, verify that

$$T(\mathbf{u}+\mathbf{v})=T(\mathbf{u})+T(\mathbf{v})\quad\text{and}\quad T(c\mathbf{u})=cT(\mathbf{u}).$$

(a) $T:\mathbb{R}^2\to\mathbb{R}^2$, where

$$T\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right)=\begin{pmatrix}x_1+2x_2\\2x_1-x_2\end{pmatrix}.$$

(b) $T:\mathbb{R}^2\to\mathbb{R}^2$, where

$$T\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right)=\begin{pmatrix}x_1+x_2\\x_1\end{pmatrix}.$$

(c) $T:\mathbb{R}^3\to\mathbb{R}^2$, where

$$T\left(\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}\right)=\begin{pmatrix}x_3\\x_1+x_2\end{pmatrix}.$$

(d) $T:\mathbb{R}^2\to\mathbb{R}^3$, where

$$T\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right)=\begin{pmatrix}x_1\\x_2\\x_1+2x_2\end{pmatrix}.$$

(e) $T:\mathbb{R}^3\to\mathbb{R}$, where

$$T\left(\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}\right)=x_1+x_3.$$

4. For each linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, find the standard matrix A such that

$$T(\mathbf{x}) = A\mathbf{x}.$$

Hint: Calculate the transformation of each standard basis vector of the domain. The resulting vectors form the columns of A.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$, where

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} y \\ -x \end{pmatrix}.$$

(b) $T : \mathbb{R}^3 \to \mathbb{R}^3$, where

$$T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} x+2y \\ x-2y \\ x+y-2z \end{pmatrix}.$$

(c) $T : \mathbb{R}^2 \to \mathbb{R}^2$, where

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+2y \\ x-2y \end{pmatrix}.$$

(d) $T : \mathbb{R}^3 \to \mathbb{R}^3$, where

$$T\left(\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}\right) = \begin{pmatrix} v_1+2v_2 \\ v_3 \\ -v_1+4v_2+3v_3 \end{pmatrix}.$$

(e) $T : \mathbb{R}^3 \to \mathbb{R}$, where

$$T\left(\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}\right) = v_1+v_3.$$

5. Let the Fibonacci sequence be defined by

$$F_0 = 0, \qquad F_1 = 1, \qquad F_n = F_{n-1}+F_{n-2} \quad (n \geq 2).$$

Complete the following steps.

(a) Determine a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$T\left(\begin{pmatrix} F_{n-1} \\ F_{n-2} \end{pmatrix}\right) = \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}.$$

Write $T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)$ explicitly.

(b) Calculate the standard matrix A for the transformation T.

(c) Use the matrix A and your preferred matrix software to calculate F_{10}.

Practice Questions – Ch. 4

1. For each matrix below, find its eigenvalues and corresponding eigenvectors.
   (a)
   $$A = \begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}.$$
   (b)
   $$A = \begin{pmatrix} 6 & -4 \\ 3 & -1 \end{pmatrix}.$$
   (c)
   $$A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}.$$
   (d)
   $$A = \begin{pmatrix} -1 & -2 \\ 6 & 6 \end{pmatrix}.$$
   (e)
   $$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

2. For each matrix below, find a diagonalisation
   $$A = VDV^{-1},$$
   where D is diagonal and the columns of V are corresponding eigenvectors.
   (a)
   $$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$
   (b)
   $$A = \begin{pmatrix} 5 & 6 \\ -2 & -2 \end{pmatrix}.$$
   (c)
   $$A = \begin{pmatrix} 6 & -1 \\ -5 & 2 \end{pmatrix}.$$

(d)

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

(e)

$$A = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 3 \\ 1 & 1 & -1 \end{pmatrix}.$$

3. For each matrix below, a diagonalisation $A = VDV^{-1}$ is provided. Use the factorisation to:
   (i) calculate A^4; and
   (ii) calculate A^{-1} and confirm that $A^{-1}A = I$.

   Show how the diagonal matrix D simplifies each calculation.

   (a)

$$A = \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

   (b)

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

   (c)

$$A = \begin{pmatrix} -1 & 0 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

4. Determine whether each set of vectors below is linearly independent.

   (a)

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} -1 \\ 3 \\ 8 \end{pmatrix}.$$

   (b)

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} -1 \\ 3 \\ 7 \end{pmatrix}.$$

(c)

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 2 \\ -3 \\ -1 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 0 \\ 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{x}_4 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ -1 \end{pmatrix}.$$

(d)

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 2 \\ -3 \\ -1 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 0 \\ 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{x}_4 = \begin{pmatrix} 2 \\ 0 \\ -1 \\ -1 \end{pmatrix}.$$

5. For each part, express the given vector as a linear combination of the provided basis vectors.
   (a) Express

   $$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

   in terms of the basis

   $$\left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}.$$

   (b) Express

   $$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

   in terms of the basis

   $$\left\{ \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix} \right\}.$$

   (c) Express

   $$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

   in terms of the basis

   $$\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\}.$$

(d) Express

$$\begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$$

in terms of the basis

$$\left\{ \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}.$$

6. A canoe-hire business operates from four sites. Clients collect a canoe from one site and may return it to any of the four sites.
The probability that a canoe collected from one site is returned to each location is shown below.

Pick-up location	Return location Site A	Site B	Site C	Site D
Site A	0.7	0.1	0.15	0.05
Site B	0.2	0.6	0.1	0.1
Site C	0.1	0.2	0.5	0.2
Site D	0.05	0.05	0.1	0.8

(a) If the business has 200 canoes, how many would you expect to find at each location in the long term?

(b) Throughout January, a school hires Site D and all the canoes located there. At the end of the month, a group of 50 tourists plans to hire canoes from Site A. Should there be enough canoes available at Site A?

7. Four websites contain links to one another, as shown below.

Site	has link to Site 1	Site 2	Site 3	Site 4
Site 1	-	Y	Y	N
Site 2	Y	-	N	N
Site 3	N	Y	-	Y
Site 4	Y	Y	Y	-

(a) Propose a sensible method for assigning a rank to each page.
Hint: Refer to §4.5.4.

(b) Express your ranking method as a matrix problem and solve it. List the pages in order from highest to lowest rank.

Practice Questions – Ch. 5

1. Find the angle between the vectors $\mathbf{v}$ and $\mathbf{w}$ using the standard inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\mathsf{T}\mathbf{y}.$$

   (a) $\mathbf{v} = (2,-3)^\mathsf{T}$ and $\mathbf{w} = (3,2)^\mathsf{T}$.

   (b) $\mathbf{v} = (4,1)^\mathsf{T}$ and $\mathbf{w} = (3,2)^\mathsf{T}$.

   (c) $\mathbf{v} = (-2,3,1)^\mathsf{T}$ and $\mathbf{w} = (1,2,4)^\mathsf{T}$.

   (d) $\mathbf{v} = (1,3,-1,4)^\mathsf{T}$ and $\mathbf{w} = (-1,1,-2,1)^\mathsf{T}$.

   (e) $\mathbf{v} = (2,1,3)^\mathsf{T}$ and $\mathbf{w} = (6,3,9)^\mathsf{T}$.

2. For $\mathbf{v} = (v_1, v_2)^\mathsf{T}$ and $\mathbf{w} = (w_1, w_2)^\mathsf{T}$ in $\mathbb{R}^2$, consider the inner product

$$\langle \mathbf{v}, \mathbf{w} \rangle = 2v_1w_1 + v_1w_2 + v_2w_1 + 2v_2w_2.$$

   Find the inner product of each pair of vectors using this definition.

   (a) $\mathbf{v} = (1,0)^\mathsf{T}$ and $\mathbf{w} = (-1,2)^\mathsf{T}$.

   (b) $\mathbf{v} = (2,-1)^\mathsf{T}$ and $\mathbf{w} = (3,6)^\mathsf{T}$.

   (c) $\mathbf{v} = (1,-2)^\mathsf{T}$ and $\mathbf{w} = (2,1)^\mathsf{T}$.

   (d) $\mathbf{v} = (0,1)^\mathsf{T}$ and $\mathbf{w} = (-2,1)^\mathsf{T}$.

3. Verify that each set below is orthogonal under the standard inner product, and determine whether it is orthonormal. If it is not orthonormal, normalise the vectors to obtain an orthonormal set.

   (a)

$$\mathbf{u} = (1,1)^\mathsf{T}, \qquad \mathbf{v} = (-1,1)^\mathsf{T}, \qquad S = \{\mathbf{u}, \mathbf{v}\}.$$

   (b)

$$\mathbf{u} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)^\mathsf{T},$$
$$\mathbf{v} = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)^\mathsf{T},$$
$$\mathbf{w} = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^\mathsf{T}, \qquad S = \{\mathbf{u}, \mathbf{v}, \mathbf{w}\}.$$

   (c)

$$\mathbf{u} = (\cos\theta, \sin\theta)^\mathsf{T}, \qquad \mathbf{v} = (-\sin\theta, \cos\theta)^\mathsf{T}, \qquad S = \{\mathbf{u}, \mathbf{v}\}.$$

(d)

$$\mathbf{u} = (1,1,0)^\mathsf{T}, \qquad \mathbf{v} = (1,-1,2)^\mathsf{T}, \qquad \mathbf{w} = (-1,1,1)^\mathsf{T}, \qquad S = \{\mathbf{u},\mathbf{v},\mathbf{w}\}.$$

4. Let

$$\mathbf{v}_1 = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)^\mathsf{T}, \qquad \mathbf{v}_2 = \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)^\mathsf{T}.$$

Find a vector $\mathbf{v}_3$ such that $\{\mathbf{v}_1,\mathbf{v}_2,\mathbf{v}_3\}$ is orthonormal.

5. Let

$$\mathbf{u} = (-6,b,2)^\mathsf{T} \qquad \text{and} \qquad \mathbf{v} = (b,b^2,b)^\mathsf{T}.$$

For what values of b are $\mathbf{u}$ and $\mathbf{v}$ orthogonal with respect to the standard inner product?

6. Verify that the matrix A is orthogonal using two different methods.

$$A = \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{2}{3} \end{pmatrix}.$$

7. Calculate the eigenvalues of each symmetric matrix and hence determine whether the matrix is positive definite.
   (a) $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$

   (b) $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$

   (c) $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$

   (d) $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}.$

   (e) $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$

8. Using the simplified rating method from Section 5.3.2, what song would a collaborative filtering system recommend if I gave:
   (a) a rating of 5 for Song 1 and a rating of 4 for Song 10; and
   (b) a rating of 1 for each of Songs 1, 7 and 8?

9. My friend's new ice-cream company is developing ten flavours and has employed eight tasters to rate them. Their ratings are shown below. If I rate Vanilla a 5 and Lemon Sorbet a 3, which three flavours would you recommend I try next?

Flavour	Tester ratings							
	T1	T2	T3	T4	T5	T6	T7	T8
Chocolate	5	5	5	3	3	5	3	2
Vanilla	4	1	2	5	5	4	2	1
Strawberry	1	4	2	1	1	5	4	2
Rum and Raisin	4	5	2	3	3	2	3	4
Salted Caramel	2	4	5	3	2	2	3	3
Pistachio	1	1	1	2	5	1	3	3
Cookies and Cream	3	5	2	4	5	2	1	5
Bubblegum	1	5	4	1	5	3	1	4
Lemon Sorbet	4	4	2	5	5	2	3	2
Mandarine Sorbet	1	2	5	4	4	5	5	2

10. Determine least-squares curves of best fit for the following data using each of the proposed models.

x_i	1	2	3	4	5	6	7	8	9	10
y_i	9.876	9.862	9.833	9.465	7.635	6.714	3.559	-0.397	-7.424	-19.429

   (a) A linear model, $y = mx + c$.
   (b) A cubic model, $y = ax^3 + bx^2 + cx + d$.
   (c) A hyperbolic model, $y = a\cosh(x/2) + b\sinh(x/2) + c$.

Practice Questions – Ch. 6

1. Determine whether each set is a vector space (with the usual operations).
   (a) The set of all polynomials of the form $ax^3 - ax + b$.

   (b) The set of all vectors $(x_1, x_2, x_3)^{\mathsf{T}}$ such that $x_1 + x_3 = 1$.

   (c) The set of all vectors $(x_1, x_2, x_3)^{\mathsf{T}}$ such that $x_1 = x_2 + x_3$.

   (d) The set of all complex numbers of the form $a + 2i$.

   (e) A student claims that

   $$S = \{(x, y)^{\mathsf{T}} \in \mathbb{R}^2 : x \geq 0\}$$

   is a vector space because the sum of any two vectors in S is also in S. Find the error in the student's argument and show that S is not a vector space.

2. Determine whether each set of vectors is linearly independent.
   (a) $\mathbf{x}_1 = (1, -1, 2)^{\mathsf{T}}$, $\mathbf{x}_2 = (-2, 3, 1)^{\mathsf{T}}$ and $\mathbf{x}_3 = (-1, 3, 8)^{\mathsf{T}}$.

   (b) $\mathbf{x}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}.$

   (c) $\mathbf{x}_1 = 1$, $\mathbf{x}_2 = e^x + e^{-x}$ and $\mathbf{x}_3 = e^x - e^{-x}$.

   (d) $\mathbf{x}_1 = 1 - x^2$, $\mathbf{x}_2 = 2x^2$ and $\mathbf{x}_3 = 3$.

3. Find a basis for each set in Question 1 that is a vector space.

4. Explain why each set cannot be a basis for the stated vector space.
   (a) $\{(1, -1)^{\mathsf{T}}, (-2, 1)^{\mathsf{T}}, (-1, 3)^{\mathsf{T}}\}$ for $\mathbb{R}^2$.

   (b) $\{1 - x, x^2 - 1, x^2 - x\}$ for the polynomials of degree at most 2.

   (c) $\{x^2 - x, 1 + x, x^2 + 1, x^2 - 1\}$ for the polynomials of degree at most 2.

   (d) $\{(1, -1, 2)^{\mathsf{T}}, (-2, 3, 1)^{\mathsf{T}}, (-1, 3, 8)^{\mathsf{T}}\}$ for $\mathbb{R}^3$.

5. For each part, express the given vector as a linear combination of the vectors in the ordered basis B. Hence, state its coordinate vector relative to B.
   (a) Find the coordinates of $(1,2)^\mathsf{T}$ relative to
   $$B = \{(1,-1)^\mathsf{T}, (-2,1)^\mathsf{T}\}.$$
   (b) Find the coordinates of $(1,0,0)^\mathsf{T}$ relative to
   $$B = \{(1,-1,2)^\mathsf{T}, (-2,3,1)^\mathsf{T}, (-1,3,1)^\mathsf{T}\}.$$
   (c) Find the coordinates of $\sinh(2x)$ relative to
   $$B = \{e^{-2x}, e^{-x}, 1, e^{x}, e^{2x}\}.$$
   (d) Find the coordinates of x^2 relative to
   $$B = \{1-x, x^2-2, x^2-x\}.$$

6. Give a basis and dimension for the column space, row space and null space of each matrix. For part (a), use the following guided steps:
   (i) Row-reduce the matrix and identify its pivot columns.
   (ii) Use the corresponding columns of the *original* matrix for a basis of the column space.
   (iii) Use the nonzero rows of the reduced matrix for a basis of the row space.
   (iv) Solve the homogeneous system to obtain a basis of the null space.
   Then apply the same method to parts (b)–(d).
   (a) $\begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \\ 4 & 7 & 8 \end{pmatrix}.$

   (b) $\begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \end{pmatrix}.$

   (c) $\begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 2 & -1 & -1 \end{pmatrix}.$

   (d) $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$

   Summarise your answers in a table of the following form.

Matrix	$\dim \mathrm{Col}(A)$	$\dim \mathrm{Row}(A)$	$\dim \mathscr{N}(A)$	$\mathrm{rank}(A) + \mathrm{nullity}(A)$
(a)				
(b)				
(c)				
(d)				

7. Three source quantities x_1, x_2, x_3 produce two sensor readings according to

$$\mathbf{y} = A\mathbf{x}, \qquad A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

   (a) Find a basis and dimension for the column space of A. What does the column space represent in this setting?
   (b) Find a basis and dimension for the null space of A. Explain in context what a nonzero vector in the null space means.
   (c) Can the two sensor readings uniquely determine all three source quantities? Justify your answer.

8. Prove that if $\mathbf{x}_1$ and $\mathbf{x}_2$ are linearly independent vectors in $\mathbb{R}^4$ and $A \in \mathbb{R}^{4\times 4}$ is nonsingular, then $A\mathbf{x}_1$ and $A\mathbf{x}_2$ are linearly independent.

9. Prove that if $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are vectors in a vector space V and

$$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}),$$

   then $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are linearly dependent.

Practice Questions – Ch. 7

1. Which of the factorisations considered in this chapter (PLU, QR, Cholesky and SVD) have the following properties or requirements? More than one factorisation may apply in each part.
   (a) Produces an orthogonal matrix.

   (b) Produces an upper- or lower-triangular matrix.

   (c) Can be applied to a non-square matrix.

   (d) Requires a real symmetric positive-definite matrix.

   (e) Provides an orthonormal basis for the column space.

2. Use software to calculate the PLU, thin QR, Cholesky and thin SVD factorisations of each matrix, where possible. If a factorisation cannot be applied, explain why. Here, *thin* means that only the $r = \text{rank}(A)$ components associated with independent directions are retained. This is also called a *rank-reduced* or *compact* factorisation.
   (a) $A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$.

   (b) $A = \begin{pmatrix} 8 & 1 & 6 \\ 3 & 5 & 7 \\ 4 & 9 & 2 \end{pmatrix}$.

   (c) $A = \begin{pmatrix} 6 & 1 & -4 \\ 1 & 6 & -5 \\ -4 & -5 & 6 \end{pmatrix}$.

   (d) $A = \begin{pmatrix} 4 & 0 & 0 \\ -1 & 4 & 1 \\ 0 & 0 & 4 \end{pmatrix}$.

   (e) $A = \begin{pmatrix} 3 & -1 \\ 1 & -1 \\ 2 & 3 \end{pmatrix}$.

   (f) $A = \begin{pmatrix} 3 & -1 & 1 & 2 \\ 1 & -1 & 0 & -2 \\ 2 & 0 & 1 & 4 \end{pmatrix}$.

3. For the matrix

$$A = \begin{pmatrix} 3 & -1 \\ 1 & -1 \\ 2 & 3 \end{pmatrix}$$

from Question 2(e), compare its thin SVD with the diagonalisation of $A^\mathsf{T} A$ and AA^T. Explain how their eigenvalues and eigenvectors are related to the singular values and singular vectors of A.

4. Use the data from Section 5.7.2 and the QR-factorisation approach to solve the least-squares problem for each model.
   (a) $y = ae^{-x} + be^{-x/2} + c$.

   (b) $y = ax^3 + bx^2 + cx + d$.